

RelateAnything: Real-Time Open-Vocabulary Relation Prediction From Any Inputs

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Open-vocabulary object detection accepts an arbitrary class list at inference, and promptable segmentation returns regions carrying no class names at all: in both, the taxonomy has left the model and become an input. Relation prediction has not made that transition. Scene-graph models are still trained and evaluated on the 50 or 56 predicates of a single annotation style, with the relation head conditioned on object class labels and therefore tied to one detector and one label space. Three obstacles account for this, and they are not primarily modelling ones: no relation corpus is both free-text and verified, an architecture conditioned on labels cannot accept a vocabulary it was not trained on, and the standard metric rewards agreement with the training corpus, so that enlarging the vocabulary is scored as a regression.

We present **RelateAnything**, a 53M-parameter relation model that takes an image and a set of regions from any source and returns scored relations over a predicate vocabulary supplied at inference as a list of strings. Object class labels are never an input, so the region source can be replaced without retraining, and the vocabulary is a bank of text embeddings rather than a learned classifier, so changing it is a matrix substitution. The model runs at 20 ms per frame on a single GPU. Training over 19,103 predicates requires two departures from standard practice: the supervision must be treated as positive-unlabeled, since at this vocabulary size an unannotated predicate is frequently true, and the text encoder must be corrected, since contrastive encoders embed antonyms such as **above** and **below** at a cosine similarity of 0.95.

To supply the supervision we build **RA-4M**, a corpus of 474k images and 4.3M relations over 10,102 free-text predicates, generated by a vision-language model against numbered box markers and filtered by a deterministic geometric check. To make the result measurable we build **OV-SGG-Bench**, whose six axes are each scored across datasets and none satisfiable by the priors that standard recall rewards: a frequency table over ground-truth object categories, which sees no pixels, exceeds a trained model on the metric by which leaderboards are ordered.

On three benchmarks evaluated cross-dataset and a fourth evaluated zero-shot, RelateAnything outperforms the strongest open-vocabulary method of comparable scale, with 2.3–3.5× the mean recall, and the margins persist when a real detector supplies the regions. Against a scene-graph model built on a 3B vision-language model it leads on both metrics on all three, at under 2% of the parameters, and trails on micro recall only when both are scored through a free-text matcher. In-domain measurement overstates cross-dataset gains by approximately 5×. The model, the corpus and the benchmark are publicly released.

Code: <https://github.com/Maelic/RelateAnything>

Models: <https://huggingface.co/collections/maelic/relateanything>

Dataset: <https://huggingface.co/datasets/maelic/RA-4M>

1 Introduction

Open-vocabulary object detection accepts an arbitrary class list at inference [Gu et al., 2022, Zhou et al., 2022, Minderer et al., 2022, Li et al., 2022b, Liu et al., 2024], including at real-time rates

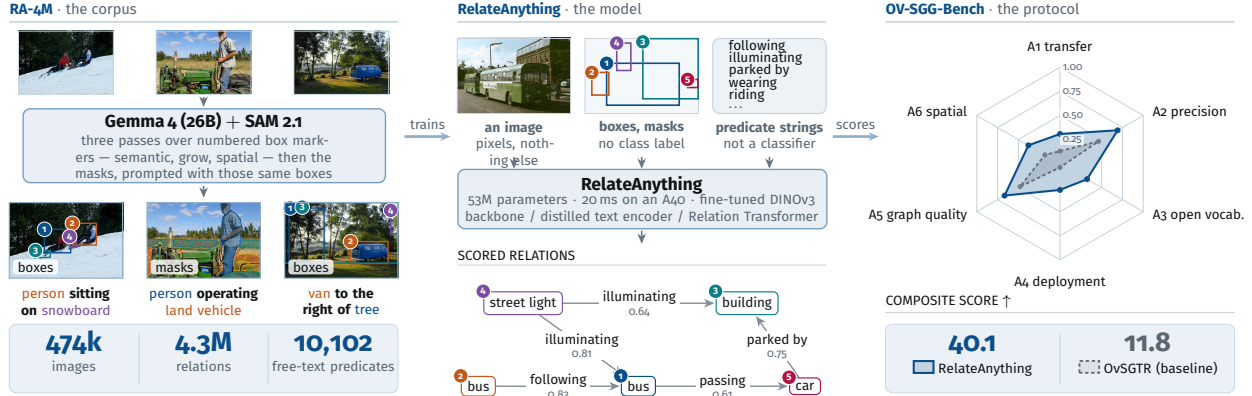


Figure 1 Overview. *Left*, RA-4M: three corpus images, the annotator, and one verified free-text relation per image. *Middle*, RelateAnything: its three inputs are separable (pixels, regions from any source, predicate strings); the regions and the five scored relations are model output, and none of the predicates is among the 50 of VG150 [Xu et al., 2017]. *Right*, OV-SGG-Bench: six chance-corrected axes (centre chance, rim perfect, rings 0.25 apart) and the composite over A1, A2, A4, A5 and A6. The baseline drawn is OvSGTR, the one system runnable on every axis and so the only one that can carry a composite; on A1 the stronger baseline is ROBIN-3B, and Tab. 3 reports the best available on each axis. A5 is Eq. (4) at equal graph length (§ 6.5).

[Cheng et al., 2024, Wang et al., 2025a], and promptable segmentation returns regions carrying no class names at all [Kirillov et al., 2023, Zhao et al., 2023]. In both cases the taxonomy has left the model and become an input, which is what allows either to be pointed at a domain it was not built for. Relation prediction has not made that transition yet. Scene-graph generation, the literature that would supply it, is still trained and evaluated on the 50 predicates of VG150 [Xu et al., 2017] or the 56 of PSG [Yang et al., 2022], drawn from one annotation style; its dominant architecture predicts objects and relations from a shared backbone and conditions the relation head on predicted object classes [Zellers et al., 2018, Tang et al., 2019, Chen et al., 2025], which fixes the label space and excludes the region sources a deployed system actually has, such as a different detector, a class-agnostic segmenter or a human annotation; and where the vocabulary is open at all, it arrives as a single caption whose predicate names are encoded jointly, which bounds it at roughly 150 strings and keeps a language model in the inference path.

Three obstacles have kept relation prediction at the scale of a benchmark vocabulary, and none of them is primarily a modelling obstacle. The first is that the supervision does not exist. Relation annotation is the most expensive and least reliable form of annotation in vision, and the machine-generated corpora that do exist either project onto a fixed class list or leave the annotator unchecked (Tab. 19 in § C). The second is that the architecture cannot use it: a relation head conditioned on object labels inherits a label space, and a vocabulary encoded as one caption cannot be enlarged. The third, and the one that has received least attention, is that the measurement cannot see it. Scene-graph recall is computed against a benchmark whose predicate vocabulary is also its training vocabulary, so it rewards agreement with a corpus rather than the relation in the image, and a model that enlarges its vocabulary is penalised for answering with a string the annotation does not use. In this work, we present a model, a corpus and a benchmark that together remove all three obstacles, and we measure the effect of each on the other two (Fig. 1).

A relation model whose inputs are separable. We present RelateAnything, a 53M-parameter relation model that takes an image and a set of regions from any source and returns scored relations between pairs of those regions, over a predicate vocabulary supplied at inference as a list of strings (Fig. 1).

Two properties of that interface account for most of the design. Object class labels are never an input, at any stage, so the region source can be replaced without retraining and the model cannot inherit a detector’s label distribution; and the vocabulary is not a learned classifier but a bank of text embeddings that no learned layer transforms, so changing it is a matrix substitution rather than a retraining. The relation model runs at 20 ms per frame on a single GPU, alongside a real-time object detector [Wang et al., 2025a]. Training it over 19,103 free-text predicates, rather than the few dozen usual in this literature, changes the learning problem in two ways that must both be treated. The supervision becomes positive-unlabeled, because at this vocabulary size a pair annotated *riding* is also, unstated, *sitting on*, and penalising the unstated predicate teaches the model to suppress correct answers. We address this with a batch-local InfoNCE loss whose positive is the annotated predicate’s synonym group and whose negatives are discounted by an estimated probability, fitted on the corpus’s multiply-annotated pairs, that they are also true. The target text space itself is also defective: contrastive text encoders embed antonyms such as *above* and *below* at a cosine similarity of 0.95, which upper-bounds what any model regressing onto those directions can express, whoever trains it. We correct the text encoder by distilling a student whose objective adds an antonym-repulsion term, so that inverse pairs separate while synonym groups stay together.

Supervision without a label budget. Such a model requires a corpus whose vocabulary is not that of a benchmark and whose annotation is dense enough that unannotated pairs are rare. An image with N objects contains $N(N - 1)$ candidate pairs, and for most of them whether the relation holds is a matter of judgement, which is why the community kept the 50 most frequent of the 36,549 predicate strings Visual Genome’s annotators wrote [Krishna et al., 2017, Xu et al., 2017], and why the discarded remainder is exactly the part an open-vocabulary model is meant to serve. A vision-language model (VLM) is what makes the alternative affordable. We generate RA-4M, a corpus of 474k images and 4.3M relations over 10,102 free-text predicates. The VLM annotates against numbered markers placed on the boxes, so that grounding is an input to the annotator rather than an inference from its output, and every proposed relation then passes a deterministic geometric gate that rejects what box geometry contradicts and leaves what geometry cannot constrain unchecked and counted. On the same images and the same boxes as the annotations it replaces, the result is $1.7\times$ denser with $107\times$ the vocabulary.

Measuring relation prediction rather than corpus agreement. Measurement has to come first, since a protocol that rewards agreement with a corpus cannot register whether either of the other obstacles has been removed. Scene-graph recall proves to be, to a large extent, a prior-matching score (§ 5.2): a frequency table over ground-truth object categories, given no pixels, outperforms a trained model on the metric that orders leaderboards while falling well below it per predicate; the vocabulary a training corpus shares with a benchmark predicts recall on that benchmark; and a model that learns which pairs an annotator wrote down learns the annotator rather than the image. Two scoring conventions amplify these effects (§ 5.3), a matcher that assigns detections to ground-truth objects without constraint and a detector operating point that goes unreported yet moves attainable recall by more than the spread between published methods. Each finding constrains the design of OV-SGG-Bench. Because shared triplet mass predicts recall, every axis is scored across datasets and none of the benchmarks contributes a training image, a setting scene-graph generation has left largely unexplored, since results there are almost always reported in-domain [Chang et al., 2023]. Because a model can profit from what an annotator chose to write down, two axes are scored against negatives that a human adjudicated, on which a confident false positive costs as much as a miss. And because each axis can be satisfied on its own by a different shortcut, whether corpus match,

uniform caution, collapse onto the frequent predicates or the operating point of the detector, the six are reported together and aggregated in a way that penalises imbalance across them.

Overview of contributions. The paper contributes (i) RelateAnything (§ 3), the relation model described above, together with the objective that makes a five-figure predicate vocabulary trainable; (ii) RA-4M (§ 4), the first relation corpus we are aware of that is both free-text and verified against the geometry of the boxes it names; (iii) OV-SGG-Bench (§ 5), the evaluation protocol and the measurements of prior and of scoring convention that motivate it; and (iv) an empirical study (§§ 6 to 9) organised around three questions, namely what scene-graph recall measures (§ 5), whether relation prediction benefits from being coupled to object prediction (§ 7), and whether in-domain measurement predicts cross-dataset transfer (§ 8). On three benchmarks evaluated cross-dataset and a fourth evaluated zero-shot, RelateAnything outperforms OvSGTR [Chen et al., 2025], the strongest open-vocabulary method of comparable scale, on every recall and precision metric, with mean recall higher by a factor of 2.3–3.5 and rare-predicate recall higher by a factor of 5–21, and the margins persist with the same structure when a real detector supplies the regions. On the question of coupling we find that relation supervision sharpens object identity in the dense features without creating relational ones, and that the relation score is 87–93% pair context against 0.1% object identity, so there is no shared representation to justify a shared backbone. On the question of measurement we find that in-domain gains overstate cross-dataset gains by approximately 5×, and that a change selected in-domain can lose out of it. Model, corpus and benchmark are public.

Evaluation setting. All results in this paper are obtained under cross-dataset transfer, on benchmarks the model was not trained on, with four marked exceptions: the HICO-DET row trained on a 5% relation share of its training split (Tab. 4) and the three fine-tuned rows of Tab. 12. § 8 shows that in-domain measurement can favour changes that reduce transfer performance.

2 Related work

Scene graphs and their benchmarks. Visual relationship detection was formulated by Lu et al. [2016] as the prediction of ⟨subject, predicate, object⟩ triplets, with a language prior over predicates in the method from the outset. Visual Genome [Krishna et al., 2017] supplied the annotations and the VG150 split [Xu et al., 2017] fixed the vocabulary at 150 object classes and 50 predicates. Later benchmarks changed the images or the region type more than the formulation: PSG [Yang et al., 2022] uses panoptic segments, GQA [Hudson and Manning, 2019] normalises the vocabulary, Open Images [Kuznetsova et al., 2020] is sparse and large, IndoorVG [Neau et al., 2024] targets indoor robotics, and HICO-DET [Chao et al., 2018] and SpatialSense [Yang et al., 2019] isolate human–object interactions and adversarially collected spatial relations [Chang et al., 2023]. Haystack [Lorenz et al., 2023] differs in construction: annotated per predicate rather than per image, built on SA-1B [Kirillov et al., 2023], and alone among these in publishing explicit negatives, which is what allows § 6.2 to measure precision against adjudicated negatives.

Closed-set generation and its evaluation. The dominant approach predicts relations over a fixed predicate set with a classifier conditioned on object labels and layout, using message passing [Xu et al., 2017], sequential context [Zellers et al., 2018], tree structure [Tang et al., 2019], or knowledge-routed graphs [Chen et al., 2019]. Zellers et al. [2018] also introduced the frequency baseline that we measure in § 5.2. Evaluation moved from recall at K [Xu et al., 2017] to mean recall [Chen et al., 2019, Tang et al., 2019] once it became clear that the skew of VG150 allows a model to score well

while predicting little beyond `on` and `has`, and subsequently to debiasing [Tang et al., 2020, Li et al., 2021, Gao et al., 2025], compositional augmentation [Knyazev et al., 2021] and label correction [Li et al., 2022a]. Each of these estimates its correction from training label statistics, that is, from a prior of the same kind as the one it corrects. None addresses the two observations of §§ 5.2 and 5.3, namely that benchmarks share their vocabulary with the training corpora and that the standard matcher [Tang, 2020] grants credit that object detection has disallowed since COCO [Lin et al., 2014]. From LVIS [Gupta et al., 2019] we adopt three evaluation practices: federated annotation, a support-based rare/common/frequent split, and precision computed over labelled cells only.

One-stage and real-time generation. SGTR [Li et al., 2022c, 2024a], RelTR [Cong et al., 2023], EGTR [Im et al., 2024] and PE-Net [Zheng et al., 2023] predict subject, object and predicate from a single set of queries, and one-stage human–object interaction detectors [Liao et al., 2020, Tamura et al., 2021] apply the same design at real-time rates. REACT [Neau et al., 2025] and REACT++ [Neau and Falomir, 2026] are, to our knowledge, the fastest published scene-graph models; both are closed-set and trained per benchmark, and § 9 compares against their published numbers. All share a backbone between objects and relations and condition on predicted object classes.

Open-vocabulary relations. Replacing the predicate classifier with similarity against text embeddings admits new predicates at inference time. Approaches include prompting a vision–language model [He et al., 2022], building on the visual–semantic space of a grounded detector [Zhang et al., 2023], composing CLIP [Radford et al., 2021] prompts without relation training [Li et al., 2023], expanding predicates with a language model [Yu et al., 2023, Chen et al., 2024a], generating the graph as a sequence [Li et al., 2024b], and refining the prompting and the pairing [Liu et al., 2025, Li et al., 2025]. OvSGTR [Chen et al., 2024b] formulated fully open-vocabulary generation over both objects and predicates and defined the OvR-SGG protocol that we reproduce in § 5.3. Its extension [Chen et al., 2025] pre-trains on MegaSG, a machine-annotated corpus, which makes it the closest available control for supervision quality and our baseline throughout. Scene-Graph ViT [Salzmann et al., 2024] and relational pre-training [Yuan et al., 2022, 2023] learn relations and objects jointly from region–text pairs. Our setting differs in three respects: no object class labels are used at any stage, supervision is free text over a five-figure vocabulary rather than a projection onto a benchmark’s set, and evaluation is conducted both with the answer set supplied and with it withheld (§ 6.3). Detectors that accept a class list at inference [Gu et al., 2022, Zhou et al., 2022, Minderer et al., 2022, Li et al., 2022b, Liu et al., 2024], including real-time ones [Cheng et al., 2024, Wang et al., 2025a], and segmenters that return class-agnostic regions [Kirillov et al., 2023, Zhao et al., 2023] are the region sources available to a relation model. A relation model that requires object labels can use none of them without a label space it was trained on, whereas one that reads regions can use all of them (§ 7.2).

Machine-generated relation supervision. Existing approaches differ in how a generated relation is attached to an instance and in whether the annotator is checked (Tab. 19 in § C). Caption parsing [Chen et al., 2023] yields ungrounded triplets, RLIPv2 [Yuan et al., 2023] grounds parsed text with a learned tagger, and GPT4SGG [Chen et al., 2023], MegaSG [Chen et al., 2025], Synthetic Visual Genome [Park et al., 2025] and the All-Seeing Project [Wang et al., 2024] prompt a multimodal model directly, accepting its output or filtering it with a second model. None verifies a generated relation against the geometry of the boxes it names; RA-4M does (§ 4), and retains every surface form rather than projecting onto a class list as GQA and MegaSG do.

Learning at vocabulary scale. Positive-unlabeled learning [Elkan and Noto, 2008, Kiryo et al., 2017, Bekker and Davis, 2020] treats an unlabeled example as a mixture of positive and negative; federated annotation [Gupta et al., 2019] is the same observation applied to a benchmark. We encounter it in the training signal (§ 3.3): a pair annotated riding is also, implicitly, sitting on. Contrastive text spaces are known to be anisotropic [Mu and Viswanath, 2018] and to exhibit hubness [Radovanović et al., 2010, Lample et al., 2018]; relations expose an additional failure that these works do not address, namely that antonyms embed almost identically.

3 The RelateAnything model

RelateAnything takes an image and a set of regions and returns scored relations between pairs of regions, drawn from a predicate vocabulary supplied at inference. We state the problem (§ 3.1), describe the architecture (§ 3.2) and discuss the two properties of the learning problem that change at a vocabulary of ten thousand predicates (§ 3.3); the evidence that coupling object and relation prediction does not help is separate (§ 7).

3.1 Problem statement

Given an image I and a set of boxes $B = \{b_i\}_{i=1}^N$, the model outputs scored triplets (i, j, p) , where p is drawn from a vocabulary $\mathcal{V}$ supplied at inference as strings. Two constraints follow.

No object labels. The model receives only pixels and box coordinates. This setting is harder than the predicate-classification protocol under which most published numbers are obtained, in which ground-truth classes are supplied and carry the prior analysed in § 5.2, and it corresponds to deployment, where labels come from a detector and may be noisy, out of vocabulary, or absent. Object categories are used during training only inside two loss terms, the pair sampler’s relatedness loss and the object–text grounding term (§ 3.3, § B.1), and never as a network input.

Vocabulary as input. There is no predicate classifier. Scores are cosine similarities between visual pair embeddings and text embeddings, and no learned transformation is applied on the text side, since a projection fitted to the training vocabulary would be undefined for strings supplied later. Replacing the vocabulary therefore amounts to substituting one matrix and requires no retraining.

3.2 Architecture

The parameters of the model are concentrated in the pair representation (Fig. 2); § A gives the full specification and the ablation behind each component.

Visual path. A DINOv3 backbone [Siméoni et al., 2025] encodes the image at 448×448 . Dense patch features are read at depths -6 , -3 and -1 , layer-normalised per tap and fused with learned weights. Per-object features are obtained by coordinate-aware soft pooling rather than by cropping: a query formed from a Fourier position encoding [Tancik et al., 2020] of box b_i cross-attends over all patch tokens, $v_i = \text{Attn}(q(b_i), F, F)$, so that an object’s representation can incorporate the context it interacts with. The backbone is fully fine-tuned at a learning rate $8\times$ lower than that of the head.

Pair representation. For an ordered pair we form

$$x_{ij} = [v_i; v_j; v_{\cup(i,j)}; v_{\cap(i,j)}; \text{MLP}(g_{ij})] W_{\text{proj}}, \quad x_{ij} \in \mathbb{R}^{512}, \quad (1)$$

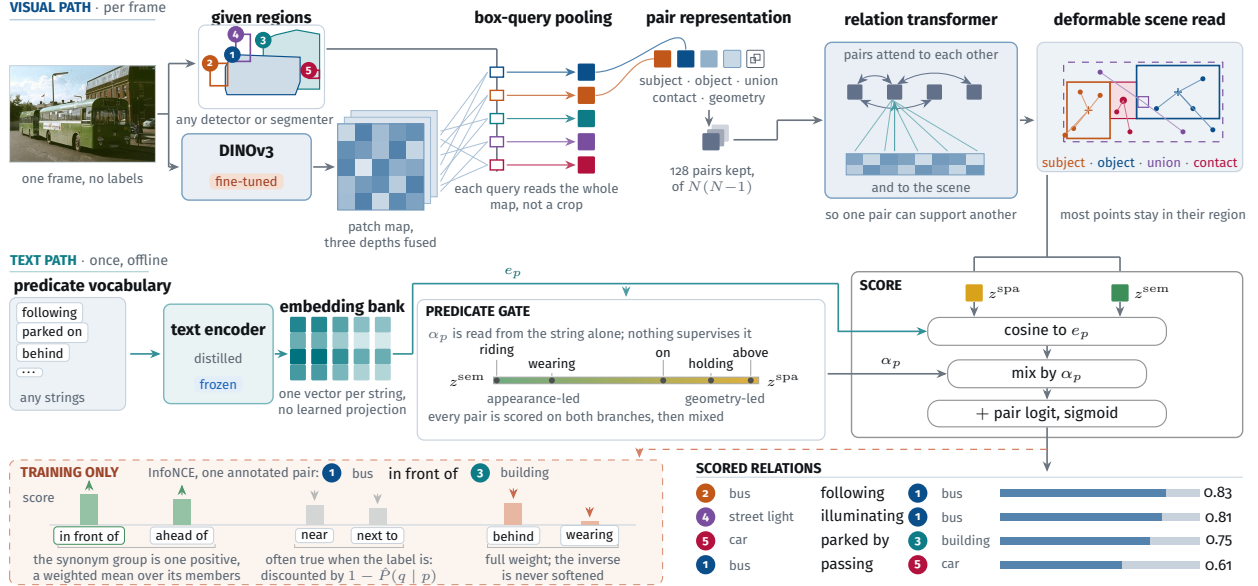


Figure 2 Overview of RelateAnything. **Band 1, the visual path.** The frame goes to the backbone and, unchanged, to whatever produces the regions; that producer is not part of the model and its class labels are discarded. One query per box pools the patch map, five pooled features form each ordered pair, and the sampled pairs are refined against one another and against the scene. **Band 2, the text path and the score.** Predicate strings supplied at inference are embedded once by a frozen encoder into a bank of unit vectors that no learned layer transforms; a gate maps each embedding to the weight α_p mixing the two branches (§ A.3). **Band 3, the loss, training only:** one annotated pair scored against six predicates in its batch, the annotated string and its synonyms forming one positive, a negative the corpus often co-annotates with the label discounted, the directional inverse never. Frame, regions and relations are those of Fig. 1.

where $v_{\cup}$ pools the union box, $v_{\cap}$ pools the *contact* region (the intersection when the boxes overlap and the gap between them otherwise), and $g_{ij} \in \mathbb{R}^{19}$ collects scale-invariant geometric features.¹ A small transformer then refines the sampled pairs with self-attention over pairs and cross-attention to the scene tokens together with tokens encoding the pair’s own box corners, followed by a block of joint self-attention over pairs and scene tokens. A final additive stage allows each pair to read the scene outside its own boxes: it predicts sampling offsets around its four anchors and reads the feature map at those locations through a zero-initialised gate [Zhu et al., 2021, Liu et al., 2022]. This stage is what allows *parked on* to be decided by the road surface and *hanging from* by the attachment point above the subject (§ A.2).

Two branches and a predicate-conditioned gate. Spatial and semantic relations rely on different evidence, so the head scores each pair twice and mixes the two cosine channels per predicate:

$$\ell_{\text{pred}}(i, j, p) = \tau [\alpha_p \cos(z_{ij}^{\text{spa}}, e_p) + (1 - \alpha_p) \cos(z_{ij}^{\text{sem}}, e_p)] + \beta, \quad (2)$$

where e_p is the text embedding of predicate p and $\alpha_p = g(e_p)$ depends on that embedding alone, so that it is defined for strings the model has never seen. The gate receives no supervision of its own. The weights it learns are strongly bimodal: projective and proximity relations are routed to the geometry branch and actions to the appearance branch (Fig. 11 in § A.3). The two channels can also be read out separately at inference, as a layout graph and a content graph (Tab. 13).

¹Fifteen box features and four mask-only slots, constant for boxes and encoding mask shape for segments.

Pair sampling. A two-stage sampler retains 400 of the $N(N-1)$ ordered pairs by geometric plausibility and then 128 by a learned relatedness score. It retains 99.79% of annotated positives; exhaustive scoring costs $1.02\times$ and yields no measurable gain. The learned stage is trained to predict which pairs carry annotations, and hence learns the annotation propensity discussed in § 5.2. Its logit enters the final score additively (§ D.1), which allows it to be removed at evaluation time and its contribution measured (§ 6.2).

3.3 Training with 10^4 predicates

Open-vocabulary relation models are usually trained on the same few dozen predicates as closed-set models, with only the head replaced. We instead train against a bank of 19,103 free-text predicate strings: RA-4M’s own 10,102, together with 9,001 strings taken from the vocabularies of three further relation corpora [Park et al., 2025, Hudson and Manning, 2019, Yang et al., 2019] whose relations we do not train on. Since nothing on the text side is learned, the bank is the answer space rather than the supervision, and making it wider than any single source costs nothing (§ C.4). Two properties of the learning problem change at this scale, and both must be addressed for the larger vocabulary to be beneficial. Tab. 15 in § B summarises the notation.

The primary objective is a batch-local InfoNCE loss [van den Oord et al., 2018] over predicate slots. For each annotated pair, the positive is the annotated predicate’s *synonym group*, each member weighted by an estimated probability of synonymy and the group aggregated as a weighted mean, and the negatives are the other predicates in the batch. Aligning to a group rather than to a single string is a trade-off that we quantify in § B.2: group positives transfer to unseen vocabularies, whereas exact-string positives improve within-group ranking at a cost on rare predicates. Four auxiliary terms, namely a per-cell sigmoid loss, a swap hinge on directional predicates, background suppression and an object–text grounding term on the pooled features, are specified in § B.1, and the full recipe is given in Tab. 14.

Positive-unlabeled supervision. With 50 predicates, treating every non-annotated predicate as a negative is approximately correct. With ten thousand it is systematically wrong: if a ⟨man, horse⟩ pair is annotated *riding*, then *sitting on* and *mounted on* are true but unstated, and penalising them teaches the model to suppress correct answers. This is the positive-unlabeled structure of federated annotation [Gupta et al., 2019], encountered at training time. We address it by estimating, for every ordered pair of predicates, the probability that q also holds when p is annotated, and by discounting each negative by that probability. The estimate is fitted on the 706k box pairs of the training mixture that carry more than one annotation, from co-annotation counts, text-space cosine and frequency, with the correction of Elkan and Noto [2008] for the fact that an unstated predicate is unlabeled rather than false. The denominator logit of negative q for a pair annotated p becomes

$$\ell_{\text{pred}}(i, j, q) + \log(1 - \hat{P}(q \mid p)), \quad (3)$$

so that a predicate that is almost surely also true contributes almost nothing to the denominator; a pair carrying several annotations combines their discounts as if independent. Two exemptions are essential: annotated predicates are never down-weighted, and directional inverses always receive full weight, since the supervision that separates *above* from *below* is the one signal that must not be softened. The positive-side synonym weights are an isotonic fit of the probability of synonymy against text-space cosine on 1,177 lexical synonym pairs, and the swap hinge of § B.1 is weighted by the estimated probability that a predicate is directional, so no semantic constant of the objective is set by hand. Object categories enter training only through the sampler’s relatedness loss, which

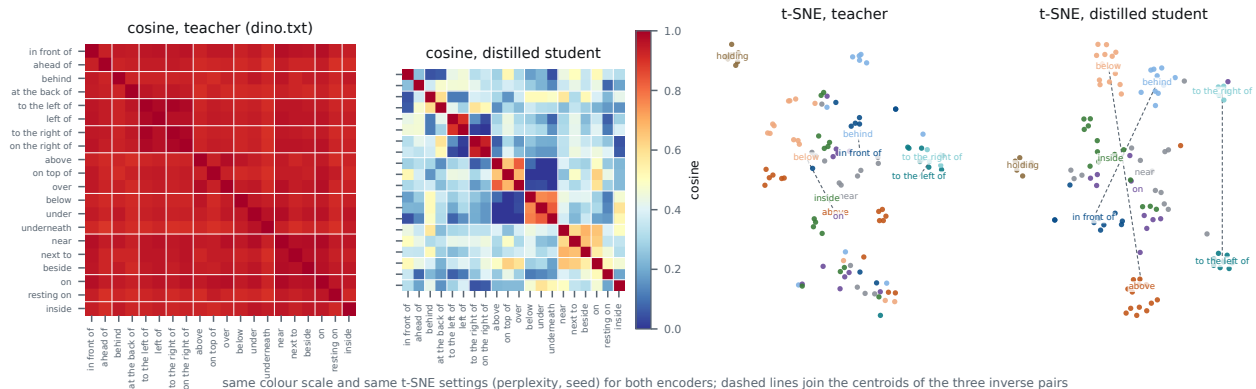


Figure 3 Direction in the text embedding space. Left two panels: pairwise cosines of twenty spatial predicates under the teacher and under the distilled student, same colour scale. Right two: ten spatial synonym families (107 surface forms) under t-SNE [van der Maaten and Hinton, 2008], same settings for both. The teacher’s matrix is a near-uniform block above 0.8 in which **above/below** is as close as **above/on top of**, and its t-SNE panel interleaves the members of each inverse pair. The student retains the synonym blocks, drives the inverse blocks to near zero and splits each inverse pair into two clusters. Statistics in Tab. 16 (§ B.2).

weights an unannotated pair by one minus the annotation rate of its category pair (floor 0.3), and through the grounding term of § B.1; both condition a loss rather than the forward pass and cannot supply a predicate prior to the network. A source-aware refinement is described in § B.1.

Direction in the text embedding space. Because the head regresses visual features onto text directions, the geometry of the text space bounds what the model can express. Contrastive text encoders fail in precisely the dimension on which relations depend: antonyms occur in near-identical contexts and are embedded almost identically. On our vocabulary the teacher encoder (dino.txt [Jose et al., 2025], 2048-d) places **above** and **below** at a cosine similarity of 0.95, **in front of** and **behind** at 0.94, and **to the left of** and **to the right of** at 0.99, which is indistinguishable from its synonym pairs at 0.96 on average (Fig. 3; Tab. 16 in § B.2). No visual encoder trained against such targets can reliably separate **above** from **below**. This limitation lies in the objective rather than in the visual model, and it applies to every method that scores relations against off-the-shelf text embeddings. Removing the mean direction [Mu and Viswanath, 2018] does not resolve it.

We therefore distil a 512-dimensional student encoder over the full CLIP BPE vocabulary [Radford et al., 2021], so that no predicate string can fall out of vocabulary. Of its five objectives (§ B.2), the antonym-repulsion hinge on known inverse pairs is the one a frozen teacher cannot supply, and a neighbourhood-preservation term is what stops the student from buying that separation cheaply: without the latter, a student reaches an equally good synonym-versus-inverse AUC at an effective dimension of 24 rather than 44. After distillation that AUC rises from 0.85 to 0.99, the mean inverse cosine falls from 0.92 to 0.09 while synonyms stay at 0.71, and hubness falls from 8.1 to 1.3. The student is used offline only; at inference the model carries the embedding bank.

One consequence of this objective affects every closed-vocabulary result below. Because the model is aligned to a synonym group, it frequently answers with a member of the correct group other than the annotated one, for instance **wearing** where VG150 reads **wears**, which exact-string metrics count as a miss. Such numbers are therefore conservative for our model, and A3 (§ 6.3) provides the corrected view.

4 RA-4M: a verified free-text relation corpus

The model described above requires a corpus whose predicate vocabulary is not that of a benchmark, whose annotation is dense enough that unannotated pairs are rare, and whose errors are of a kind that can be detected. None exists, so we generated one: 474,413 images, 4,282,531 relations, 10,102 distinct free-text predicates and 9.03 relations per image, at a cost of 104 GPU-hours. The pipeline, the statistics and a comparison with existing machine-annotated corpora are in § C.

Motivation. Relation annotation is among the most expensive and least reliable forms of annotation in vision. An object box has one correct answer, whereas a relation is a claim about a pair, of which an image with N objects contains $N(N - 1)$, and for most pairs whether the claim holds is a matter of judgement. When Visual Genome’s annotators were free to write what they saw they produced 36,549 distinct predicate strings [Krishna et al., 2017]; the community retained the 50 most frequent [Xu et al., 2017], and the discarded strings are precisely what an open-vocabulary model is intended to cover. A vision–language model annotates no single relation better than a person, but it annotates more densely and uniformly at marginal cost, and its errors are of a kind a geometric check can detect. We therefore use it as the annotator, not the predictor.

4.1 Generation and geometric verification

Generation runs an open-weight vision–language model [Gemma Team, 2026] on images with *numbered markers*, one coloured dot carrying the box index at the centre of each box, so that grounding is an input to the annotator rather than an inference from its output. Every proposed relation then passes through a deterministic geometric gate (Alg. 1 in § C), which rejects 11.3% of raw candidates; Fig. 14 breaks this figure down by gate. To our knowledge, no other machine-annotated relation corpus verifies its annotator (Tab. 19).

The design principle is that a gate fires only when box geometry logically constrains the predicate. A rejection is then a true negative up to box-annotation error, whereas predicates geometry cannot constrain pass through unchecked and are counted as residual risk; a gate that also guessed on those would replace a known gap in coverage with an unknown error rate. Verification rejects a relation or swaps its roles; the predicate string stays the annotator’s own, except in the balancing step below. After gating, 0% of containment and 0.1% of contact relations have disjoint boxes, and the corpus has no duplicate triples and no self-loops. Direction repair is the one non-rejecting gate: for left/right and above/below the roles are swapped to match the boxes and the surface form kept. Since the annotator prefers the left and upper object as subject, each verified left/right, above/below and front/behind relation is then restated from the other endpoint with probability one half, roles swapped and the predicate replaced by its inverse (**to the left of** becomes **to the right of**); the geometric top-up is balanced the same way. This leaves the corpus direction-balanced, a property the model depends on and no source corpus has.

The residual error classes are the geometrically unconstrained ones. Gaze is the largest: roughly 22% of **looking at/watching** annotations have disjoint boxes and cannot be verified. Wrong-instance grounding, prior-driven labels on very large boxes, contact predicates between overlapping but unrelated boxes (**person wearing tent** passes the contact gate because the boxes do overlap), and mislabelled source boxes all propagate into the corpus. One of these classes has a measurable downstream effect, which we trace in § H.3.

4.2 Statistics and training mixture

Relative to the source annotations on the same images and the same boxes, RA-4M is $1.7\times$ denser (9.03 versus 5.29 relations per image), has $107\times$ the vocabulary and 1.4 nats higher predicate entropy (Tab. 20), and reproduces 73 of the 94 source predicate classes as exact strings. Since the images and boxes are identical, the comparison isolates the annotation schema. The distribution remains long-tailed (the top ten predicates carry 57% of the mass), but the head predicates differ (Fig. 15 in § C): **behind**, **in front of**, **wearing**, **to the right of** and **to the left of**, spatial and balanced between inverses by construction, replace **near** and **on**. The mean object degree rises from 1.88 to 3.20, and the share of images whose graph forms a single connected component rises from 70.9% to 82.6%.

holding, **grasping**, **carrying** and 535 further members of the same family are retained as distinct strings, because a model whose deployment vocabulary is arbitrary must be trained on targets that populate the text space densely; canonical groups exist only within the loss and the gates (§ C.4). Every evaluation split is excluded by image identity, using explicit $\text{VG} \leftrightarrow \text{COCO}$ id tables and perceptual near-duplicate search, which removes 8,687 corpus images and 32,456 raw Visual Genome images (§ C.4).

Training mixture. RA-4M is not used alone. We refer to one trained model as a *tower*, that is, a backbone, a pair head and a training mixture, and we evaluate two towers throughout. The *zero-shot tower* is trained on RA-4M together with the leakage-filtered raw Visual Genome annotations, using all of their free-text predicate strings rather than the 50 of VG150 (40,615 images, 758,547 relations, 17,742 predicate strings); the mixture is 91.9%/8.1% by image and 84.6%/15.4% by relation. This tower has seen no image from any evaluation split used in this paper and none of the benchmarks’ label sets; its Visual Genome share is raw free-text annotation on images outside every evaluation split. The *released tower* additionally includes the HICO-DET training split at 21.0% of images, which amounts to a 5.0% relation share because HICO-DET annotates 1.94 relations per image against 9.02 for the leakage-filtered RA-4M, and uses the source-aware negatives of § B. Its mixture is 72.7/6.3/21.0 by image and 80.6/14.4/5.0 by relation over the three sources (Tab. 14); Tab. 17 lists each share alongside the shared triplet mass it produces. Both towers are scored against the same 19,103-string predicate bank, so the zero-shot tower carries 108 of HICO-DET’s 116 predicate strings as columns without having been trained on its relations. Unless a row is marked *zero-shot*, the released tower is the model reported in every table; on HICO-DET, where it is not zero-shot, the zero-shot tower is reported alongside it. The two towers differ by less than one point on every axis other than HICO-DET itself (Tab. 29 in § G).

5 OV-SGG-Bench: analysis of scene-graph evaluation and a cross-dataset protocol

We analyse what scene-graph recall rewards, identifying three priors that a standard architecture can exploit by construction (§ 5.2) and two scoring conventions whose effect exceeds the differences they are used to measure (§ 5.3), and then define the six-axis protocol on which § 6 reports (§ 5.4). The model and corpus of the preceding sections serve here only as measurement instruments; no finding below depends on their design.

5.1 Metrics and evaluation modes

R@K denotes scene-graph recall, the fraction of an image’s annotated $\langle$ subject, predicate, object $\rangle$ triplets appearing among the K highest-scoring predicted triplets, averaged over images; since every triplet counts equally, it is dominated by frequent predicates. mR@K is the mean of the per-predicate R@K values, so a rare predicate contributes as much as on; F1@K is their harmonic mean; wR@K weights each predicate’s recall by the logarithm of its inverse frequency in the evaluated split; and the rare, common and frequent buckets partition predicates by annotation count in that split (below 50, 50 to 500, above 500), following LVIS [Gupta et al., 2019]. Federated average precision (fAP) and P-AUC are computed over labelled cells only and used on A2 alone (§ 6.2).

Under *graph-constrained* scoring each subject–object pair contributes only its highest-scoring predicate to the K slots, whereas unconstrained scoring lets one pair occupy several; the difference is 11.6 R@50 on VG150 and 19.0 on PSG for our model. All numbers in this paper are graph-constrained, as are the baseline’s. Results come in one of two modes: with *ground-truth boxes* the regions are given and only the relations are scored, whereas in *detection mode* (SGDet) they come from a detector, so the score also reflects detector recall, an unreported free parameter (§ 5.3).

5.2 Three priors rewarded by recall

Shared triplet mass. The VG150 test split uses the same 50 predicate strings as its training split, so a VG150-trained model encounters no vocabulary novelty at all; nevertheless, *open-vocabulary* methods are ranked on exactly this number on the OvR-SGG leaderboard [Chen et al., 2024b], which we reproduce in § 6.7. We quantify this confound as the *shared triplet mass*, the fraction of a corpus’s relation instances whose $\langle$ subject category, predicate, object category $\rangle$ triple the benchmark also annotates (Tab. 17 in § C). The triple rather than the predicate, because a predicate string is not an annotation: on between a person and a horse and on between a book and a table are different acts, and two corpora can agree on the string while never agreeing on the pair it is asserted of. Our mixture shares 53.4% of its mass with VG150 on the predicate alone and 12.8% on the triple. No image and no annotation is shared; leakage is excluded separately by image identity (§ C.4). The confound is concentrated in-domain and is very large there: the baseline’s fine-tuning corpus reproduces 90.9% of its relation mass as triples VG150 also annotates, against our 12.8%, whereas on PSG it sits at 8.6% against our 10.7%. Because the triple charges for the two object names as well, it also charges for taxonomy and domain mismatch, and Tab. 18 prints the components separately. The statistic is predictive. An arm trained with a larger share of raw Visual Genome reached 54.3 R@50 on VG150, the best zero-shot figure we are aware of, while being the worst model we trained on every tail metric.

Object-category prior. It has been known since Zellers et al. [2018] that a lookup table over object category pairs is a strong scene-graph baseline. Tab. 1 shows its advantage to be confined to the metric by which methods are usually ranked: in per-edge top-1 accuracy the frequency baseline exceeds our trained model by 15% on all three benchmarks, while on the per-predicate average of the same predictions our model exceeds it by 29–86%. The baseline returns each category pair’s majority predicate, which a micro metric rewards and a macro metric penalises. Two readings should be avoided. The table does not show that visual input is unnecessary, since the edge-by-edge join (Fig. 20 in § E) finds the model correct where the baseline is wrong on 6.9–12.8% of edges; nor that our model is weak, since the baseline receives oracle object categories we never see. The conclusion is narrower: a metric a pixel-free lookup table can win does not measure relation understanding, and it is the metric that orders leaderboards. A model given object class labels obtains this prior

Table 1 A frequency table exceeds the trained model on the micro protocol and falls below it on the macro protocol in every cell. `FREQ` [Zellers et al., 2018] returns the most frequent training predicate for a $\langle \text{subject}, \text{object category} \rangle$ pair from ground-truth labels and no pixels; `ours` is the zero-shot tower, which receives no labels. Per-edge top-1 accuracy over ground-truth pairs and boxes. *only ours*: edges we predict correctly and `FREQ` does not; *union*: the oracle combination.

benchmark	edges	micro (per edge)			macro (per predicate)			join	
		FREQ	ours	Δ	FREQ	ours	Δ	only ours	union
VG150	152,535	68.4	57.7	-15.6%	18.9	35.1	+85.7%	6.9	75.3
PSG	13,623	50.9	43.3	-15.0%	20.7	31.6	+52.4%	12.8	63.7
IndoorVG	29,175	67.9	57.3	-15.5%	29.8	38.4	+28.9%	8.2	76.2

for free, and because it is estimated from one corpus’s label statistics it is also the component that does not survive a change of corpus.

Annotation propensity. The third prior lies in what the annotation omits. A few pairs per image are annotated and the rest left empty, but an empty pair is not a negative one: on PSG test at its maximum-F1 threshold, 78% of the edges a `RelateAnything` checkpoint emits fall on pairs the corpus never annotated in either direction, against 11% matching an annotated triple (§ D.5). A model trained to predict which pairs carry annotations therefore learns *annotation propensity*, which on these corpora reduces to a contact prior. Our architecture contains such a term by construction, the pair-relatedness score of § 3.2, so its contribution can be measured, and its sign differs by benchmark family: retaining it improves recall on the annotation-derived benchmarks and degrades truth judgement on the adjudicated negatives of `SpatialSense` (§ D.3 in § D). Recall against sparse annotation and truth judgement against adjudicated negatives are therefore different quantities.

A protocol that is robust to these three priors should report a vector of measurements rather than a single number, state the shared triplet mass behind every recall cell, and include at least one axis scored against adjudicated negatives and one that a frequency prior cannot help.

5.3 Two scoring conventions with effects larger than those measured

Whereas § 5.2 concerned what the benchmarks ask, this section concerns how they score. We encountered both conventions while reproducing `OvR-SGG` [Chen et al., 2024b], the open-vocabulary relation leaderboard, which holds 15 of the 50 VG150 predicates out of training and scores in detection mode. The reproduction is faithful: running the released checkpoints through our inference code and scoring them with a transcription of the baseline’s metric recovers all twelve published numbers to within 0.35 points (Tab. 24 in § D.7). The observations below concern the protocol itself.

Duplicate credit in the matcher. The standard detection-mode matcher, inherited from the reference implementation [Zellers et al., 2018, Tang, 2020] and still used by current open-vocabulary methods, credits a predicted triplet when its subject and object boxes each reach $\text{IoU} \geq 0.5$ with the corresponding ground-truth boxes and the label triple matches. No assignment constraint is enforced, so a ground-truth object covered by d detections offers d opportunities to recover the same relation; on VG150 test, each recovered ground-truth box is covered by 2.99 detections on average. Object detection has not permitted this since `COCO` [Lin et al., 2014], where each ground-truth instance may be matched by at most one detection and the remainder count as false positives. Restoring the constraint reduces the published model’s R@50 by 12.0% and its novel R@50 by 13.5%, consistently across both released checkpoints. This amounts to 2.45 points of Base+Novel R@50 ,

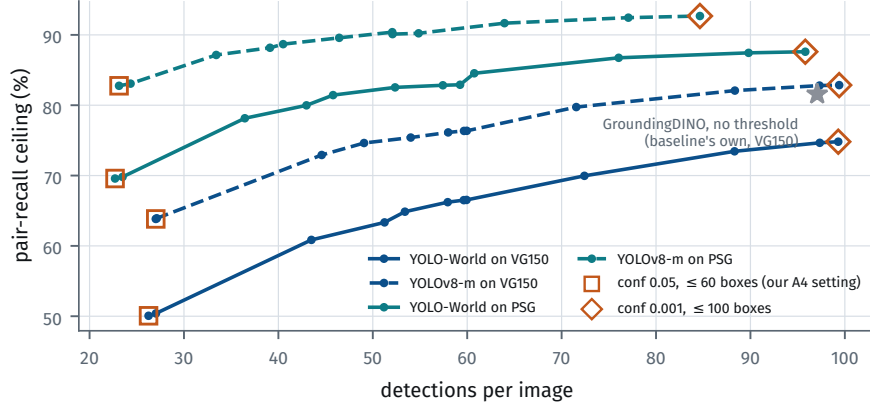


Figure 4 The pair-recall ceiling as a function of the detector operating point. One curve per detector and benchmark, weights fixed, swept over confidence threshold and box cap; markers are the two operating points of Tab. 23 (§ D). The 4.3-point spread between all published OvR-SGG methods fits inside the vertical extent of any single curve. The baseline’s GroundingDINO [Liu et al., 2024] retains every non-background query, the source of its apparent advantage.

compared with the 2.39 points gained by upgrading the same method’s backbone from Swin-T to Swin-B, which constitutes a separate leaderboard entry. The correction applied here is the most generous available, matching each ground-truth object to the highest-IoU detection of the correct class; a class-agnostic assignment costs more (25.5 down to 15.2 R@50).

Detector operating point. Detection-mode recall is bounded above by *pair recall*, the fraction of ground-truth relations for which both endpoints are detected. Pair recall is quadratic in object recall, bounds every model evaluated on the same boxes identically, and depends not on the detector itself but on its confidence threshold and box cap, parameters that are rarely reported. Holding the detector weights fixed and varying only these two settings moves the ceiling by 12–49% relative across four detector/benchmark combinations (Fig. 4); on VG150 the change is 19 points, against a 4.3-point spread between all published methods on the leaderboard. A threshold can also invert a comparison between detector families. The baseline’s GroundingDINO [Liu et al., 2024] appears stronger than a YOLO detector, with a ceiling of 0.82 against 0.64, because it applies no threshold and retains 97 boxes per image against the YOLO detector’s 27; at a matched budget the YOLO detector reaches 0.83. Our own A4 is measured at a confidence threshold of 0.05 against a ceiling of 0.696, whereas the same detector reaches 0.876 at a matched budget. Detection-mode results should therefore state the matcher, the detector operating point and the resulting pair-recall ceiling; § 10 lists the reporting practices this implies.

5.4 The six axes

OV-SGG-Bench reports a vector of six axes, summarised in Tab. 2. Each axis answers a question that no other axis can, and each is listed together with the way in which it fails when read alone, which is the reason for reporting a vector. Two axes are prior-blind by construction: A2 is scored against negatives that a human adjudicated, so annotation propensity cannot help, and A6 is balanced by design, so the category prior cannot help. Every claim in this paper about relational competence, as opposed to corpus match, rests on these two axes.

On A3, a prediction is counted as correct when it matches the ground truth under a synonym matcher at a cosine threshold calibrated to one embedding space. A threshold fitted in one space

Table 2 The six axes of OV-SGG-Bench; the third column states why no single axis suffices.

	what it measures	how it fails alone
A1 transfer	closed-vocabulary recall on ground-truth boxes, four sources of differing shared triplet mass	can be satisfied by corpus match
A2 precision	Haystack [Lorenz et al., 2023]: federated AP and P-AUC against explicit negative annotations; the only axis on which a confident false positive on a rare predicate is visible	invariant to uniform score depression: a model that never asserts a relation still ranks well
A3 open vocab	the full 19,103-string vocabulary deployed, synonym-tolerant matching	rewards head collapse: on and in are members of most synonym sets
A4 deployment	detection-mode graphs on a <i>shared</i> open-vocabulary detector, against the measured pair-recall ceiling	bounded by the detector
A5 graph quality	the information in the relations that a vision-language judge without ground truth accepts one at a time (Eq. (4)); the only axis that can credit a true but unlabelled relation	reported only when the judge’s controls pass; shown whole graphs, the judge prefers short repetitive ones, and that verdict is reported alongside
A6 spatial	SpatialSense [Yang et al., 2019]: adversarially collected, balanced true/false triples, chance 50%	a boxes-only baseline scores 68.8 <i>without the image</i> : a high score indicates resistance to priors rather than visual reasoning

and reused in another accepted 0.6% of true synonyms and reduced the axis to zero (§ H.4).

Composite. The axes are combined into a single scalar, OVS, over A1, A2, A4, A5 and A6. Each cell is chance-corrected, A1 entering as F1@50 averaged over its four benchmarks, and the axes are combined by a harmonic mean, which penalises imbalance, following the practice in generalised zero-shot learning. Two cells are normalised by a measured quantity rather than a chance level, A4 by the pair-recall ceiling of the shared detector and A5 by the information in the images’ own annotations, so that each reports the share of what is recoverable that a model delivers. A3 is reported but excluded, since the only baseline runnable on it is scored through a matcher whose choice moves the result by more than the systems differ (§ 6.3). The composite does not replace the vector; it exists to keep model selection from optimising one axis. Withholding each axis in turn leaves the ordering of the two systems unchanged, at ratios between 2.0 and $3.7\times$ (Tab. 21).

Baselines. Each axis is reported against the strongest system we could run on it. OvSGTR [Chen et al., 2025] pre-trained on MegaSG, the corpus whose images we re-annotate, is the closest control for supervision quality and the only system runnable on all six; we run its released checkpoint through our evaluator in its relation-open-vocabulary mode, the only configuration with no frequency prior and a head that accepts an arbitrary vocabulary. One residual asymmetry favours it throughout: its protocol consumes object labels, whereas ours never does. ROBIN-3B [Park et al., 2025] and four general multimodal models answer in free text instead, which makes them the only systems runnable on A3 and, in ROBIN’s case, the stronger baseline on A1; they need a matcher, whose choice § 6.3 shows to move the result by more than the systems differ. The remaining open-vocabulary methods of § 2 lack released weights or an inference path that accepts our regions and vocabulary, and appear on the leaderboard of § 6.7 instead; the omission reflects what we could run, not a judgement.

Table 3 The six axes of **OV-SGG-Bench**, released model against the best baseline available on each axis, one evaluator for all. Each baseline cell names its system: ROBIN-3B [Park et al., 2025] leads OvSGTR [Chen et al., 2025] on A1 and is the only one of the two runnable on A3, and OvSGTR is the only system run on A2, A4, A5 and A6. That column is an envelope over two systems rather than a system, so the composite is computed for OvSGTR alone. A5 is Eq. (4); A6 is the macro mean of per-predicate AUC, pooled in parentheses; the HICO-DET cell uses the zero-shot tower.

axis	measure	best baseline	RelateAnything	note
A1 transfer	VG150, F1@50 (mR@50)	19.6 (15.2)	36.9 (28.2)	ROBIN; triplet mass 13%
	PSG, F1@50 (mR@50)	27.9 (22.9)	34.7 (30.6)	ROBIN; triplet mass 11%
	IndoorVG, F1@50 (mR@50)	22.7 (21.5)	37.8 (29.5)	ROBIN; triplet mass 7%
	HICO-DET, F1@50 (mR@50)	7.9 (4.5)	18.7 (12.7)	OvSGTR; zero-shot tower
A2 precision	mean fAP (rare-predicate fAP)	52.1 (44.6)	72.6 (70.7)	OvSGTR; pair coverage 89% vs 100%
A3 open vocabulary	mR@50 over 19,103 strings, VG150 / PSG / IndoorVG	20.9 / 25.1 / 25.4	34.5 / 28.3 / 34.6	ROBIN; OvSGTR caps at ≈ 150 strings, and ROBIN is scored through a matcher
A4 detector	PSG on a shared YOLO-World, wR@50 (mR@50)	4.0 (6.2)	20.0 (20.5)	OvSGTR; pair-recall ceiling 69.6
A5 graph quality	true bits per image (share of annotation)	13.4 (0.48)	18.6 (0.67)	OvSGTR; equal length, 199 images; annotation 27.9 bits; verdicts in § 6.5
A6 spatial	SpatialSense, macro AUC (pooled AUC)	59.1 (61.7)	69.0 (67.5)	OvSGTR; chance 50.0; boxes-only baseline 68.8
OVS composite	harmonic mean of the chance-corrected axes	11.8	40.1	OvSGTR alone; A1, A2, A4, A5, A6

6 Results

Tab. 3 summarises all six axes, discussed in turn below. No result is in-domain, the regime in which a corpus-matched prior is correct and the priors of § 5.2 are invisible: every cell is measured on a split whose images and annotations were excluded from training, by a tower that never trained on the benchmark’s label set, the sole exception being the marked HICO-DET row of Tab. 4. One evaluator scores every system, and the shared triplet mass is printed beside each cell (§ 5.2); no images or annotations are shared.

6.1 A1: cross-dataset transfer on four benchmarks

As Tab. 4 shows, RelateAnything outperforms OvSGTR on every metric of every source, including micro recall, though it receives no object labels and the baseline is given ground-truth ones. The margins vary: head recall is $1.04\text{--}1.40\times$ the baseline’s, mean recall $2.3\text{--}3.5\times$ and rare-bucket recall $5\text{--}21\times$, the ratio being undefined on VG150 where the baseline scores exactly 0.0. This is the pattern the prior-matching account of § 5.2 predicts: VG150 is OvSGTR’s own benchmark, where 90.9% of its triplet mass is shared against our 12.8%, and its headline number there is carried by 21 frequent predicates, the part of the distribution recoverable without visual input. OvSGTR is not the strongest baseline on this axis, however. ROBIN-3B leads it on F1@50 on all three benchmarks both were run on, $19.6 / 27.9 / 22.7$ against $16.5 / 13.5 / 20.2$, which is why Tab. 3 carries ROBIN

Table 4 Axis 1: cross-dataset transfer. Test splits, ground-truth boxes, graph-constrained, one evaluator, OvSGTR [Chen et al., 2025] throughout. *triplet mass* is the share of *our* training relations whose subject–predicate–object triple the benchmark also annotates (§ 5.2); the baseline’s is 90.9%, 8.6% and 10.4% on VG150, PSG and IndoorVG. No images or annotations are shared. *rare* is recall on the rare support bucket. The baseline receives ground-truth object labels; RelateAnything receives none. On HICO-DET the released model has seen a 5% relation share of the training split and is reported alongside the zero-shot tower.

Source	triplet mass	OvSGTR [Chen et al., 2025]				RelateAnything (ViT-S/16+)			
		R@50	mR@50	F1@50	rare	R@50	mR@50	F1@50	rare
VG150	13%	39.9	10.4	16.5	0.0	53.3	28.2	36.9	42.7
PSG	11%	28.7	8.8	13.5	1.7	40.1	30.6	34.7	23.9
IndoorVG	7%	48.1	12.8	20.2	4.0	52.7	29.5	37.8	21.6
HICO-DET (zero-shot tower)	—	34.0	4.5	7.9	0.2	35.3	12.7	18.7	4.3
HICO-DET (+5% HICO-DET train)	—	34.0	4.5	7.9	0.2	45.2	31.4	37.1	23.9

in the A1 cells; it too stays behind us on both metrics everywhere (Tab. 26). Our own micro recall warrants the same scepticism, which is why Tab. 4 carries the triplet-mass column. The strongest systems we run are not on this axis at all: the in-domain specialists of § 9 lead us on F1@K on PSG and IndoorVG once a detector supplies the boxes (Tab. 12).

6.2 A2: precision on adjudicated negatives

Recall does not penalise hallucination. Since an unlabelled pair is not a negative pair, a model can assert false relations at no cost in recall. A2 addresses this by scoring against negatives that an annotator adjudicated rather than negatives assumed by the protocol.

A2 is evaluated on Haystack [Lorenz et al., 2023], the only scene-graph test set we are aware of that publishes explicit negatives, that is, cells an annotator inspected and marked as not carrying the predicate, as distinct from cells left blank. Annotating per predicate rather than per image reaches rare classes that the PSG test split barely contains and makes its positives sparse: 2,870 positives against 23,174 explicit negatives over 11,368 images. It reuses PSG’s 56 predicates and panoptic format, so a model runs on it unchanged, and its images come from SA-1B [Kirillov et al., 2023], so it shares no images or annotation ancestry with VG150, IndoorVG, PSG or the Visual Genome component of our mixture. Scoring is federated in the sense of LVIS [Gupta et al., 2019], over the 26,044 labelled cells only, on ground-truth boxes; we report Haystack’s own per-predicate ROC-AUC (P-AUC) together with a federated AP that ranks within each predicate.

The results are given in Tab. 6. RelateAnything improves mean federated AP by 39% and rare-predicate fAP by 59%, over 55 predicates with support of at least five, 31 of which are rare, and raises P-AUC from 83.9 to 93.7. Both models express all 56 predicates, but reach different fractions of the cells. The baseline scores every ordered pair; our relatedness sampler reaches 89.0% of the 26,044 labelled cells, and we do not force the rest through it, since an unsampled pair is one the deployed system emits nothing for. Those cells are scored zero: a correct rejection for a negative, a miss for a positive. We note that the same predictions, scored by counting every unlabelled pair as a false positive, yield an AP of 0.035, whereas scored against explicit negatives they yield an fAP of 0.56; the difference lies entirely in what the protocol assumes about unannotated pairs.

Table 5 Free-text systems on the open-vocabulary axis. Ground-truth regions for every model, graph-constrained, one scorer; ours is the 53M ViT-S/16+ tower. PSG test, 2,179 images: one set of generations per model scored under the matchers in circulation, so the rows of a system differ from one another only in how its free text is mapped onto the benchmark’s words. The general models are reported under our A3 rule alone. Our A3 rows use the $\tau = 0.60$ matcher shared by every system here rather than the calibrated threshold of Tab. 6, hence the different PSG figures. *Pair cov.* is the share of annotated pairs a system reaches at all; ours at top- K as elsewhere. The cross-dataset comparison, closed-set prompt and SGM leg are in § E.4.

Model	Setting	R@20	R@50	mR@20	mR@50	pair cov.	rel/img
<i>PSG test, vocabulary withheld — one row per matcher, same generations</i>							
RelateAnything	exact	12.3	13.6	12.4	13.0	99.7	top- K
RelateAnything	A3	31.7	36.8	29.1	31.3	99.7	top- K
ROBIN-3B	exact	25.8	26.2	19.9	20.0	45.6	9.4
ROBIN-3B	SBERT argmax	30.4	35.7	21.1	22.9	77.4	34.0
ROBIN-3B	A3	35.1	37.9	24.1	25.1	65.6	18.9
Qwen3-VL-32B	A3	15.0	15.6	5.1	5.2	35.5	13.8
Qwen3-VL-8B	A3	10.2	10.2	3.3	3.3	23.6	9.7
InternVL3.5-8B	A3	9.5	9.6	4.5	4.5	23.1	8.3
GLM-4.6V-Flash	A3	8.9	9.1	2.1	2.2	23.4	11.2

6.3 A3: the full vocabulary with the answer set withheld

A1 supplies the benchmark’s vocabulary to the model, which a deployed open-vocabulary model does not have. A3 removes this assistance: the model answers from its full 19,103-string vocabulary and is scored with synonym-tolerant matching. Recall does not collapse (56.0 R@50 on VG150, against 53.3 with the benchmark’s own 50 strings supplied), and the median rank of the ground-truth string among 19,103 candidates is 1 on VG150 and IndoorVG and 10 on PSG (Tab. 6).

Applicability to the baseline. OvSGTR scores relations against text, but receives it as a single caption encoding the predicate names jointly, so its vocabulary is bounded by the encoder’s 512 word pieces, about 150 strings, and splitting 19,103 strings across 140 captions does not recover the measurement, since each name’s embedding depends on the others in its caption. Ours are embedded independently and offline. This axis is one a model must be designed for.

Free-text systems. A model that answers in free text can be evaluated on this axis by construction, which admits a family of baselines that the other axes cannot accommodate: ROBIN-3B [Park et al., 2025], a scene-graph model built on a 3B vision–language model, and general multimodal models prompted for the same output: Qwen3-VL [Bai et al., 2025a], InternVL3.5 [Wang et al., 2025b] and GLM-4.6V-Flash [GLM-V Team et al., 2025]. Tab. 5 evaluates them on ground-truth regions, so that relation prediction is measured independently of localisation; prompts are in § E.4.

The matcher is a third scoring convention whose effect exceeds the differences it is used to measure. Tab. 5 is one set of generations per model read up to three ways: mapping free text onto the benchmark’s words is worth +11.7 R@50 to ROBIN, from 26.2 exact to 37.9 at $\tau = 0.60$, and +23.2 to us, since a model answering from 19,103 strings is the one exact matching penalises hardest (§ B.2.2). The choice decides the mean-recall ordering of the two purpose-built systems: ROBIN leads under exact matching, 20.0 against 13.0, and we lead under every synonym-tolerant matcher, 31.3 against 25.1 at $\tau = 0.60$. Micro recall does not turn over, ROBIN staying ahead throughout, so

Table 6 Axes 2–4 in detail. *A2* (top left): precision on the explicit negatives of Haystack [Lorenz et al., 2023]; fAP ranks within a predicate, P-AUC is Haystack’s own per-predicate ROC-AUC. *Pair coverage* is the share of labelled cells whose pair a model scores at all: complete for the baseline, which scores every ordered pair, and 89.0% for ours, whose sampler prunes and whose unsampled cells score zero. Both express all 56 predicates. *A3* (top right): all 19,103 training strings, synonym-tolerant matching at the calibrated threshold; *MRR* and *median rank* score the ground-truth string against that vocabulary. *A4* (bottom): PSG test, graph-constrained, IoU matching, on a shared YOLO-World detector [Cheng et al., 2024] prompted with each benchmark’s categories at confidence 0.05 and at most 60 boxes per image; the pair-recall ceiling bounds both models identically (Fig. 4), and the baseline additionally consumes the detector’s predicted labels.

A2 precision (Haystack, ground-truth boxes)				A3 open vocabulary (19,103 strings)					
Metric	OvSGTR [Chen et al., 2025]	RelateAnything (ViT-S/16+)	Δ	Source	R@50	mR@50	rare	MRR	median rank
mean fAP (support ≥ 5)	52.1	72.6	+39.2%	VG150	56.0	34.5	39.6	0.68	1 / 19,103
fAP, rare predicates	44.6	70.7	+58.6%	PSG	30.5	28.3	20.8	0.39	10 / 19,103
fAP, common predicates	59.5	76.0	+27.8%	IndoorVG	53.3	34.6	34.6	0.65	1 / 19,103
mean predicate ROC-AUC (P-AUC)	83.9	93.7	+11.7%						
pair coverage (labelled cells scored)	100.0	89.0	-11.0%						

A4 detection mode (PSG test, shared YOLO-World boxes)				
Model	R@50	mR@50	rare	wR@50
OvSGTR [Chen et al., 2025] (uses detector labels)	17.7	6.2	1.2	4.0
RelateAnything (ViT-S/16+)	25.4	20.5	17.0	20.0
pair-recall ceiling (both models)	69.6	—	—	—

the two metrics disagree in the direction § 5.2 predicts. A single row is therefore a choice of scorer rather than a measurement of a model, and we report the band.

Two things survive the matcher. One is which pairs a system proposes: pair coverage is 99.7% for us, 45.6–77.4% for ROBIN and 23.1–35.5% for the general models, and a pair never named cannot carry the right predicate; scale buys mostly this, 8B to 32B adding +11.9 points of it. The other is what happens off PSG. Run on three benchmarks, each system under its own protocol (Tab. 26), we lead ROBIN on both metrics on all three, by 1.9 $\times$ on VG150 micro recall and 2.2 $\times$ on IndoorVG against 1.1 $\times$ on PSG, the benchmark on which such systems are usually compared.

6.4 A4: detection-mode graphs on a shared detector

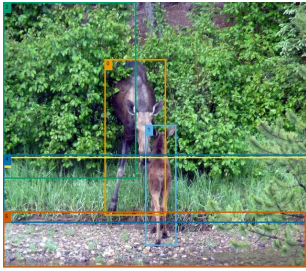
The transfer results above use ground-truth boxes; in Tab. 6 a shared open-vocabulary detector supplies the regions and both models are re-scored against the measured pair-recall ceiling of 0.696. Every margin of Tab. 4 survives with the same structure (mR@50 3.3 $\times$, rare bucket 14 $\times$), so the relation model is unaffected by the change of box source and what it learned transfers. The dominant loss is the detector’s. Under a one-to-one matcher that ignores the detected class, retention relative to ground-truth boxes is 62–63% on PSG and 31–32% on IndoorVG across all three towers; when the matcher also requires the detected class to agree, as in § 5.3, it is 46% and 19% (Fig. 21 in § E). A relation cannot be recovered unless both endpoints are detected, and neither a better relation model nor more relation annotation raises pair recall. Both models are scored on identical boxes, so the comparison is unaffected by the operating point; the absolute numbers are specific to a confidence threshold of 0.05.

6.5 A5: graph quality judged without ground truth

A5 addresses a question none of the other axes can: whether, without ground truth, the emitted graph describes the image well. A vision–language judge from a different model family than the

Table 7 A5: Eq. (4) and its decomposition, PSG test, the shared YOLO-World boxes of A4, one judge, restricted-vocabulary arm. Each relation the judge accepts is credited with its surprisal under the PSG training marginal over the 56 predicates both systems emit; the last column is the product. The judge is asked only whether the relation is true. The two blocks differ in rendering as well as depth, and each carries its own corruption-acceptance control (§ D.8.1). The headline cell of Tab. 3 is the matched block.

model	asserted/img	accepted/img	bits/accepted	true bits/img
<i>matched depth: the top 10 pairs of every system</i>				
RelateAnything, vocab = pack	9.8	4.07	4.57	18.6
OvSGTR [Chen et al., 2025]	9.8	4.33	3.09	13.4
judge control at this depth: false acceptance 0.201 on specific corruptions ($n = 369$), 0.275 pooled ($n = 556$)				
<i>deployed depth: each system emits what its own scores justify</i>				
RelateAnything, vocab = pack	18.3	6.53	4.39	28.7
OvSGTR [Chen et al., 2025]	10.3	4.64	3.04	14.1
judge control at this depth: false acceptance 0.135 on specific corruptions ($n = 659$), 0.196 pooled ($n = 986$)				



OvSGTR

6 relations emitted; 13.9 bits if all were true

o cow on 5 dirt	2.3 b
1 dog on 5 dirt	2.3 b
1 dog on 3 grass	2.3 b
o cow on 3 grass	2.3 b
o cow on 4 grass	2.3 b
1 dog on 4 grass	2.3 b

RelateAnything

10 relations emitted; 38.9 bits if all were true

1 dog standing on 5 dirt	3.8 b
o cow standing on 4 grass	3.8 b
o cow standing on 5 dirt	3.8 b
o cow standing on 3 grass	3.8 b
1 dog standing on 3 grass	3.8 b
1 dog standing on 4 grass	3.8 b
1 dog looking at o cow	5.7 b
o cow standing on 2 tree	3.8 b
o cow beside 1 dog	2.5 b
1 dog standing on 2 tree	3.8 b



OvSGTR

10 relations emitted; 23.3 bits if all were true

o tree on 1 sky	2.3 b
2 building on 1 sky	2.3 b
4 banner on 1 sky	2.3 b
2 building on 3 stop sign	2.3 b
3 stop sign on 1 sky	2.3 b
3 stop sign on 2 building	2.3 b
4 banner on o tree	2.3 b
2 building beside o tree	2.5 b
o tree on 2 building	2.3 b
o tree on 3 stop sign	2.3 b

RelateAnything

10 relations emitted; 36.4 bits if all were true

4 banner in front of 1 sky	4.6 b
4 banner in front of o tree	4.6 b
4 banner in front of 2 building	4.6 b
4 banner in front of 3 stop sign	4.6 b
1 sky in front of o tree	4.6 b
o tree beside 2 building	2.5 b
o tree beside 3 stop sign	2.5 b
2 building on 3 stop sign	2.3 b
3 stop sign on 2 building	2.3 b
o tree standing on 1 sky	3.8 b

Figure 5 Two graphs on the same boxes. Two PSG test images from the A5 run, the shared YOLO-World boxes of A4 drawn and numbered, and the top ten relations each system emits from the benchmark’s 56 predicates. Object names are the detector’s, shown for the reader only. **on** is set in grey; each predicate carries its surprisal in bits under the PSG training marginal, which Eq. (4) credits when the judge accepts it. The baseline’s graphs consist mostly of **on**; ours convey more per relation and include claims a judge rejects.

annotator is shown the image with numbered boxes and asked about a system’s relations (§ D.8).

The judge yields three verdicts, two of which favour the baseline. Shown whole graphs it prefers the baseline’s at deployed length and ours at equal length, and shown single relations it accepts the baseline’s at a slightly higher rate; but each relation it accepts from the baseline conveys about a third less information, because most of them are **on**. The three verdicts are not in conflict, and the axis is scored by the third.

Per-relation judgement. The judge is shown one relation at a time, with the subject and object boxes drawn on the image, and asked whether the claim holds for that photograph; a fixed share of

relations are first corrupted by substituting another predicate from the same image, and the run reports how often a corruption is accepted. Truth alone favours **on**, which is almost always true and almost never informative, so a system answering **on** everywhere would be accepted more often than either model here. Asking the judge to rate informativeness does not resolve this: it rated the baseline’s **on** relations at 1.05 on a scale of 0–3 against 0.72 for its other predicates, the reverse of the intended ordering. The redundancy of the five-hundredth **on** is a property of the distribution it occurs in, which a judge shown one relation at a time cannot observe.

Measuring informativeness. The information carried by a predicate is its surprisal under a reference distribution, and the axis is defined as the total surprisal of the relations the judge accepted, per image:

$$I = \sum_{r : \text{judge}(r) = \text{true}} -\log_2 p_{\text{ref}}(\text{pred}(r)). \quad (4)$$

The two factors compensate for each other’s weakness. Surprisal alone would reward rare predicates emitted at random, which the judge rejects as false; truth alone would reward **on**, priced at 2.31 bits against 7.04 for **riding**. Neither can be increased by repetition. The reference is the PSG training marginal, shared by both systems and restricted to the 56 predicates both emit, so no smoothing constant influences the comparison.

Tab. 7 gives the decomposition. At matched depth, that is, the top ten pairs of each system, the judge accepts slightly more of OvSGTR’s relations than of ours, 4.33 against 4.07 per image. Each accepted relation is, however, worth 3.09 bits against our 4.57, so the axis reads 13.4 bits per image for the baseline against 18.6 for ours, and 14.1 against 28.7 at deployed depth. A reference-free estimator that charges a system for the size of the vocabulary it uses agrees on the ordering (§ D.8.1). Since bits per image have no upper bound, **Tab. 3** and **Fig. 1** divide by the same quantity evaluated on the annotation itself: the ground-truth relations of those 199 images carry 27.9 bits under the same reference, and at matched depth we deliver 0.67 of that against 0.48 for OvSGTR. The ratio is not bounded by one (at deployed depth ours is 1.03) because the annotation is sparse, which is why the matched-depth cell is the one that enters the composite.

Corruption control. Each depth carries its own: the judge accepts 0.135 of specific corruptions at deployed depth and 0.201 at matched depth, a difference due to the rendering rather than the depth (§ D.8.1). The headline cell is thus the arm with the weaker control. We report it nonetheless, because that rate is an upper bound rather than a false-positive rate, a swapped predicate sometimes being true of its new pair, and because the ordering holds under both renderings and depths.

Whole-graph comparison. Shown two whole graphs, in both presentation orders and with several votes each, the judge chose the baseline’s in 49 of 58 decisive comparisons in the restricted-vocabulary arm and 35 of 46 in the full-vocabulary arm, with all controls passing (**Tab. 25** in § D.8). Two factors confound the verdict. Our graphs are longer, 18.3 relations against 10.3, and the baseline’s was never the longer of the two; a separate run in which each system contributes its top ten pairs reverses the outcome, 40 of 64 decisive comparisons in our favour. The graphs also differ in composition: 66% of the baseline’s relations are **on**, and averaged per graph its modal predicate accounts for 0.85 of the relations against 0.52 for ours (**Fig. 5**). A graph-level verdict cannot separate the two readings, that our additional relations are false or that the judge prefers shorter text, and a verdict at equal length says nothing about what the additional relations are worth. The axis is therefore scored by **Eq. (4)**, the only verdict of the three that does not depend on graph length.

Table 8 OvR-SGG leaderboard (VG150 test, SGGDet, 35 base / 15 novel predicates). Our rows use the *baseline’s* matcher, Swin-T detector boxes, predicted labels and ranking rule. The two rows marked *novel set seen* have the 15 novel strings in their training vocabulary, so their Novel columns are not held-out; the row above them is the retrained model satisfying the protocol. Mean recall and the class count are measured here for both re-run systems and reported by no published row (n/r). Bold: the better of the two measured here.

Method	Backbone	Base+Novel R		Novel R		mean recall @50		classes hit
		@50	@100	@50	@100	B+N	Novel	
VS3 [Zhang et al., 2023]	Swin-T	15.6	17.3	0.0	0.0	n/r	n/r	n/r
OvSGTR [Chen et al., 2025]	Swin-T	20.5	23.9	13.5	16.2	n/r	n/r	n/r
RAHP [Liu et al., 2025]	Swin-T	20.5	25.7	15.6	19.9	n/r	n/r	n/r
OvSGTR + MegaSG [Chen et al., 2025]	Swin-T	25.4	29.7	17.0	21.1	n/r	n/r	n/r
OvSGTR [Chen et al., 2025]	Swin-B	22.9	26.6	16.4	19.7	n/r	n/r	n/r
INOVA [Li et al., 2025]	Swin-B	24.8	29.3	20.0	24.7	n/r	n/r	n/r
OvSGTR [Chen et al., 2025]	re-run here	20.4	23.8	13.2	15.9	4.1	1.9	13/50 (3/15 novel)
RelateAnything (ViT-S/16+), novel 15 held out	their boxes	22.4	26.5	11.8	15.0	14.9	7.2	40/50 (8/15 novel)
RelateAnything (ViT-S/16+), zero-shot	novel set seen	27.3	32.0	22.8	27.0	n/r	n/r	n/r
RelateAnything (ViT-S/16+)	novel set seen	27.9	32.6	24.1	28.5	n/r	n/r	n/r

6.6 A6: spatial relations against adversarial negatives

A6 is the axis on which a frequency prior cannot help. Without it, changes that moved recall by tens of percent while leaving the projective predicates near chance (Fig. 23 in § E) would have looked like improvements in relational understanding. The sigmoid auxiliary and the deformable read together are what moved them, raising in front of from 0.58 to 0.68, above from 0.54 to 0.60 and to the left of from 0.56 to 0.64 on the released tower, for a macro AUC of 0.690 and a pooled AUC of 0.675. This axis remains our weakest.

The baseline row evaluates OvSGTR on the same 2,758 cells with the same metric implementation, from ground-truth boxes and at full coverage: macro AUC 0.591, pooled 0.617. We are ahead on all nine predicates, and on three the baseline is at or below chance (above 0.495, to the right of 0.489, under 0.515), the insensitivity to projective structure the axis was designed to detect. Neither model reaches the 68.8% of the original authors’ boxes-only baseline, which does not see the image: we reach 62.5% and the baseline 59.0%. An AUC near 0.69 is therefore not evidence of visual spatial reasoning, and § E lists the probes bounding this claim. We read the remaining gap as a limit of supervision rather than capacity, since capacity does not affect it (§ G).

6.7 The OvR-SGG leaderboard

Since § 5.3 reproduces OvR-SGG exactly, we can also report on the leaderboard itself, scored by the baseline’s matcher rather than ours (Tab. 8): VG150 in detection mode, 15 of 50 predicates held out of training, our rows scored with a transcription of the baseline’s metric on its own detector boxes, predicted labels and ranking rule. § D.7 documents the two incompatible predicate splits in circulation and why the table retains the uncorrected matcher.

Composition of the novel split. Recall at K on this split is a micro average, and the 15 novel predicates carry 55% of the test relations because *on*, *of* and *in* are among them and account for 93% of the novel mass by themselves. Novel recall is therefore, to within a few points, recall on three of the most frequent strings in Visual Genome, and the baseline scores non-zero on 3 of the 15. Mean recall makes this visible, which is why Tab. 8 adds it for the two systems we evaluate ourselves: the baseline reaches 4.1 Base+Novel and 1.9 Novel mR@50 against our 14.9 and 7.2, alongside a Base+Novel micro recall differing by two points.

Table 9 The share of each model’s output that is a lookup of its object-label pair. VG150 test, both models on the *same* boxes and predicted labels (the baseline’s Swin-T detector), top-50 pairs per image. *self-lookup*: share of predictions reproduced by a table of the model’s own most frequent predicate per label pair, fitted on the other half of images; $H(p \mid c_s, c_o)$: conditional entropy of the prediction given the label pair (nats); *freq. agreement*: share equal to the training majority predicate. Bottom block: labels shuffled or constant.

Model	self-lookup	$H(p \mid c_s, c_o)$	H ratio	freq. agreement	distinct p
OvSGTR Swin-T [Chen et al., 2024b]	87.7	0.41	0.17	65.1	20 / 50
OvSGTR Swin-T + MegaSG [Chen et al., 2025]	89.9	0.37	0.14	74.3	31 / 50
RelateAnything (ViT-S/16+), zero-shot	68.6	1.24	0.30	52.8	50 / 50
RelateAnything (ViT-S/16+)	69.7	1.19	0.30	53.9	50 / 50
<i>self-lookup (%) when the object labels handed to the model are true / shuffled across boxes / a fixed pseudo-label, 5,000 images</i>					
OvSGTR Swin-T	85 / 85 / 73				
RelateAnything (ViT-S/16+)	67 / 67 / 52				

Holding out the novel predicates. Our released tower has seen all 50 predicate strings of VG150, since they lie within its training vocabulary, so its rows do not satisfy the protocol on the novel half and are marked accordingly. We therefore retrained the tower with the 15 novel predicates and their synonym groups removed from the training mixture, the analogue of the rare-class deletion required by OV-LVIS [Gu et al., 2022] (§ D.7). This model reaches 22.4 Base+Novel and 11.8 Novel R@50, which places it ahead of the Swin-T entries without MegaSG pre-training on Base+Novel, behind the MegaSG-pretrained and Swin-B entries, and behind every published row but one on Novel recall. We make no claim to lead on those columns. The gap to the full-vocabulary rows, -5.5 and -12.3 R@50, measures how much of novel performance under this protocol consists of supervision on *on*, *of* and *in*. On the leaderboard’s own micro metric the held-out model is mid-table; on mean recall over the same predictions it exceeds the baseline by a factor of nearly four. This disagreement illustrates the effect described in § 5.2.

7 Analysis: decoupling object and relation prediction

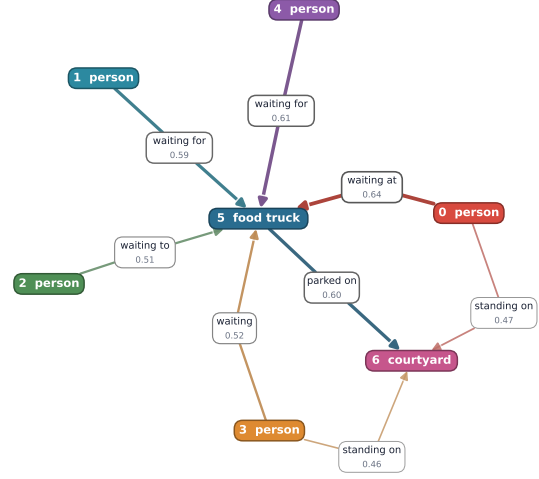
The dominant scene-graph architecture predicts objects and relations from a single backbone and conditions the relation head on predicted object classes [Xu et al., 2017, Zellers et al., 2018, Tang et al., 2019, Chen et al., 2025]. RelateAnything does neither. We first measure what the label input costs the strongest baseline (§ 7.1) and show the region source to be interchangeable (§ 7.2), then ask whether coupling would have helped: whether relation supervision produces relational features a detector could share (§ 7.3), and where in the input the relation logit comes from (§ 7.4).

7.1 Label dependence of the relation head

Tab. 9 measures label dependence on the predictions both models produce for the leaderboard of § 6.7. For the baseline, 88–90% of predictions are reproduced by a table of its own most frequent predicate per label pair, the label pair explains all but 0.4 nats of its output entropy, and only 20–31 of the 50 predicates occur. For our model the lookup reproduces 69–70%, the residual entropy is three times larger, and all 50 occur. The lower block reports an intervention: shuffling the labels leaves the baseline’s output statistics unchanged, so its predicate is a function of the labels rather than the image, whereas our model cannot observe them at all. Our residual 69% is the part of relation prediction determined by category (people wear clothes, cups rest on tables), recovered

8 relations at $p \geq 0.45$

4,585-tag prompt-free regions · 19,103 predicates · score = sigmoid(0.56·(pred+rel)-1.96), calibrated



10 relations at $p \geq 0.45$

1-tag prompt-free regions · 19,103 predicates · score = sigmoid(0.56·(pred+rel)-1.96), calibrated

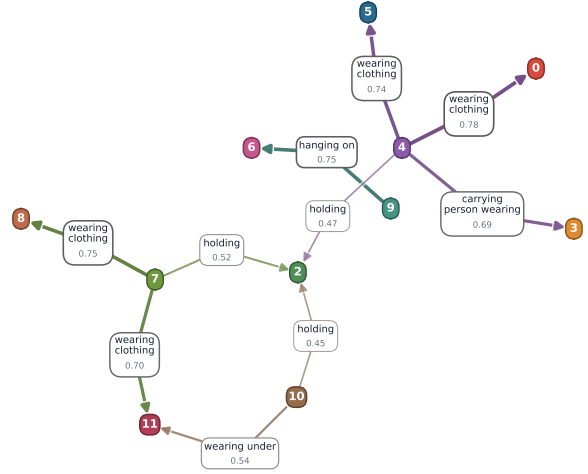
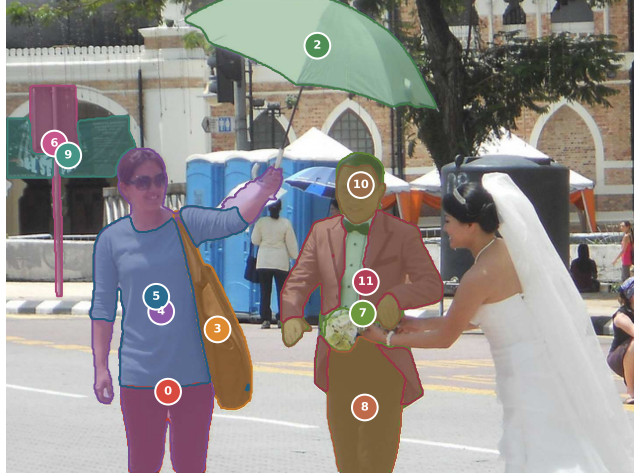


Figure 6 One relation model behind two unrelated region sources. Both rows use one checkpoint at the same calibrated operating point ($p \geq 0.45$, at which it emits PSG’s own density of 6.0 edges per image, § D.5) over the same 19,103-predicate vocabulary; only the region source differs. Top: YOLOE [Wang et al., 2025a] run prompt-free over 4,585 tags returns seven named segments, from which the model reads eight edges; *food truck* and *courtyard* are names no scene-graph vocabulary contains. Bottom: FastSAM [Zhao et al., 2023] has a single class, so its twelve segments carry an index and no name, and the model still returns *wearing clothing*, *holding* and *carrying person wearing*. Names in the top row are drawn for the reader and never reach the model. Both panels use a proxy-scale development checkpoint; they illustrate the interface rather than the operating quality.

from pixels rather than supplied as input.

7.2 Region sources

Because the model receives regions rather than detections the box source is interchangeable, and § 6.4 showed every margin of Tab. 4 surviving a real detector with the same structure; because

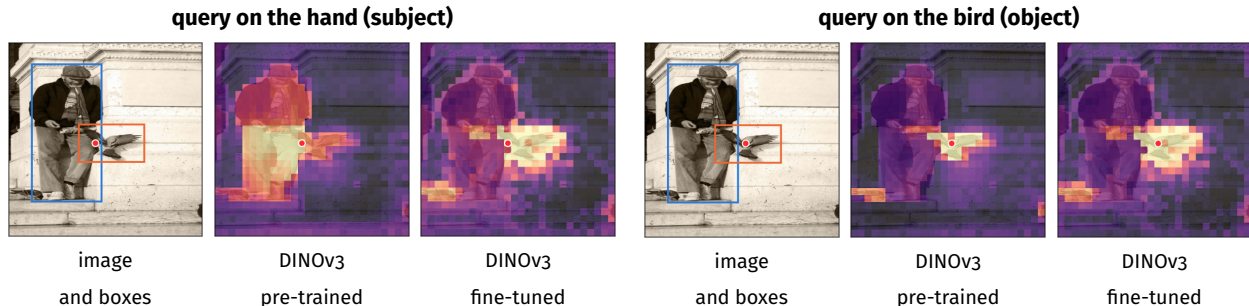


Figure 7 The interaction region is an object feature. Cosine similarity to the query patch (red dot), last state, per-image mean-centred. Left three panels: the query is the subject patch nearest the object within their intersection, the hand. Right three: the query is on the object, the bird. Before fine-tuning the hand query highlights the whole person; after, it highlights hand and bird, and an interaction region has emerged. Querying the bird yields the same region, and retrieving the *verb* from it fails (Tab. 28 in § F): the region encodes which object is handled, not what is being done with it.

labels are never an input, the detector’s vocabulary is likewise unconstrained. Fig. 6 shows the consequence: one checkpoint reads relations from an open-vocabulary detector prompted with 4,585 tags and from a class-agnostic segmenter supplying no names, and the edges returned in the second case are of the same kind as in the first (see also Fig. 22 in § E, on 1,198-class LVIS boxes [Gupta et al., 2019, Wang et al., 2025a] passed through unchanged). The failure mode a region-reading model inherits is visible there too: a detector returning eleven furniture boxes yields a dense layout graph with few interactions, since the relation model cannot be more selective than its boxes.

7.3 Effect of relation supervision on the dense features

For a single backbone to serve both tasks, relation supervision would have to induce features that encode relations. The probes in Tab. 28 (§ F) indicate that it does not. The backbone is fully fine-tuned, and fine-tuning changes it substantially and predictably (Fig. 26): similarity to the pre-trained model falls with depth, to a linear CKA [Kornblith et al., 2019] of 0.58 at the last state, while two runs under different recipes stay at a CKA of at least 0.94 with each other, so the destination is set by the task. What changes is within-object rather than between-object structure, visible directly in the patch features (Fig. 9 in § A).

Patch similarity. Querying from a subject’s patches and measuring cosine similarity across the image, we find that fine-tuning raises class selectivity from 0.242 to 0.281 and leaves relation selectivity essentially unchanged, from 0.133 to 0.138. The probability that a relation partner is more similar to the subject than an unrelated object of the same class decreases from 0.469 to 0.385, against a chance level of 0.500. Fine-tuning on relations thus made the dense features less able to distinguish a partner from an arbitrary object of the same category.

The interaction region. A weaker form of the hypothesis is that relational structure emerges only where the objects meet. That region does change: the contact advantage over a distant subject patch rises from 0.356 to 0.438, and the feature shifts from covering the whole person to covering hand and object (Fig. 7). What it encodes, however, is object identity. Cross-image 1-NN retrieval of the object class from that feature reaches 0.761 against a chance level of 0.068, whereas retrieval of the verb reaches 0.269 against a majority-class chance of 0.424, that is, below chance, and fine-tuning does not change this at any depth (Fig. 25 in § F). The finding is specific to our recipe and this

Table 10 **Input-channel lesion ladder**, zero-shot tower, per-edge top-1 accuracy over ground-truth pairs, test splits ($n = 152.5\text{k} / 13.6\text{k} / 29.2\text{k}$ edges). Each row removes one channel and re-runs inference; the top block is absolute accuracy for the intact model, the rest relative changes. The variance decomposition is exact in fp32 (maximum residual 2×10^{-5}).

channel removed	micro Acc@1 (% Δ vs. full)			macro Acc@1 (% Δ)		
	VG150	PSG	IndoorVG	VG150	PSG	IndoorVG
full model	57.7	43.3	57.3	35.1	31.6	38.4
– object→text channel	-0.1	-0.0	-0.1	-4.4	-0.2	+0.6
– box geometry	-16.8	-23.6	-6.0	-36.1	-33.3	-29.6
– pixels (shuffled image)	-57.5	-68.3	-44.3	-87.2	-91.1	-85.3
– all inputs	-99.3	-90.5	-98.5	-94.3	-94.4	-93.0
<i>exact variance decomposition of the semantic logit:</i>						
pair context	92.0%	87.4%	93.1%			
subject / object features	0.1 / 0.0%	0.1 / 0.0%	0.1 / 0.0%			

family of objectives (§ 10), but it indicates that a backbone trained the way scene-graph backbones currently are holds no relational representation for a detector to share.

7.4 Attribution of the relation logit

Given that the relation is not encoded in the object features, we ask where it originates. Tab. 10 answers by lesion and by exact decomposition, and the two agree: the object-identity channel contributes almost nothing. The head contains an optional path by which subject and object features address the predicate vocabulary directly, gated by a learned scalar initialised at 0.1 (a separate gate from the predicate-conditioned α_p of Eq. (2), not lesioned here). Training drives it to 0.014–0.044 across all three towers, the subject and object terms account for 0.1% and 0.0% of the semantic logit’s variance, and removing the path costs 0.1% of micro accuracy on every source, affecting only macro accuracy on VG150, by −4.4%. Pair context accounts for nearly all the remainder, 87–93% of the variance. Scoring against the features of a different image costs 44–68% of micro and 85–91% of macro accuracy, and removing box geometry 6–24%. The micro–macro gap is the asymmetry of Tab. 1 again: a prior supplies head predicates; visual evidence contributes in the tail.

Summary. Relation supervision does not create relational features (§ 7.3), the relation logit does not depend on object identity (§ 7.4), and object labels supply a prior that inflates the metric by which methods are ranked and does not transfer (§ 5.2, § 7.1). There is thus no shared representation that would justify a shared backbone, and there is a measurable cost to the shared input. The detector can consequently be replaced without retraining the relation model, and the relation model cannot inherit the detector’s label distribution.

8 Analysis: in-domain versus cross-dataset measurement

We evaluate the same models in-domain and under transfer, and test three quantities that might be expected to predict transfer: alignment of the training distribution with the target vocabulary, quantity of annotation when that annotation is sparse, and backbone capacity. Each uses a single-variable arm read against the 1.3% noise floor of § H. A prior the model violated that no axis could

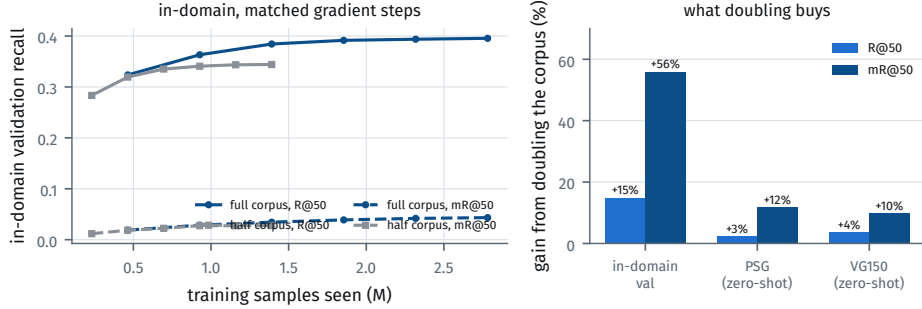


Figure 8 Doubling the corpus, read in-domain and under transfer. Left: in-domain validation recall against training samples seen, so the half-corpus arm is compared at matched gradient steps; the penalty for repetition grows during training and that arm saturates by its third pass. Right: the relative gain of the full corpus at the end of training, in-domain and on two zero-shot benchmarks.

detect is analysed in § H.3.

8.1 Corpus scale read in-domain and under transfer

Both arms below hold everything else fixed and are evaluated in-domain on the corpus’s own validation split and zero-shot on VG150 and PSG test. Doubling the corpus yields +15% in-domain micro recall (R@50) and +56% in-domain macro recall (mR@50); under transfer the same change yields +2.5 to +3.8% and +9.9 to +11.9% (Fig. 8), so the in-domain slope overstates the transfer slope by approximately 5×. The added images teach the corpus’s annotation style, which the in-domain split shares by construction, so in-domain measurement credits a corpus-bound gain.

The second arm changes sign. Removing an automatically derived left/right excess from the mixture improves the in-domain metrics at every epoch, by +2.0% micro and +27.0% macro, but reduces transfer macro recall by between −5.9 and −0.3% across the two benchmarks. Two consequences follow. Decisions on data scaling must be made on transfer measurements; the extrapolated transfer gain from a further 500k annotated images was about one point absolute, and we therefore did not generate them. The same applies to checkpoint selection: ranking arms by their final epoch predicts out-of-distribution behaviour better than best-epoch selection ($\rho = 0.77$ against 0.66).

Distribution alignment. A natural explanation of the mixture result is that transfer tracks the share of training mass that falls on the target benchmark’s predicates. This is not the case. Dropping the automatically derived edges raises the training mass on the 56 PSG test predicates from 27.3% to 39.4% and reduces left/right from 29.5% to 7.4%, as verified on the loaded relations. Transfer macro recall nonetheless decreased, finishing 3.8 points below the mixture it replaced.

Annotation quantity under sparse annotation. An Open Images V6 extension [Kuznetsova et al., 2020] of 62,589 images, audited for evaluation overlap, was tested at a deliberately amplified 25% relation share so that the sign of its effect would exceed the noise floor. Its effect was negative on every axis relative to the matched arm without it (the deformable-read arm of Tab. 30 in § H): about −6% on the development composite and −40.2% on the worst cell, HICO-DET tail recall. Open Images carries 4.33 relations per image against 9.03 for RA-4M, so its unannotated pairs enter training as false negatives under the positive-unlabeled structure of § 3.3. Sparse annotation is therefore worse than none, and the threshold at which this occurs is set by the loss, not the data.

Table 11 End-to-end cost against the baseline under one protocol, batch 1, median latency, eager PyTorch for both. *params* counts the whole system including the detector; 53.2M of our 231M belong to the relation model (Tab. 29). This compares deployed configurations rather than architectures: OvSGTR’s deformable-attention kernel has no half-precision implementation, so it is timed at tf32 against our bf16, at its native 800/1333 resize against our 448 px, scoring every pair among 98 boxes where we sample 128 pairs from 20; the box budget is the largest of the three differences.

System	params	boxes/img	batch 1, p50 ms			FPS
			A40	A100	H100	A40
OvSGTR Swin-T [Chen et al., 2025]	177M	98	194.0	179.9	128.1	5.1
OvSGTR Swin-B [Chen et al., 2025]	237M	98	228.5	195.1	134.3	4.3
RelateAnything (ViT-S/16+) + YOLO-World	231M	20	25.0	35.0	25.6	40.0

Backbone capacity. A matched ladder of three backbones, trained twice, once with the released recipe and once without the HICO-DET share, is reported in § G. Capacity does not raise the composite: on neither ladder is the composite monotone in model size, and the largest tower, with +114% parameters over the released one, changes it by -0.1% . On both ladders the smallest towers perform best on the spatial axis. What scaling changes is the source of the evidence: dependence on box geometry decreases with size while dependence on pixels increases. The advantage that the largest tower holds on individual axes does not survive the deployment operating point, at which 5–7 edges are emitted per image and most of them fall on boxes that no ground-truth object matches. We therefore release ViT-S/16+ and conclude that the component to improve next is the detector.

9 Inference cost

Open-vocabulary scene-graph models are generally reported without their computational cost. OvSGTR [Chen et al., 2025] scores every pair among about 98 detections at 800 pixels in fp32 at five frames per second on an A40. We report our cost against that baseline under one protocol and against the fastest closed-set real-time models we know of on a shared evaluator (§ I).

Comparison with OvSGTR. Tab. 11 places the two systems under one protocol. Our system is $7.8\times$ faster end to end on an A40 while having more parameters than the Swin-T baseline, because batch-1 cost is not dominated by parameter count. The baseline scores about 9,500 pairs among 98 boxes at 800/1333 in fp32, whereas we score at most 128 sampled pairs among 20 boxes at 448 px in bf16. Under the same box budget the two would be considerably closer.

Comparison with closed-set real-time models. Tab. 12 places the released tower alongside the two specialists as reported in their own papers, on the same per-dataset YOLOv8-m weights and at the K over which each source table averages. REACT++ is the faster model, at 21–23 ms against 35–36 ms under a protocol that runs our head eager and unfused, and both specialists reach higher micro recall on every benchmark they were trained on, while ours is higher on macro everywhere but PSG. The two combine into an F1@K above REACT on all three, above REACT++ on VG150, 0.2 below it on IndoorVG and 3.7 below it on PSG, for a model that has seen no annotation from any of them, answers from an arbitrary vocabulary and receives no object labels. Fine-tuned on the benchmark it takes the highest micro recall in the table on all three, at a cost in macro recall on VG150 and IndoorVG. The specialist recipe, one-stage decoding over a fixed predicate set, is available neither to an open vocabulary nor to a swapped detector.

Table 12 Generalist against specialist in the same latency class. REACT [Neau et al., 2025] and REACT++ [Neau and Falomir, 2026] are closed-set one-stage models trained on each benchmark; ours is trained once and zero-shot on every row but the fine-tuned one. Their rows are as published (REACT Tables 2–4, REACT++ Tables 2 and 3, plain non-DCS rows), each benchmark read at the K its source table averages over, F1@K quoted as printed. Our rows use the standard SCDet evaluator [Tang, 2020] (graph-constrained, seeds 0–2, maximum seed σ 0.006) on those papers’ per-dataset YOLOv8-m weights, averaged over the same K . Bold: best per column and benchmark. Latency measured here for every row, batch 1 end to end on one A40 in eager mode; dashes mark rows we did not time.

Model	in-domain	R@K	mR@K	F1@K	ms (A40)
<i>VG150 test, YOLOv8-m boxes, 22,371 images; $K = 50/100$</i>					
RelateAnything (ViT-S/16+), zero-shot	no	26.0	15.4	19.3	35.9
RelateAnything (ViT-S/16+)	no	26.7	15.2	19.4	35.9
RelateAnything (ViT-S/16+), fine-tuned	yes	30.2	11.1	16.3	35.9
REACT [Neau et al., 2025]	yes	27.4	12.9	17.6	—
REACT++ [Neau and Falomir, 2026]	yes	28.9	13.2	18.2	21.3
<i>PSG test, YOLOv8-m boxes, 2,177 images; $K = 20/50/100$</i>					
RelateAnything (ViT-S/16+), zero-shot	no	25.6	22.8	24.1	35.7
RelateAnything (ViT-S/16+)	no	26.5	23.2	24.7	35.7
RelateAnything (ViT-S/16+), fine-tuned	yes	36.4	24.9	29.5	35.7
REACT [Neau et al., 2025]	yes	30.3	19.8	23.9	41.3
REACT++ [Neau and Falomir, 2026]	yes	33.6	24.7	28.4	20.9
<i>IndoorVG test, YOLOv8-m boxes, 4,403 images; $K = 20/50/100$</i>					
RelateAnything (ViT-S/16+), zero-shot	no	25.2	22.2	23.6	35.3
RelateAnything (ViT-S/16+)	no	25.9	21.8	23.7	35.3
RelateAnything (ViT-S/16+), fine-tuned	yes	32.2	18.8	23.7	35.3
REACT [Neau et al., 2025]	yes	30.9	18.0	22.8	—
REACT++ [Neau and Falomir, 2026]	yes	28.1	20.7	23.9	22.6

Deployment configuration. Three measurements from § I determine how the model is run. At batch 1 the relation head is bound by kernel dispatch rather than arithmetic: the three towers cost 19–20 ms on an A40 across a $2.5\times$ range of FLOPs. Compilation is the largest single gain, $1.5\text{--}1.8\times$ on every GPU, and brings the released tower with its detector and decoding to 20 ms per frame. Scoring 19,103 strings instead of 50 costs under a millisecond at batch 1 and 11–22% of throughput when batched. Because the text encoder lies outside the inference graph and the detector is not part of the model, the same graph runs on a CPU: an fp16 build reaches 7 frames per second on eight threads with 0.955 top-1 agreement against fp32 (Tab. 36).

10 Discussion and limitations

Held-out concepts. Every unmarked result is cross-dataset: the images, object distribution and annotation style of each benchmark are unseen. The predicate strings are not: every benchmark predicate occurs in the training vocabulary. § 6.7 evaluates the held-out variant on the split that defines it; the general variant, holding concepts out across all four benchmarks, has not been run, although the architecture permits it since the predicate matrix is never learned. Several conclusions rest on one seed; a difference within the 1.3% noise floor of § H is named as such.

Corpus precision. Every precision claim about RA-4M is structural: the gate rejects 11.3% of candidates, a rejection is a true negative up to box error, and predicates geometry cannot constrain are named and counted rather than checked (§ 4.1). We have no human-audited precision estimate, a sample of generated relations read and scored by a person. That is the measurement a reader should expect before relying on 4.3M machine-generated annotations, and the one to add first.

Scope of the decoupling result. § 7 shows that our relation supervision does not produce relational dense features, not that none could: nothing in our recipe provides a patch-level relational objective, which the probes suggest is what a shared backbone would require.

The A5 verdicts. A5 is scored by Eq. (4), the information in the relations a vision–language judge accepts, and our model leads on it. The same judge returns two verdicts favouring the baseline (§ 6.5): shown whole graphs at deployed length it prefers the baseline’s, and shown one relation at a time it accepts slightly more, which a second judge repeats (§ D.8). Per-relation precision is a genuine shortcoming: at any depth the baseline overreaches less, and a deployment that cannot tolerate a false edge should prefer a shorter graph. Methodologically, a judge cannot perceive repetition, so informativeness must be measured against a reference distribution, not elicited.

Detector recall. The dominant loss between a benchmark number and a deployed system is object-detector recall: pair recall is quadratic in object recall and no relation model can exceed it, while a larger relation model buys accuracy in a part of the ranking deployment never emits (§ G.1).

Reporting practices. Each of these is inexpensive and changed a conclusion here: the shared triplet mass between corpus and benchmark, beside every recall number (§ 5.2); the matcher and whether it enforces one-to-one assignment; the detector’s threshold and box cap, with their pair-recall ceiling (§ 5.3); whether recall is graph-constrained; and one axis on adjudicated negatives (§ 5.4).

11 Conclusion

Detection and segmentation became taxonomy-free; relation prediction did not. We have argued that the obstacles were the supervision, the architecture and the measurement, and we have removed one of each. RA-4M supplies 4.3M relations over 10,102 free-text predicates, verified against the geometry of the boxes they name. RelateAnything takes regions from any source and a vocabulary supplied at inference, and on three benchmarks evaluated cross-dataset and a fourth evaluated zero-shot it exceeds the strongest open-vocabulary method of comparable scale by $2.3\text{--}3.5\times$ in mean recall, at 20 ms per frame. OV-SGG-Bench scores six axes across datasets that no prior can satisfy.

The measurement obstacle is the most consequential, because it is invisible from inside the standard protocol. Scene-graph recall as currently reported is largely a prior-matching score: a pixel-free frequency table wins the leaderboard metric, benchmarks share the training corpus’s annotation style, and scoring conventions move results by more than the differences between methods. Two further findings follow from measuring across datasets. Coupling object and relation prediction has no representational basis in our probes, since relation supervision sharpens object identity without creating relational features and the relation score is almost entirely determined by pair context. And in-domain measurement overstates transfer gains by about $5\times$ and can favour changes that reduce transfer.

Corpus, model and protocol are public. The cheapest practice we recommend is to report the shared triplet mass between corpus and benchmark beside each recall number: how much of a model’s supervision asserts the same relations, of the same kinds of object, as the benchmark.

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A Architecture and training details

A.1 Visual path

The backbone, the feature taps and the pooling are specified in § 3.2. Two choices there were settled by ablation (§ H). The three taps are layer-normalised individually; without this the fusion is dominated by the larger activation scale of the last layer. The backbone is fully fine-tuned at a learning rate $8\times$ below the head’s, which at this scale outperforms both a frozen backbone and LoRA adapters [Hu et al., 2022], although the ordering of those two alternatives depends on the checkpoint selection rule and inverts under best-epoch selection.

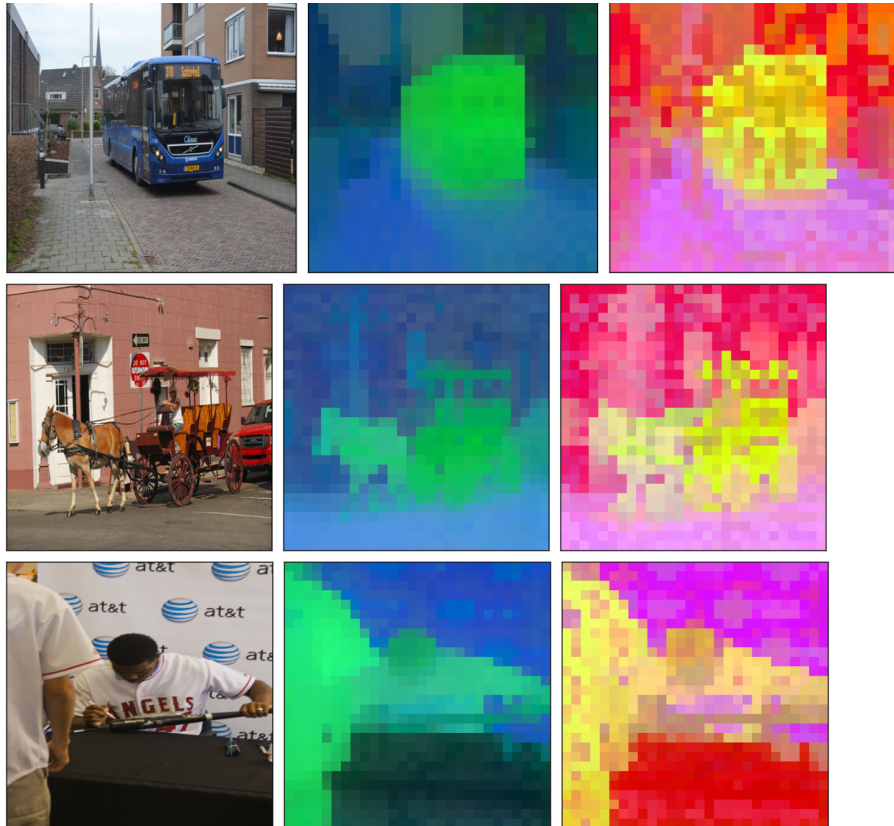


Figure 9 Last-state patch features before and after relation fine-tuning, projected by a single PCA fitted jointly per row (columns: image, pre-trained, fine-tuned). Fine-tuning sharpens object-versus-background boundaries and part texture within objects; it does not introduce structure that spans an object pair. The effect is quantified in Fig. 26 (§ F).

A.2 Deformable scene read

The read samples $8 \text{ heads} \times 4 \text{ points}$ around each of a pair’s four anchors. Three properties of the stage matter. It is additive behind a zero-initialised gate, so that training starts from the model without it and the gate’s own gradient decides whether the read is used; a module that perturbed an already converged system would measure the perturbation rather than the mechanism. Offsets are unbounded, in units of the anchor’s half-extent, because the anchors are an initialisation prior rather than a constraint, and clamping would prevent exactly the reads outside the box that

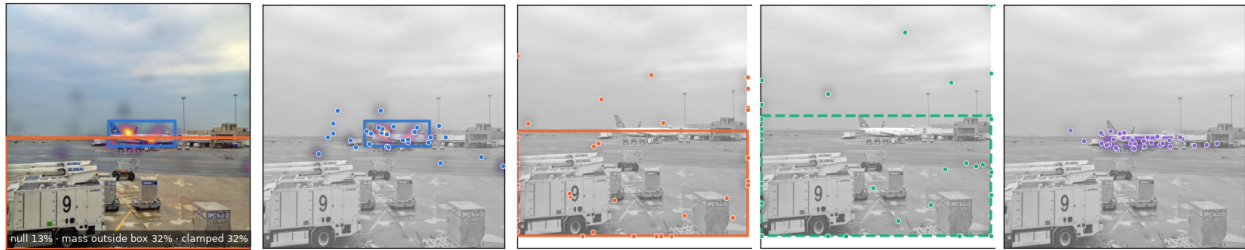


Figure 10 Sampling locations of the deformable read, for one PSG-val pair, airplane parked on road at score 0.99. Left: attention mass splatted at half-patch width. Then, per anchor (subject, object, union, contact), the sampled points. The subject anchor carries 56% of the mass and reads the runway around the aircraft; the object and union anchors sample mostly outside their boxes and carry 6% each.

Table 13 The two-graph decode: one forward pass scored separately over the benchmark’s spatial and semantic predicates, split by the corpus type map. Test splits, ground-truth boxes, graph-constrained.

benchmark	layout graph (spatial predicates)			content graph (semantic predicates)		
	#pred	R@50	mR@50	#pred	R@50	mR@50
VG150	7	63.1	30.6	43	48.6	30.3
PSG	4	60.5	53.9	52	40.5	30.3
IndoorVG	6	61.2	35.1	31	42.0	30.2

motivate the stage. Points are initialised on rings of distinct angles and radii per head; a variant that zero-initialised the offset predictor placed every point of an anchor at one location, where identical gradients cannot differentiate them.

Visualised on 300 PSG and 300 HICO-DET images (Fig. 10), the trained read is dominated by the subject and contact anchors (PSG shares: subject 0.43–0.48, contact 0.17–0.21, object 0.12–0.17, union 0.04–0.05). The union anchor contributes least because its box approaches the whole image, so that 1.5 half-extents falls outside the frame and 42–53% of its reads are clamped at the border. Direction is carried by the null slots rather than by geometry: a person→tie **wearing** pair scored 0.99 places 61% of its mass on the contact anchor, whereas the same pair with the roles reversed, scored 0.04, places 52% on the null slots. The eight heads weight one location rather than sampling eight distinct ones (divergence 0.05–0.07).

A.3 Pair head and predicate-conditioned gate

The gate of Eq. (2) receives no supervision, and the weights it learns over the training vocabulary are bimodal and semantically ordered (Fig. 11). The median α_p is 0.002 and 12% of predicates exceed 0.5. Projective relations lie at the geometric extreme (**behind** 1.000, **below** 0.9998, **above** 0.970) and actions at the other (**carrying** 0.0004, **parked on** 0.0008, **riding** 0.012), with contact predicates between them (**holding** 0.82, **on** 0.64, **wearing** 0.22), as is to be expected of predicates that require both geometry and appearance.

Because the two branches are scored before they are mixed, one forward pass yields two graphs: a layout graph over the benchmark’s spatial predicates and a content graph over its semantic ones (Tab. 13). The layout graph is the easier problem on every source, and the content graph carries the vocabulary.

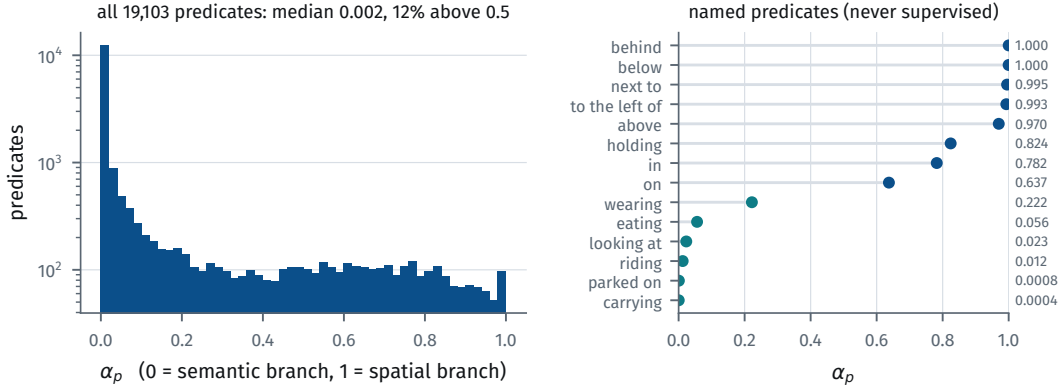


Figure 11 The predicate-conditioned gate over the vocabulary, computed from the text embedding alone. Left: distribution of α_p over all 19,103 training strings (log scale). Right: selected predicates. Projective and proximity relations are routed to the geometry-dominated branch, actions to the appearance branch, and contact relations in between; none of this is supervised.

A.4 Training recipe

Table 14 The released training recipe. The run takes 5h04m on four A100 GPUs.

Setting	Value
backbone	DINOv3 ViT-S/16+ (LVD-1689M)
input resolution	448, multi-scale 0.5,1.5 over 7 rungs
feature taps	-6, -3, -1, layer-normalised
model width	$d = 512$
backbone adaptation	full fine-tune
deformable read	4 points, 8 heads, 2 null slots
pair budget	400 by geometry $\rightarrow$ 128 learned
epochs	12
batch size	32 per GPU $\times$ 4 GPUs
learning rate	head 4×10^{-4} , backbone 5×10^{-5}
dropout / augment	0.2 / photometric 0.3, no geometric
loss weights	InfoNCE 0.5 $\cdot$ sigmoid 0.25 $\cdot$ swap 0.5 $\cdot$ background 0.05 $\cdot$ grounding 0.1
sampler loss weights	geometry pre-scorer 1.0 $\cdot$ relatedness 1.0
negative discount	estimated $\hat{P}(q p)$, Eq. (3)
positive aggregation	weighted mean over the synonym group
InfoNCE negatives	512 per anchor
source-aware negatives	HICO-DET train
EMA	0.9998
mixture	RA-4M 72.7%, raw Visual Genome 6.3%, HICO-DET train 21.0% by image

Two elements of Tab. 14 depart from common practice. The resolution of a batch is drawn from seven values in $[0.5, 1.5] \times 448$; its mechanism is regularisation rather than resolution (§ H), and it is what allows the released model to be evaluated at other resolutions without retraining (Tab. 31). Geometric augmentation is excluded, because a horizontal flip inverts every left/right relation and the corpus contains these in balanced quantity, so that flipping without rewriting the labels would introduce noise on the axis that most requires supervision. Photometric augmentation, which commutes with geometry, is retained.

B Training objective

Table 15 Notation used in §§ 3 and 3.3 and in this appendix.

symbol	meaning
$I, B = \{b_i\}$	the image, and the regions supplied with it
$\mathcal{V}, p$	the predicate vocabulary supplied at inference, and one predicate in it
v_i	pooled visual feature of region i
$v_{\cup}, v_{\cap}$	pooled features of the union box and of the contact region
g_{ij}	19 scale-invariant geometry features of the ordered pair (i, j)
x_{ij}	the 512-d pair representation assembled from the five above
$z_{ij}^{\text{spa}}, z_{ij}^{\text{sem}}$	the pair’s geometry-branch and appearance-branch embeddings
e_p	text embedding of predicate p , from the frozen encoder
$\alpha_p = g(e_p)$	per-predicate gate mixing the two branches, Eq. (2)
τ, β	learned scale and bias on the predicate logit
$\ell_{\text{pred}}, \ell_{\text{pair}}$	the predicate logit and the pair-relatedness logit
c_s, c_o	subject and object <i>categories</i> , used only in the sampler and grounding losses
$\hat{P}(q p)$	estimated probability that q also holds when p is annotated, Eq. (3)

B.1 Auxiliary objectives and source-aware negatives

Four terms accompany the InfoNCE objective of § 3.3. None is novel in isolation. A per-cell *sigmoid auxiliary* treats each (pair, predicate) cell as an independent binary problem, with the negative cells of a pair weighted so that their total mass equals that of its positives; together with the deformable read it is what moved the projective spatial band (§ 6.6), and it is what makes the scale of the output head trainable. A *swap hinge*

$$\mathcal{L}_{\text{swap}} = \max(0, m - c(i, j, p) + c(j, i, p)) \quad (5)$$

on directional predicates, where c is the mixed cosine inside Eq. (2) and $m = 0.05$, makes the subject–object order contribute to the loss; each predicate’s term is weighted by its estimated probability of being directional (§ 3.3). *Background suppression* lowers the top- k scores of unannotated pairs, at a small weight because an unannotated pair is not necessarily negative; a gated variant of this term is the traced cause of the prior violation of § H.3. A light *object–text grounding* term aligns pooled object features with category text embeddings, which is supervision on the features rather than an input. The two stages of the pair sampler carry binary losses of their own, a plain one on the geometry stage and a focal one on the relatedness stage; all weights are listed in Tab. 14.

Source-aware negatives. Mixing corpora of different annotation scope makes the positive-unlabeled problem source-specific. HICO-DET annotates verbs and never spatial relations, so a pair annotated **holding** in that source is not evidence against **on**; InfoNCE nevertheless treats it as such, and the measured effect is a degradation of the spatial axis. Restricting the negatives to predicates that the source could have annotated, by a per-source mask applied on top of Eq. (3), recovers +12% on that axis, which is 39% of what the HICO-DET share costs there, and gives the best composite of the family. It is used in the released model and not in the zero-shot one (§ 4.2).

B.2 Predicate text space and the synonym penalty

Table 16 Predicate text-space geometry before and after distillation, over the 19,103-string training vocabulary. *syn* and *inv* are mean cosine similarities over 1,967 synonym and 770 inverse pairs; AUC separates the two populations; NN₁ is the share of predicates whose nearest neighbour lies in their own synonym group; effective dimension is the participation ratio; hubness is the skew of the 10-NN in-degree distribution.

encoder	dim	syn cos	inv cos	AUC _{syn/inv}	NN ₁ in group	eff. dim	hubness
dino.txt teacher	2048	0.96	0.92	0.854	0.36	87	8.1
+ mean removal / ABTT	2048	0.49	0.12	0.828	0.41	142	0.9
student, 768-d, no neighbourhood term	768	0.93	0.50	0.980	0.18	24	1.5
student, 512-d, shipped	512	0.71	0.09	0.989	0.29	44	1.3

B.2.1 The distillation objective

The student of § 3.3 is trained against the frozen teacher under six terms. *Relational distillation* matches the student’s pairwise-cosine matrix to the teacher’s over each minibatch; being defined on cosines it is independent of the two dimensionalities, and it is what the downstream synonym and inverse masks read. An *absolute anchor*, at a small weight, regresses the student onto a principal-component projection of the teacher target and fixes the global frame. *Neighbourhood preservation* is a row-wise KL divergence between the teacher’s and the student’s softmax over batch similarities. *Antonym repulsion* is a hinge pushing the cosine of a known inverse pair below a margin, and *synonym cohesion* a hinge holding canonical-group pairs above one; a softer hinge at a lower margin holds modified spatial forms such as **far above** near their base predicate without merging them. The teacher targets are de-hubbed before distillation by removing their three leading principal directions [Mu and Viswanath, 2018], without which every off-diagonal cosine sits near 0.80 and the relational term is dominated by that bulk.

Two of the six terms carry the result (Tab. 16). Antonym repulsion is the term a frozen teacher cannot supply, since the teacher places **to the left of** and **to the right of** at 0.99 and does not separate inverses from synonyms at all (0.92 against 0.96). Neighbourhood preservation is what prevents the student from buying that separation cheaply: without it, a student reaches an equally good synonym-versus-inverse AUC (0.980 against 0.989) but collapses to an effective dimension of 24 rather than 44, and its nearest neighbour lies in its own synonym group for 18% of predicates rather than 29%. The shipped student places inverses near orthogonal (0.09) while holding synonyms together at 0.71.

B.2.2 Alignment to a synonym group

The objective of § 3.3 aligns a pair embedding to a region of the text space, the synonym group of the annotated predicate, rather than to a single point. The cost is that the model frequently answers with a member of the correct group other than the annotated one, which exact-string metrics count as a miss. Training against the group as a whole pushes the non-winning surface forms to ranks 90–338, and the surviving form is not necessarily the annotated one.

The failure is easily misdiagnosed as a directional one. Fig. 12 shows the substitutions on PSG. Directional strings are the clearest case: the model answers **on the right of** where an annotation reads **to the right of**, which exact-string recall counts as a miss, whereas measuring direction directly by subject–object swap places the model’s directional accuracy at 0.97. Every closed-vocabulary number in this paper uses exact matching and is therefore conservative for our model in a way that it is not for closed-vocabulary baselines; A3 gives the corrected view.

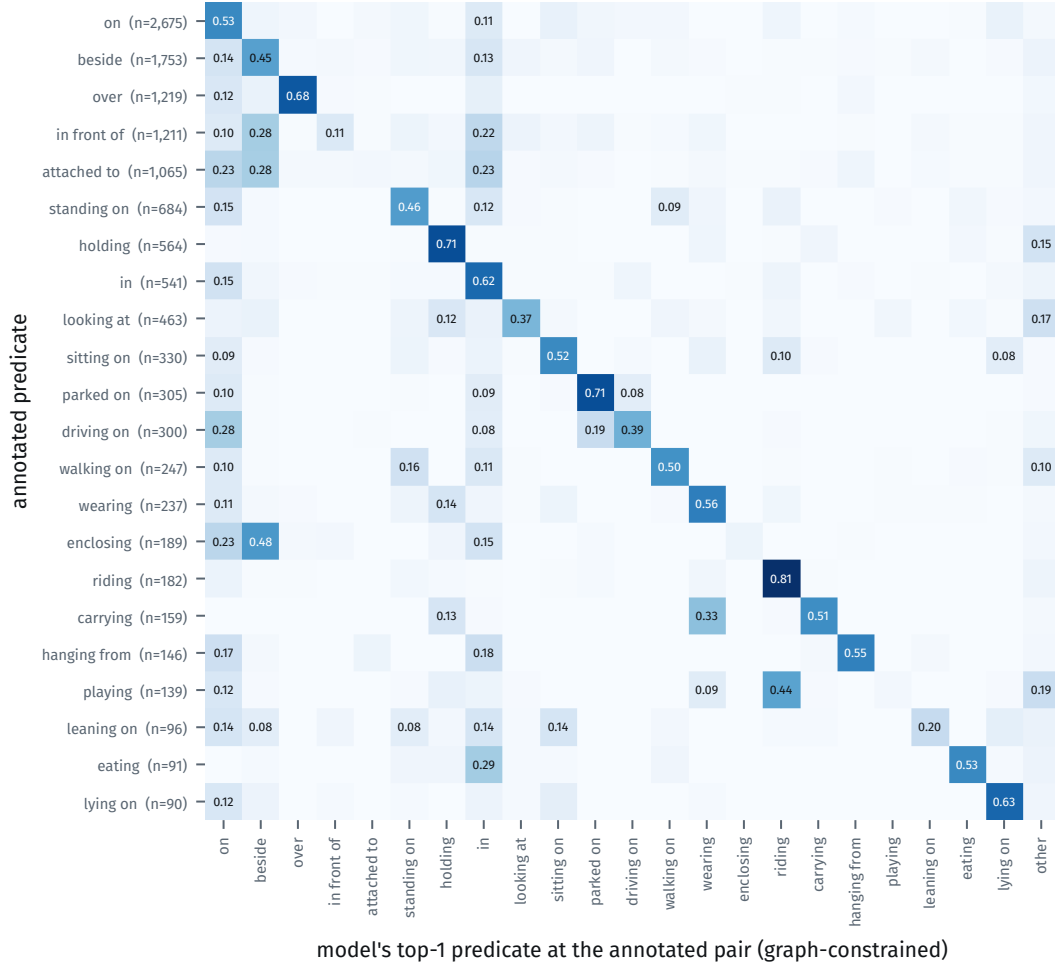


Figure 12 Confusion of exact-string top-1 predictions on PSG test, for the 22 most frequent annotated predicates: the model’s graph-constrained top-1 prediction at every annotated pair, with rows normalised. The off-diagonal mass is not random: **in front of** is predicted as **beside** (0.28) and **in** (0.22), **enclosing** as **beside** (0.48), **attached to** as **beside** (0.28) and **on** (0.23), and **carrying** as **wearing** (0.33). These are neighbours or entailments of the annotated string, and are scored as misses.

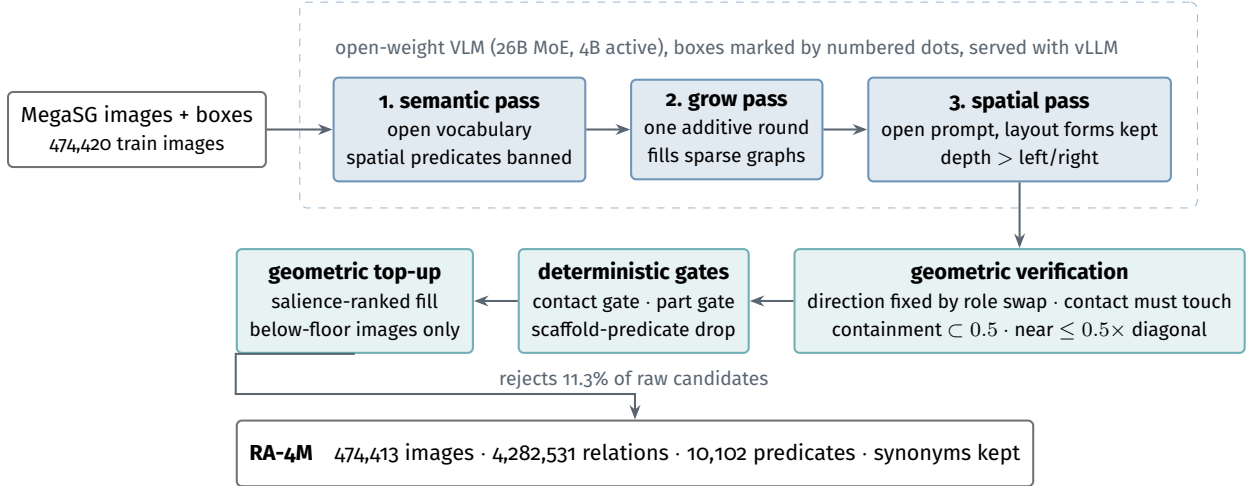


Figure 13 The generation pipeline. Three passes of an open-weight vision–language model propose relations against numbered box markers; a deterministic layer verifies what geometry can verify, rejects what it contradicts and passes what it cannot constrain. A salience-ranked geometric top-up fills the few images that remain below the spatial-density floor.

C Corpus details

C.1 Source images and generation

We annotate the image set of the MegaSG release [Chen et al., 2025], 500k COCO-style images drawn from COCO, Objects365 and web sources, taking its object boxes verbatim and replacing its relation annotations entirely. Fixing the box set isolates the contribution of relation supervision and lets every generated graph be joined back to the source annotations for inspection (Fig. 16). Of the 499,388 images with at least two boxes, 474,420 form the training split and 24,968 the validation split; seven training images lose every candidate to the gates, which is why Tab. 20 counts 474,413.

Generation uses an open-weight vision–language model [Gemma Team, 2026] (26B parameters, mixture-of-experts, 4B active per token) served with vLLM [Kwon et al., 2023], in three passes per image (Fig. 13). The image is presented with a numbered marker at the centre of each object box, so that the model refers to instances rather than to categories. The *semantic pass* proposes actions and interactions with a fully open vocabulary and an explicit prohibition on spatial predicates; its prompt (§ J) is written for coverage rather than precision. The *grow pass* runs one additive round on the same image to fill sparse graphs; it contributes most of the corpus’s density and most of its errors, and the body-part gate exists because of it, its characteristic failure being **man wearing human nose**. The *spatial pass* proposes layout relations under an open prompt that asks for depth, contact and proximity ahead of the left/right relations that dominate purely geometric annotation; every proposal is checked against box geometry, and proposals whose predicate does not reduce to one of 21 layout forms (left/right, above/below, front/behind, inside, near and thirteen contact phrasings) are dropped, since actions belong to the semantic passes. Images that remain below a spatial-density floor receive a salience-ranked geometric top-up, which contributes 18.4% of spatial relations.

Two design comparisons contradicted the expected outcome. A verify-before-assert prompt, which asks the model to justify each relation before emitting it, was evaluated on a fixed 1,000-image calibration sample and rejected: it raised precision by reducing semantic predicate diversity from

404 distinct types to 135, whereas the deterministic gates reach the same precision at no cost in diversity. Conversely, a purely geometric spatial layer was free but monotone (6 surface forms, 58% left/right, almost no depth), whereas propose-then-verify yields 20 surface forms at 32.5% depth, 29% left/right, 14% proximity and 13% contact, with left/right and front/behind balanced to within 0.4% at full scale.

C.2 Verification gate

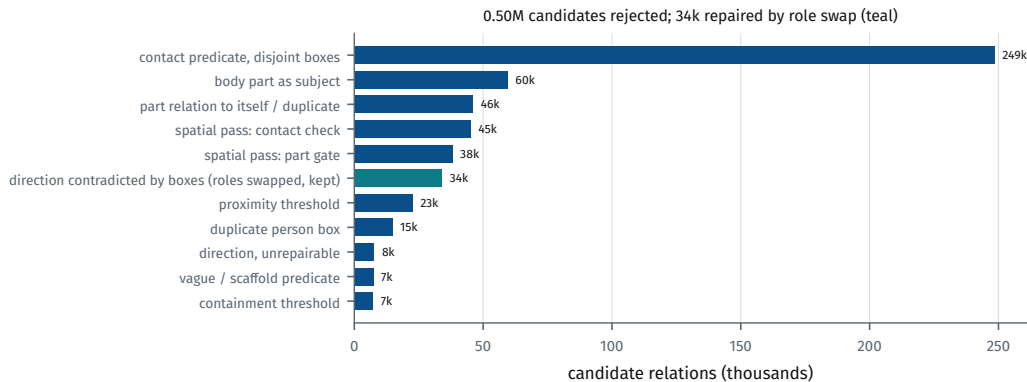


Figure 14 Candidates rejected by each gate over the full generation run. Contact predicates on disjoint boxes dominate; the direction gate is the one that repairs rather than rejects.

C.3 Statistics and comparison

The use of models to write relation annotations is not new (Tab. 19), and existing approaches differ chiefly in how a relation is grounded. The earliest parses image captions with a language scene parser, which yields triplets that are ungrounded, since the parser cannot determine which of four chairs the caption refers to, and that inherit caption bias [Chen et al., 2023]. RLIPv2 [Yuan et al., 2023] scales this to 1.9M images by generating captions with BLIP and assigning the parsed relation texts to region pairs with a learned tagger, so that grounding becomes a prediction of the same kind the downstream model is expected to learn and tagger errors are indistinguishable from supervision. GPT4SGG [Chen et al., 2023] and MegaSG [Chen et al., 2025] instead prompt a multimodal model directly, on 113k COCO images and 1M images respectively, which grounds relations in the image but accepts the model’s output without verification.

In rank–frequency, RA-4M lies between the 94-class closed vocabulary of its source and the 36,549 unconstrained strings of raw Visual Genome: it has a long free-text tail of 4,505 hapax predicates, but an order of magnitude more mass in ranks 10–100, which is where non-generic but learnable predicates are found.

C.4 Design decisions

Surface forms. Synonyms are not canonicalised in the emitted corpus, and canonical groups exist only within the loss (§ B.2.2) and the gates. The reason is that a model whose deployment vocabulary is arbitrary must be trained on targets that populate the text space densely: collapsing **grasping** onto **holding** at annotation time teaches the model that **grasping** does not exist. This is what makes the normalisation of GQA to roughly 300 predicate classes [Hudson and Manning, 2019], and the

Algorithm 1: Geometric verification of one generated relation. The canonical map $c(\cdot)$ groups surface forms for gating only; the emitted string is the annotator’s own. After verification, kept left/right, above/below and front/behind relations are restated from the other endpoint with probability one half (§ 4.1).

Input: relation (s, p, o) with boxes b_s, b_o , from the spatial pass or a semantic pass; canonical map $c(\cdot)$

Output: KEEP, KEEP WITH ROLES SWAPPED, or REJECT

```

1 if  $s$  and  $o$  are both person boxes with  $\text{IoU}(b_s, b_o) \geq 0.7$  then return REJECT
2 if the relation is from the spatial pass then
3   if  $s$  or  $o$  is a body part then return REJECT
4   if  $c(p)$  is not one of the 21 layout forms then return REJECT
5   if  $c(p)$  requires contact (on, sitting on, hanging from, ...) then
6     if  $b_s$  and  $b_o$  do not touch then return REJECT
7   if  $c(p)$  is containment then
8     if  $|b_s \cap b_o| < 0.5 |b_s|$  then return REJECT
9   if  $c(p)$  is proximity and  $b_s, b_o$  do not touch then
10    if  $\text{gap}(b_s, b_o) > 0.5 \max(\text{diag}(b_s), \text{diag}(b_o))$  then return REJECT
11  if  $c(p)$  is left/right or above/below and the box centres contradict  $p$  then
12    if  $p$  has an inverse form then return KEEP WITH ROLES SWAPPED
13    return REJECT (support phrasings such as atop)
14 else
15   if  $c(p) \neq \text{part of}$  then
16     if  $s$  is a body part then return REJECT
17     if  $o$  is a body part with at least 60% of its area inside  $b_s$  then return REJECT
18   if  $c(p)$  requires contact or containment (wearing, holding, sitting on, ...) then
19     if  $b_s \cap b_o = \emptyset$  then return REJECT
20 return KEEP

```

projection of MegaSG onto 94, unusable for our purpose. The cost is the exact-string penalty quantified in § B.2.2.

Leakage control. Every evaluation split used anywhere in this paper is excluded from training by image identity. The same photograph exists under two names, since Visual Genome keys images by VG id and MegaSG by zero-padded COCO id, so a raw filename intersection between an evaluation split and the corpus returns zero and is incorrect; the collisions appear only after mapping through an explicit $\text{VG} \leftrightarrow \text{COCO}$ table of 51,498 entries, in both directions. Removing perceptual near-duplicates of the VG150 and PSG validation and test images removes 8,687 corpus images, and the same procedure applied to the raw Visual Genome source removes 32,456; IndoorVG, which was added later, contributed a further 506 and 12. All exclusions are applied by explicit id list.

Second source. The mixture adds the raw Visual Genome relation annotations, with all of their free-text predicate strings rather than the 50 of VG150, restricted to the images that survive the leakage filter: 40,615 images, 758,547 relations and 17,742 predicate strings, of the 36,549 raw Visual Genome carries before filtering. Human free-text annotation has a different style and a different error profile from machine annotation, and a small admixture measurably improves transfer. Mixture fractions are applied per image whereas the loss is computed per relation, so only the relation share describes the supervision, and the two differ (Tab. 17).

Table 17 Shared triplet mass, and the mixture over which it is computed: the share of a training source’s relation *instances* whose $\langle$ subject category, predicate, object category $\rangle$ triple the benchmark also annotates. The two share columns differ because the sources differ in density, a share by image being multiplied by the source’s relations per image; HICO-DET’s 21.0% of images is 5.0% of relations at 1.94 relations per image against 9.02 for the leakage-filtered RA-4M, and the relation share is what the loss observes. $\#pred$ counts the distinct predicate strings among the relations loaded for training after the leakage filter and is therefore below the corpus totals of Tab. 20. The 19,103 on the mixture rows is the deployed bank, which is not the union of these three sources but the wider answer space described below, and one bank serves the whole tower family. No image overlap is involved; leakage is handled separately (§ C.4). VG150 train, the baseline’s fine-tuning corpus, is the comparison row of Tab. 18.

Training source	#pred	share of mixture		shared triplet mass (%)			
		by image	by relation	VG150	PSG	IndoorVG	Haystack
RA-4M	9,848	72.7	80.6	10.5	9.4	7.0	4.5
raw Visual Genome [Krishna et al., 2017], leakage-filtered	17,352	6.3	14.4	19.0	4.0	3.6	0.3
HICO-DET [Chao et al., 2018] train	116	21.0	5.0	31.9	50.1	8.5	24.3
released tower, the mixture scored everywhere below	19,103	100.0	100.0	12.8	10.7	6.6	4.9
zero-shot tower no HICO-DET; 91.9/84.6, 8.1/15.4 by image/relation	19,103	100.0	100.0	11.8	8.6	6.5	3.9

Table 18 Shared triplet mass split into its two components, over the same cells as Tab. 17. *pred* is the share of a mixture’s relations whose predicate string the benchmark also uses, *obj* the share whose two object-category strings both appear in its taxonomy, and *triple* the share whose whole triple the benchmark annotates. The components are printed because the triple charges for all three strings at once, so a low value can mean either annotation-style or taxonomy mismatch. The baseline row is the clear case of the second: against IndoorVG it keeps 95.7% of its predicate mass and 12.3% of its object-pair mass, because IndoorVG is indoor-only and VG150’s relation mass falls on **sky**, **building** and **street**. Against its own benchmark it loses nothing on either component, which is the asymmetry § 5.2 is about.

Training corpus	VG150			PSG			IndoorVG			Haystack		
	pred	obj	triple	pred	obj	triple	pred	obj	triple	pred	obj	triple
released tower mixture	53.4	44.4	12.8	38.7	36.7	10.7	49.5	26.8	6.6	38.7	30.7	4.9
zero-shot tower mixture	53.0	43.6	11.8	37.2	33.9	8.6	49.4	27.4	6.5	37.2	27.6	3.9
VG150 train the baseline’s	100.0	100.0	90.9	57.3	22.1	8.6	95.7	12.3	10.4	57.3	7.1	0.6

Table 19 Machine-generated relation supervision: how existing corpora are produced, and where RA-4M differs. *Grounding* is how a generated relation is attached to specific object instances; *verification* is whether the annotator’s output is checked. Literature rows are quoted from the cited papers; dashes denote quantities the papers do not state.

Corpus	annotator	images	predicates	grounding	verification
caption parsing	language parser	—	—	none (ungrounded)	none
GQA [Hudson and Manning, 2019]	normalised VG	113k	310	human boxes	none
RLIPv2 [Yuan et al., 2023]	BLIP + R-Tagger	>1.9M	free text	learned tagger	none
GPT4SGG [Chen et al., 2023]	GPT-4	113k	free text	prompted	none
MegaSG [Chen et al., 2025]	Gemini 1.5 Flash	1M	94 (release)	prompted	none
RA-4M (ours)	open 26B VLM, 3 passes	474k	10,102 free text	numbered box markers	deterministic geometry

Rejected sources, retained vocabularies. Two corpora were considered as relation sources and rejected. The scene graphs of GQA are the Visual Genome annotations normalised to roughly 300 predicate classes [Hudson and Manning, 2019], which removes the surface-form diversity discussed above, and raw Visual Genome covers 99.2% of the GQA images in any case. Synthetic Visual Genome [Park et al., 2025], at 49.2 relations per image, was rejected because half of its object names fall outside the union taxonomy, which invalidates the per-category-pair annotation statistics the

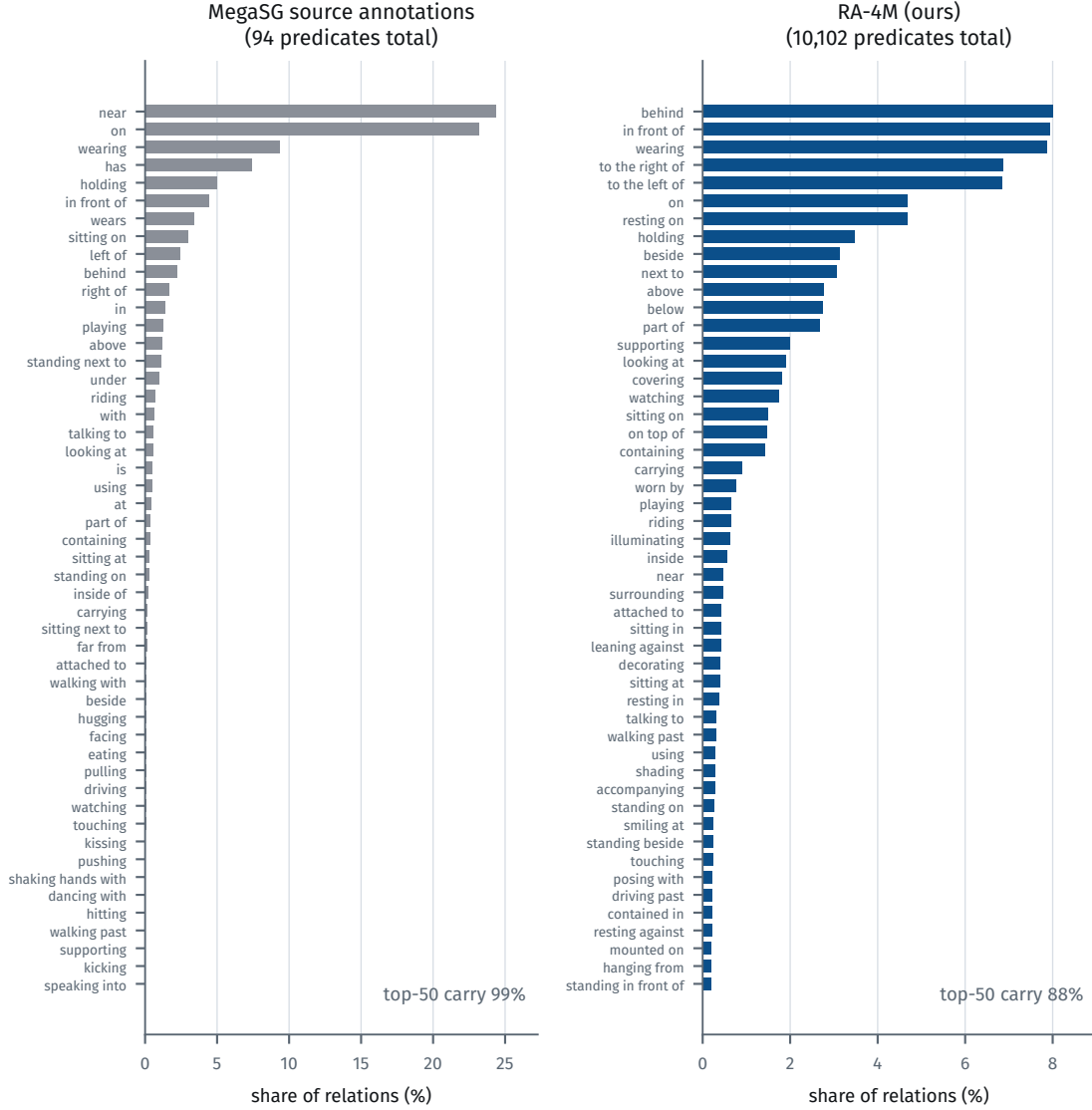


Figure 15 Top-50 predicates by relation share, MegaSG source annotations (left) against RA-4M (right), on the same images and the same boxes. The head of the source distribution is *near/on* at 24.3% and 23.2%; ours is spatial and direction-balanced (*behind* 8.0% versus *in front of* 7.9%, *to the right of* 6.9% versus *to the left of* 6.8%), with no predicate above 8%. The x -scales differ: the difference lies in concentration as well as in vocabulary size.

sampler loss depends on (§ 3.3).

Their predicate *strings* are kept even so, and this is the distinction the design turns on. Supervision and answer space are separate: a string may be a column of the bank without any relation in the corpus asserting it, since nothing on the text side is learned (§ 3.1). The deployed bank is therefore RA-4M’s 10,102 strings followed by the 9,001 further strings that GQA, SpatialSense [Yang et al., 2019] and Synthetic Visual Genome contribute, 19,103 in all, which is wider than the mixture it scores and independent of it.

Table 20 The corpus against its source annotations and raw Visual Genome. Entropy is computed over the predicate distribution, in nats. The MegaSG release’s own annotations are machine-generated (Gemini 1.5 Flash [Chen et al., 2025]) and projected onto 94 predicate classes.

Corpus	images	relations	rel/img	predicates	entropy
RA-4M (this report)	474,413	4,282,531	9.03	10,102	3.99
MegaSG source annotations (same images)	474,420	2,510,905	5.29	94	2.56
raw Visual Genome [Krishna et al., 2017]	108,077	2,315,906	22.10	36,549	3.81
after our leakage filter (secondary source)	40,615	758,547	18.70	17,742	—



Figure 16 The same images and boxes under both annotation layers. Top: the source layer asserts **cabinet on speaker**, whereas ours reads the support relations in the correct direction and adds semantics the closed schema cannot express (**camera capturing monitor**). Bottom: the source layer contains the inverted **carriage pulling horse**, ours **horse pulling carriage**, **bench part of carriage** and **person sitting on bench**. Predicates in blue are semantic-pass outputs, those in grey verified spatial-pass outputs.

D Protocol details

D.1 Relation score

The deployed relation score is

$$s = \sigma(a(\ell_{\text{pred}} + w\ell_{\text{pair}}) + b), \quad (6)$$

in which the pair-relatedness logit ℓ_{pair} is fused additively; one definition is shared by the evaluator and the deployed pipeline. Additive and multiplicative fusion rank differently. The annotation-derived benchmarks favour the multiplicative form, because multiplication up-weights ℓ_{pair} , the annotation-propensity term; on adjudicated negatives the ordering inverts (additive 0.911 AUC, multiplicative 0.904, relatedness alone 0.748). We report the additive form throughout.

D.2 Sensitivity of the composite

Table 21 Leave-one-axis-out analysis of the composite. OVS is recomputed with each axis withheld in turn, from the same chance-corrected per-axis scores that [Tab. 3](#) aggregates.

axis withheld	OvSGTR	RelateAnything	ratio
none the composite as reported	11.8	40.1	3.39×
A1 transfer	11.6	43.3	3.74×
A2 precision	10.0	36.2	3.63×
A4 detector	22.9	45.3	1.98×
A5 graph quality	10.0	36.4	3.66×
A6 spatial	10.9	40.6	3.73×

[Tab. 21](#) recomputes the composite with each axis withheld. The ordering of the two systems does not depend on any single axis, including A5, whose scoring [§ 6.5](#) discusses at greatest length. Withholding A4 narrows the gap most, because A4 is the axis on which the baseline is furthest behind.

D.3 Annotation propensity in the score

The pair-relatedness score of [§ 3.2](#) is the annotation-propensity term of [§ 5.2](#). It is present by construction, so its contribution can be measured on both benchmark families, and its sign differs between them. On the annotation-derived benchmarks, retaining it improves recall. On adjudicated negatives, removing it at evaluation time improves the result: every projective predicate gains between +0.03 and +0.05 AUC and only the two contact predicates lose, which is the expected behaviour of a contact prior ([Fig. 17](#)). Recall against sparse annotation and truth judgement against adjudicated negatives are therefore different quantities, and one term can improve the first while degrading the second.

D.4 Calibration

The model ranks well but on an uncalibrated scale ([Fig. 18](#)). Its logit scale and bias are trained against a mass-balanced objective, that is, an implicit 50/50 prior, whereas a real frame is 0.2–4% positive, so scores accumulate near 1 and a threshold has no fixed meaning; the ranking behind those scores nonetheless separates the top decile from the bottom by three orders of magnitude in

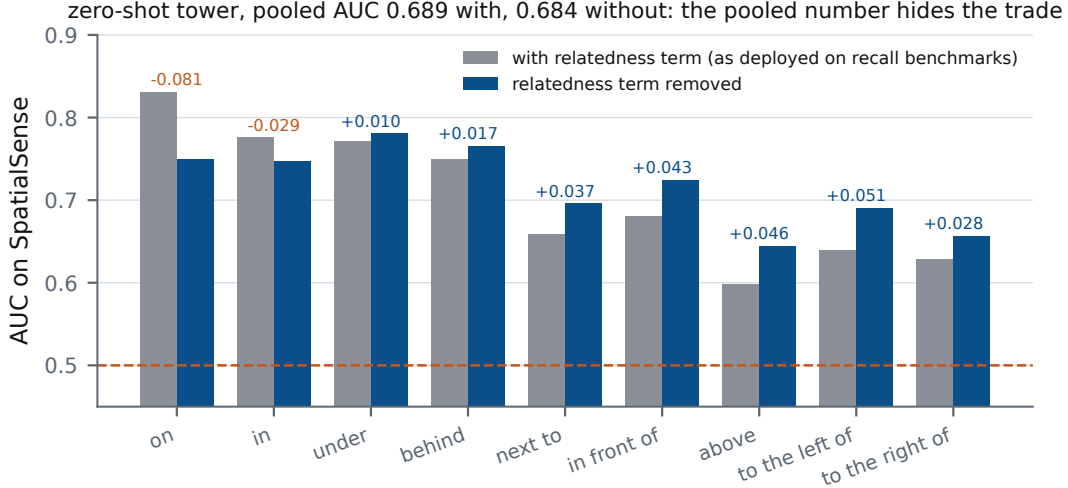


Figure 17 Effect of the relatedness term on adjudicated spatial negatives. Per-predicate AUC on the adjudicated true/false pairs of SpatialSense with the learned pair-relatedness logit fused into the score, as on every recall benchmark, and with it removed. The pooled AUC moves by 0.005 and reflects none of this. The probe uses the zero-shot tower, so its pooled AUC differs from the 0.675 that § 6.6 reports for the released tower.

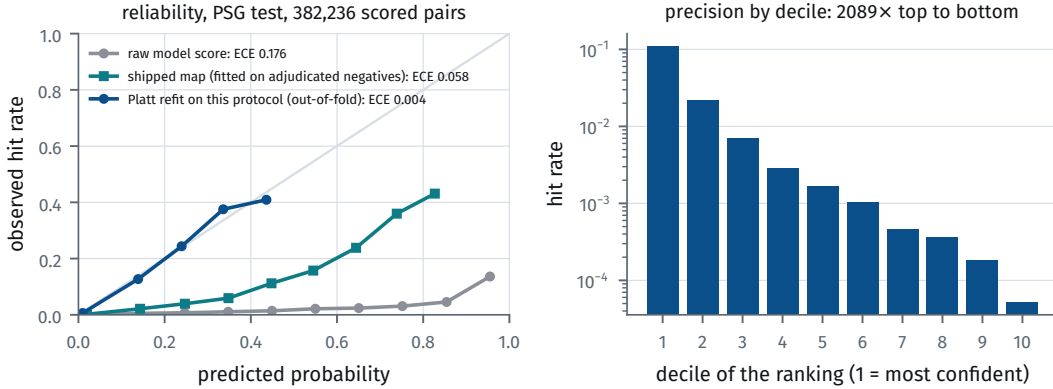


Figure 18 Calibration of the relation score. Left: reliability diagram of the relation score on all 382k scored PSG-test pairs, where a hit is an annotated triple. The raw score is over-confident (ECE 0.176); two Platt [Platt, 1999] parameters fitted out-of-fold under this protocol reduce this to 0.004; the released map, which is fitted on adjudicated negatives, lies above the diagonal here because an unannotated true relation counts as a miss (ECE 0.058). Right: hit rate by decile of the ranking, which spans three orders of magnitude.

precision. The two parameters (a, b) of Eq. (6) are obtained by a Platt fit [Platt, 1999], which is monotone, so every ranking metric is unchanged and only the meaning of a threshold changes.

What the fit is made against determines what the number means. Fitted on annotation hits, the score estimates the probability that a relation appears in an annotation and reaches an ECE of 0.004 under the same protocol. The released map is instead fitted on adjudicated negatives and estimates the probability that a person judges the relation true, which is the relevant question at deployment; it is conservative in the useful direction, reporting a precision of 0.77 at the released operating point where the adjudicated precision is 0.90. Per-predicate calibration was evaluated and rejected, as it fits in-domain and does not transfer.

D.5 Output at the operating threshold

Table 22 Output at three thresholds of the calibrated score. PSG test: 2,173 images, 13,077 annotated relations, 6.02 per image, every candidate pair scored in the deployment configuration of § I. Recall, P_{ann} and P_{cal} are percentages. P_{ann} counts an unannotated prediction as wrong and is therefore a lower bound; P_{cal} is the mean calibrated score over the retained set, which estimates precision without reference to the annotation. The head is the mask-gated variant rather than the released one, so the absolute values are not release figures; what § 5.2 uses is the decomposition below.

Criterion	τ	edges/img	recall	P_{ann}	P_{cal}
max F1 (calibrated P)	0.24	18.2	33.5	11.1	41.2
max F1 (annotated P)	0.41	7.9	23.4	17.9	53.2
count-matched (6.0 /img)	0.45	6.1	20.1	19.8	56.1
AUC 0.929 · AP 0.090 · ECE 0.061 over all 377,986 scored pairs					

Because (a, b) are fitted against adjudicated cells rather than against annotation, a threshold on the resulting score is a probability rather than a rank cut, and the qualitative figures in this paper specify a threshold rather than a top- k . Tab. 22 reports what such a threshold costs and returns. Three defensible criteria disagree by a factor of three in output volume: maximising F1 against the calibrated precision emits 18.2 edges per image, maximising it against the annotated precision emits 7.9, and matching PSG’s own annotation density emits 6.1 at $\tau = 0.45$. The last is the operating point used for the qualitative panels of § 7.2, so those are produced at the corpus’s density rather than at a budget chosen to suit them.

The two precision columns differ by a factor of 2.8 at that threshold, and the gap can be counted. Of the 39,516 edges emitted at the calibrated-F1 threshold, 4,376 (11.1%) match an annotated triple, 4,201 (10.6%) fall on a pair that PSG annotated and assign a different predicate to it, and 30,939 (78.3%) fall on a pair that PSG never annotated in either direction. Only the middle group is a predicate error in the ordinary sense; the largest group is unscored rather than wrong. This is the propensity prior of § 5.2 in counted form, and the reason § 5.4 scores one axis against adjudicated negatives. The predicate errors are more often a vocabulary difference than a mistake: among the 8,577 edges on an annotated pair, 51.0% carry PSG’s own predicate, and the remainder concentrate on near-synonyms and hypernyms, namely in for an annotated on (231 cases), on (148) and beside (137) for attached to, and beside for in front of (132). A recall metric counts every one of these as a miss.

D.6 Detector operating point

Tab. 23 is the table underlying Fig. 4: identical detector weights per row, two operating points, and the pair-recall ceiling each implies.

D.7 OvR-SGG reproduction

Tab. 24 is the reproduction underlying § 5.3: the baseline’s released checkpoints, run through our inference code and scored with a transcription of its metric, recover its twelve published numbers to within 0.35 points, and the third row of each block changes only the matcher.

Scope of the leaderboard rows. Our model has never seen a VG150 relation annotation, so on Base+Novel it performs a strictly harder task than the published rows. The premise of the split,

Table 23 Pair-recall ceiling at two detector operating points. Each row uses identical detector weights; only the confidence threshold and box cap differ. *ceiling* is pair recall, the fraction of ground-truth relations for which both endpoints are detected, which is the upper bound on relation recall for any model given those boxes. The baseline’s GroundingDINO applies no threshold and retains every query whose label is not background, which is why it appeared to be the stronger detector.

Detector	Bench	conf 0.05, ≤ 60 boxes			conf 0.001, ≤ 100 boxes		
		box/img	obj rec	ceiling	box/img	obj rec	ceiling
YOLOv8-m	VG150	27.0	69.5	63.8	99.4	85.2	82.9
YOLO-World	VG150	26.2	58.6	50.1	99.3	78.7	74.8
YOLOv8-m	PSG	23.1	77.8	82.8	84.7	86.3	92.7
YOLO-World	PSG	22.7	66.4	69.6	95.8	80.3	87.6
GroundingDINO (baseline’s own)	VG150	no threshold applied			97.1	83.9	81.6

Table 24 Reproduction of the OvR-SGG leaderboard, and the effect of the matcher. VG150 test, SgDet, graph-constrained. Published rows are the baseline’s own; reproduced rows re-run its released checkpoints through our inference code and score them with a transcription of its metric. The third row of each block changes only the matcher, by adding the one-to-one ground-truth-to-detection assignment that object detection has required since COCO.

Model / matcher	Base+Novel			Novel		
	R@20	R@50	R@100	R@20	R@50	R@100
OvSGTR Swin-T [Chen et al., 2025], as published	15.8	20.5	23.9	10.2	13.5	16.2
reproduced, their matcher	16.0	20.4	23.8	10.1	13.2	15.9
+ one-to-one assignment	13.9	18.0	21.2	8.6	11.4	13.8
OvSGTR Swin-T + MegaSG [Chen et al., 2025], as published	19.4	25.4	29.7	12.2	17.0	21.1
reproduced, their matcher	19.5	25.5	29.9	12.4	17.2	21.3
+ one-to-one assignment	17.2	22.6	26.7	10.8	14.9	18.6

however, is that the 15 novel predicates were unseen in any annotation, and RA-4M contains those strings. Open-vocabulary detection establishes where the line is drawn. OV-LVIS [Gu et al., 2022] deletes the annotations of the 337 rare LVIS categories from the training set and retrain on the remaining 866, so the split is a property of the training set rather than a column of the results table. What the protocol does not require is that the novel concepts be unseen everywhere, since the classifier for the held-out categories is built from an image–text model whose web corpus contains their names, and Detic [Zhou et al., 2022] trains on ImageNet-21k image-level labels for an overlapping vocabulary while removing the rare-class boxes. Our supervision for the novel predicates is not of the permitted kind, because it is grounded and instance-level, which corresponds to retaining those boxes. The compliant version removes those predicates and their synonym groups from the training mixture and retrain; it is the novel 15 held out row of Tab. 8, and the full-vocabulary rows below it are marked as non-compliant because the difference between them measures how much of novel performance under this protocol is supervision on three frequent strings. Neither depends on a single box set: the two full-vocabulary variants differ by less than the gap to the next entry, and the numbers move by less than 0.1 when the boxes of the baseline’s stronger checkpoint are substituted. Scoring every candidate pair rather than our deployed pair budget, which covers 43% of them, adds 2.1 Base+Novel and 3.7 Novel R@50.

Predicate split and matcher. Two incompatible predicate splits are in circulation. We use that of OvSGTR [Chen et al., 2024b] throughout, with 35 base and 15 novel predicates, because every row of Tab. 8 uses it and because it is the only one published as an enumerated list and can therefore be reproduced exactly. PGSG [Li et al., 2024b] instead “randomly select[s] 50% predicate categories as novel” (30% in its own supplement), names none of them, and reports novel-class mR@K where the leaderboard reports R@K. Recent work tabulates both under one ranking with a footnote [Min et al., 2025]; we do not, because a 25/25 partition changes the training supervision, the test denominator and the metric simultaneously, and the resulting columns are not commensurable. The whole table is scored with the uncorrected matcher, the duplicate-crediting one that § 5.3 argues against, since this is the only setting in which our rows and the published rows have the same meaning. Applying the correction to the baseline’s rows (Tab. 24) reduces Base+Novel by 11–13% and Novel by 13–15% at every K , and applying it to ours as well leaves the ordering unchanged.

D.8 Graph-quality oracle (A5)

A1–A4 score predictions against annotations and inherit their blind spots: a relation that is true but unlabelled counts as a false positive, and a graph can be no better than the corpus it is compared against. A5 removes the ground truth from the question. A vision–language judge is shown the photograph with its numbered boxes and asked about the relations a system emits. The judge is Qwen3-VL-8B [Bai et al., 2025a]. It is not a Gemma model, because RA-4M was written by one and a Gemma judge would favour our own output distribution, and it is not a reasoning variant, because the verdict is parsed from the last JSON object within a 300-token budget and a long chain of thought would exhaust it. The rubric, the prompt and the rendering are released with the benchmark.

Table 25 The whole-graph oracle, PSG test, the shared YOLO-World boxes of A4, judge Qwen3-VL-8B. Top: graph shape, which requires neither ground truth nor a judge. Bottom: the oracle verdicts as our win rate among decisive comparisons, stratified by which graph was longer. *pack* denotes our head reparameterised to the benchmark’s 56 predicates and *train* the full vocabulary. The controls certify the judge rather than any single comparison, and are pooled across runs of one judge configuration.

<i>graph shape, no ground truth and no judge</i>					
model	rel./graph	distinct pred.	modal share	vocab used	at cap
RelateAnything, vocab = pack	18.3	4.5	0.52	56	81%
OvSGTR [Chen et al., 2025]	10.3	1.7	0.85	26	24%
RelateAnything, vocab = train	18.9	8.4	0.33	939	88%
<i>oracle verdicts, ours against the baseline</i>					
arm	compared	decisive	win rate	longer: win rate (n)	equal: win rate (n)
vocab = pack	140	58	0.16	0.10 (101)	0.40 (39)
vocab = train	141	46	0.24	0.16 (106)	0.62 (35)
vocab = pack, top 10 each	142	64	0.62	—	0.62 (142)
controls: scramble accuracy 1.00 (56 decisive of 103), padding resistance 1.00 (5 of 30), primacy 0.19					
judge Qwen/Qwen3-VL-8B-Instruct, 6 votes per comparison, majority 0.67, both presentation orders					
matched-depth run, same judge: scramble accuracy 1.00 (24 decisive of 30), padding resistance 1.00 (15 of 28), primacy 0.23					

Whole-graph comparison and its controls. In the whole-graph arm reported in Tab. 25, two graphs are judged in both presentation orders, with several sampled votes per order, all votes pooled, and a supermajority required for a winner. Six controls are enforced. The share of votes cast for whichever graph was shown first is reported as the primacy rate, where 0.5 is unbiased; a single

greedy decode measures 0.70 on this task. Graph length is measured rather than suppressed: each model emits every pair scoring within 70% of its best pair on that image, up to 20, so that a model describing more of the scene is permitted to do so, and the judge’s own preference for longer text is measured by a *padding* control that compares a graph against itself with fabricated relations added. A *scramble* control compares a graph against a copy with its predicates permuted across pairs; a judge that cannot prefer the intact graph is not measuring anything, and its verdicts are discarded. Vocabulary breadth is a confound, so two arms are run, one with our head reparameterised to the benchmark’s predicates, the same range as the baseline, and one with the full vocabulary.

Certification. A run is admitted only if the pooled controls satisfy three conditions at once: scramble accuracy of at least 0.75, primacy of at most 0.60, and at least ten decisive control comparisons. The last matters because a judge that answers “tie” to a scrambled graph has demonstrated nothing, and a high accuracy over three decisions is not evidence. The reported judge satisfies all three (1.00, 0.19, 56 of 103). A smaller judge of the same family, Qwen2.5-VL-7B [Bai et al., 2025b], was also evaluated and reaches the same verdict at a lower scramble accuracy (0.95) and a lower decisive fraction, which is why the larger one is reported.

D.8.1 Per-relation oracle (A5b)

The pairwise oracle returns one verdict per pair of graphs and so cannot indicate whether a losing graph lost because its additional relations were false or because the judge prefers shorter text. A5b scores each asserted relation separately: the judge sees one relation at a time and answers whether the claim is true of the photograph.

Rendering. Two renderings were evaluated. The plain rendering presents the judge with the same numbered image used by the pairwise oracle, carrying up to a hundred identically coloured boxes; the highlighted rendering draws a separate image per relation with the subject and object boxes in different colours. Which is better is decided by the control: a fixed share of judged relations have their predicate replaced by another drawn from the same image, and the run reports how often the judge accepts the corruption. That rate is 0.201 under the plain rendering and 0.135 under the highlighted one, which indicates that the plain rendering suffered from a grounding problem. The rate is an upper bound rather than an exact false-positive rate, since a swapped predicate is sometimes true by coincidence.

Depth. Two depths were evaluated and neither supersedes the other. The deployed graph is what each system emits when it asserts as much as its own confidence spread justifies. The matched graph is the top ten pairs of each system with no threshold, which removes length as a variable: at equal depth, the number of accepted relations equals precision multiplied by depth.

Why informativeness is measured rather than elicited. The judge was initially asked a second question, how much a relation conveys beyond what the two object names already imply, because truth alone rewards a degenerate system: a model answering *on* for most pairs is usually correct and conveys almost nothing. That question did not survive its own audit. Over the 200-image run the two questions were nearly the same axis, with $P(\text{info} = 0 \mid \text{false}) = 0.998$ under both renderings and $P(\text{info} \geq 2 \mid \text{true})$ at 0.92 and 0.96, so a rate combining them was very nearly precision multiplied by a constant; and repetition went unpriced, the judge rating the baseline’s *on* relations at 1.05 on the 0–3 scale against 0.72 for its other predicates. The defect is not in the wording of the rubric. The redundancy of a relation is a property of the distribution in which it occurs, and a judge shown one relation at a time cannot observe that distribution. Informativeness is therefore measured as

surprisal, $-\log_2 p(\text{pred})$, summed over the relations the judge accepted, as in Eq. (4).

Estimators. The referenced estimator uses the PSG training marginal, which is shared by both systems, so neither is scored against its own habits, and restricted to the 56 predicates both emit, so no smoothing constant affects the comparison; the out-of-vocabulary mass is reported alongside and never folded in. Tab. 7 reports the restricted-vocabulary arm, the like-for-like comparison. An open-vocabulary arm was run as a control for vocabulary breadth: at matched depth the judge accepts its relations at the same rate, and it reaches 22.6 reference-free bits per image against 16.0. Since 73% of its predicates fall outside the reference, Eq. (4) cannot be evaluated for it without a smoothing constant determining the comparison, and it is excluded rather than scored on a biased subsample.

The reference-free estimator is a Krichevsky–Trofimov [Krichevsky and Trofimov, 1981] description length over each system’s own stream of accepted predicates: the first on is expensive and the five-hundredth nearly free, and the code charges for vocabulary size, which answers the objection that a larger predicate list creates headroom by itself. It is stable in the threshold at which rare predicates are pooled, yielding 25.0 / 24.8 / 23.7 bits per image for the restricted arm over merge thresholds of 1, 3 and 10 occurrences against 7.2 for the baseline at all three, and at matched depth 16.0 against 8.0. It therefore agrees with Eq. (4) on the ordering, at a slightly wider ratio. The same asymmetry appears without any judge or reference, in the predicate distributions themselves: entropy 4.00 against 2.14, and 1.9 distinct accepted predicates per graph against 1.5. Tab. 7 reports none of these, because the axis is defined as Eq. (4) and a table of competing estimators would invite the reader to choose one.

Normalisation. Eq. (4) is unbounded, so Tab. 3 and Fig. 1 divide it by the same quantity evaluated on the ground-truth relations of the same images, which yields the share of the information a human annotator recorded that a system delivers. Two properties bound this reading. The ground truth counts every annotated relation, including those whose endpoints the shared detector never found, so both systems carry the detector’s recall identically and the denominator does not depend on its operating point. And the ratio can exceed one because the annotation is sparse: at deployed depth our model reaches 1.03, asserting 18.3 relations per image against 6.33 in the annotation and being credited for true relations that were never annotated. The matched-depth figure, at which both systems are below one, is the one reported.

E Additional results

E.1 Per-predicate results

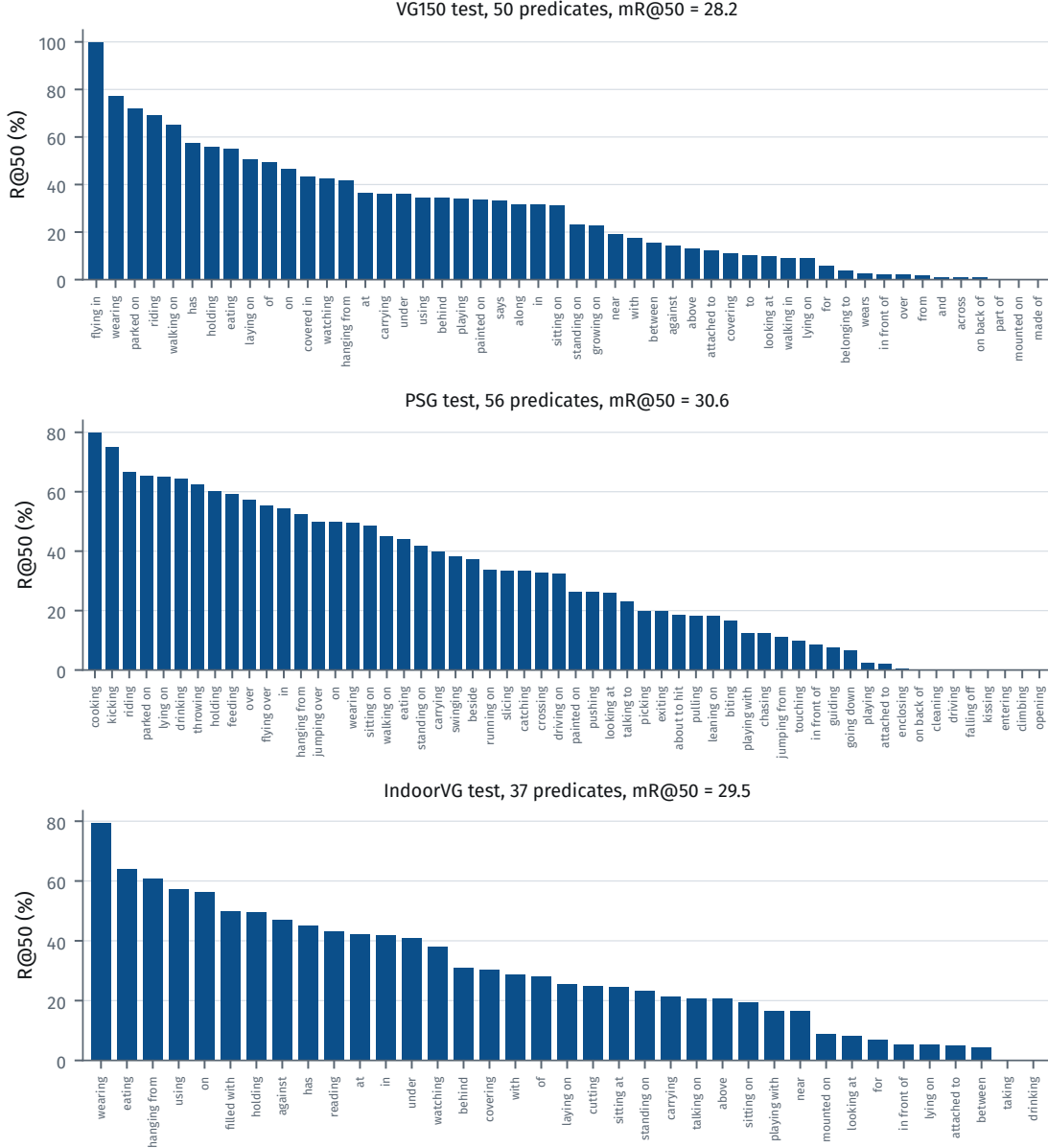


Figure 19 Per-predicate R@50 of the released model on the three closed-vocabulary transfer benchmarks (test splits, ground-truth boxes, graph-constrained, exact-string matching). Predicates at zero are, almost without exception, members of a synonym group whose other surface form the model predicts instead (§ B.2.2).

Fig. 19 gives every per-predicate recall underlying the mean-recall columns of Tab. 4. Two of the patterns visible there are emission rules rather than recognition failures. HICO-DET annotates several verbs per pair and scores each of them while the model emits only one, so the general verb occupies the single slot: **holding** absorbs **wielding**, **carrying** and **reading**, and **riding** absorbs **straddling** and **sitting on**, so that a specific verb the model ranks second counts as a miss. The remedy is a

multi-label emission rule for multi-label sources rather than additional capacity. On PSG, where the model emits exactly as many edges as the image has annotations, the misses are the informative part: the annotation records **attached to** between umbrellas where the model predicts **hanging from** and **in front of**. Unannotated true relations receive no credit, and near-synonyms count as wrong.

E.2 Priors join and detector retention

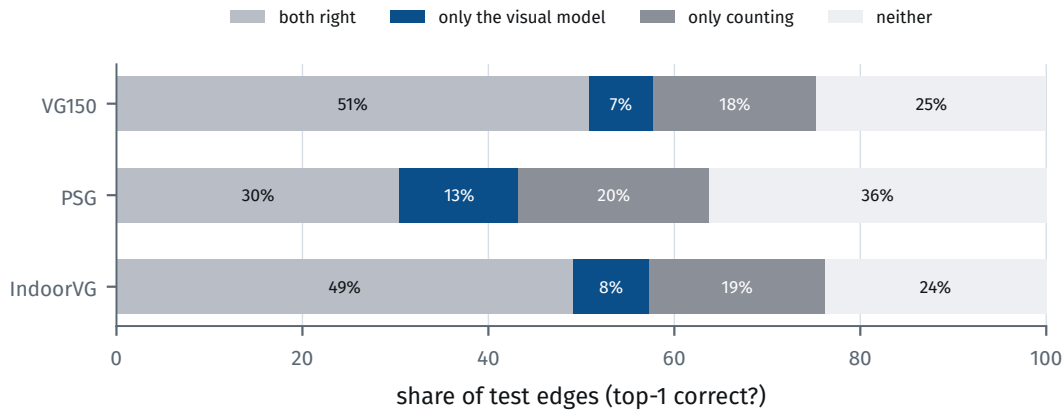


Figure 20 Edge-by-edge join of the visual model against the frequency baseline, top-1 correct or not, on the same edges as Tab. 1. The frequency baseline is correct where the model is wrong about $2.4\times$ as often as the reverse, but the model-only share is 7–13% of all edges, so the two are correct on partly different edges rather than the model being a weaker version of the prior.

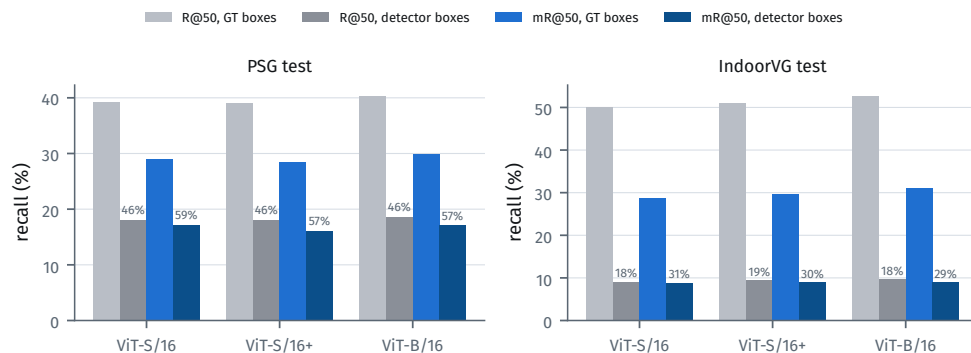


Figure 21 Retention under a real detector for each tower, YOLO-World at a confidence threshold of 0.05, with one-to-one, class-aware (strict) matching as in § 5.3 and a pair budget of 500 on both sides; percentages are retention relative to ground-truth boxes. The micro ladder survives detection, the macro ladder flattens, and IndoorVG, whose categories the detector covers poorly, loses 70–80% of the model’s performance to detection.

Fig. 20 is the edge-by-edge join underlying Tab. 1, Fig. 21 shows what each tower retains under a real detector (§ 6.4), and Fig. 22 shows the released model behind a detector prompted with the LVIS vocabulary (§ 7.2).

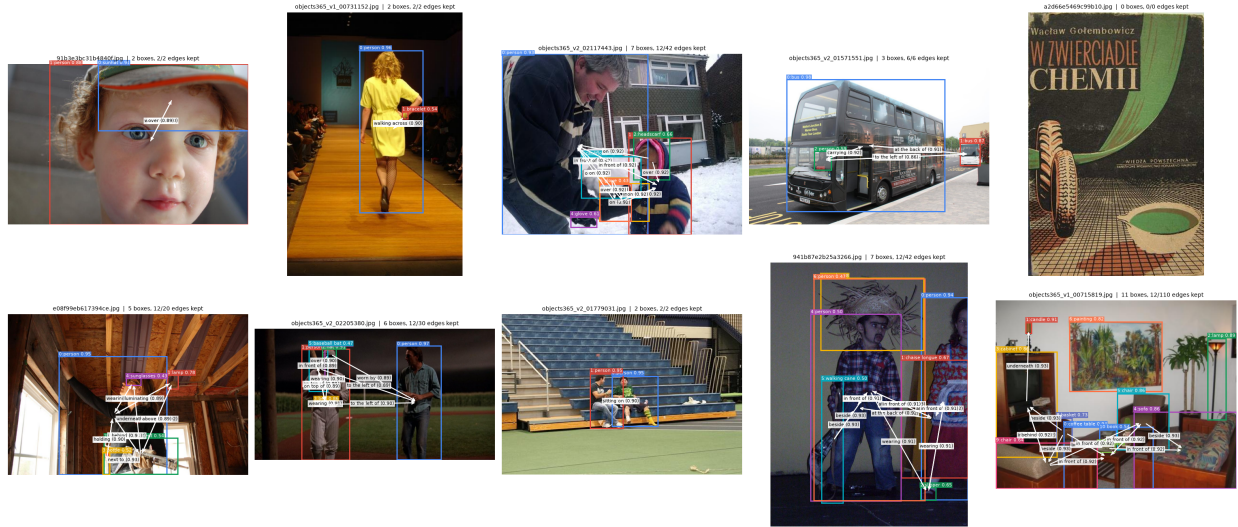


Figure 22 Regions from a detector prompted with the LVIS vocabulary, which the relation model never saw. YOLOE [Wang et al., 2025a] with all 1,198 LVIS classes [Gupta et al., 2019] on MegaSG validation images, with the boxes passed to the released model unchanged and a 35-predicate vocabulary. The names on the boxes are the detector’s and are shown for the reader only; the edges are those retained at the released operating point.

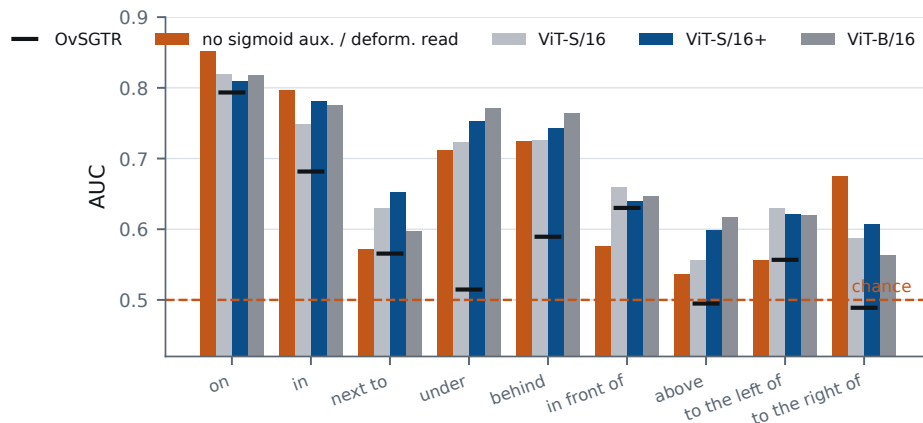


Figure 23 Axis 6: per-predicate AUC on SpatialSense, whose adversarially collected negatives defeat co-occurrence priors. The ablated arm is the recipe without the sigmoid auxiliary and the deformable read; its projective predicates (*above*, *to the left of*, *in front of*) lie at 0.54–0.58, and *to the right of* is the one projective predicate it already handles. The released family raises that set. Black dashes mark the A6 baseline, OvSGTR on the same cells under the same metric implementation, which is at or below chance on *above* and *to the right of*.

E.3 Bounds on the spatial-axis claim

Fig. 23 gives the per-predicate view of A6 and Fig. 24 the grounding probes. Three facts bound the claim made in § 6.6. The boxes-only baseline of the benchmark’s authors scores 68.8 without seeing the image, so a pooled AUC near 0.69 is not by itself evidence of visual spatial reasoning. A counterfactual spatial-grounding probe varies between 0.01 and 0.81 as a function of the backbone learning rate alone, which invalidates any grounding claim resting on a single arm, including ours. And an appearance-versus-geometry probe refutes the spatial-shortcut hypothesis in the opposite

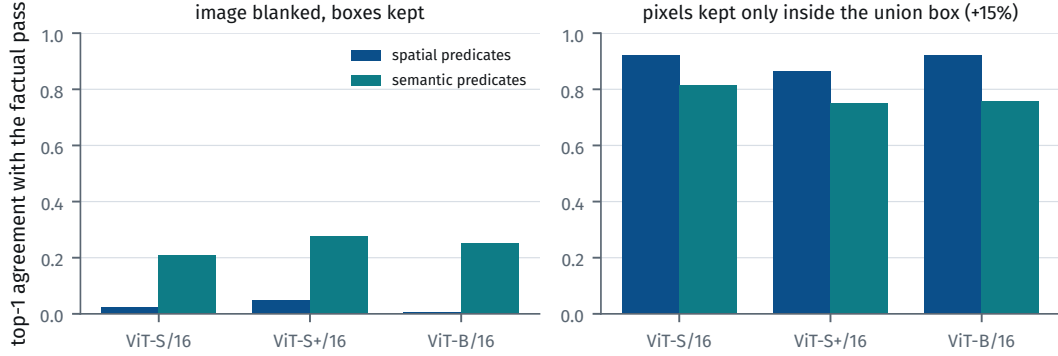


Figure 24 Grounding probes across the ladder, PSG val, with the sampler frozen so that the factual and counterfactual passes score the same pairs. Left: with the image blanked, spatial predicates retain 1–5% of their top-1 decisions and semantic ones 21–28%, which indicates that the model reads geometry from the boxes and appearance from the pixels. Right: with pixels retained only inside the union box, 75–92% of top-1 decisions survive; the scene context added by the deformable read rarely changes the answer on PSG.

direction, since object appearance moves the predicate by +34 points relative to the control. The supervision-ceiling reading of § 6.6 is tested against capacity in § G.1.

E.4 Closed-set and localisation legs of the VLM comparison

Table 26 Free-text systems across three benchmarks, each under its own protocol: ours answers in the benchmark’s own words, ROBIN-3B in its own under its published SBERT rule. Ground-truth regions, graph-constrained, on the image subsets ROBIN was run on (VG150 5k, IndoorVG in full, PSG test). *Pair cov.* is the share of annotated pairs a system reaches at all.

Benchmark	Model	Setting	R@20	R@50	mR@20	mR@50	pair cov.	rel/img
VG150	RelateAnything	closed set	43.6	51.2	23.1	27.1	—	top- <i>K</i>
	ROBIN-3B	SBERT argmax	23.3	27.3	12.4	15.2	61.4	25.5
PSG	RelateAnything	closed set	34.7	40.1	28.4	30.6	—	top- <i>K</i>
	ROBIN-3B	SBERT argmax	30.4	35.7	21.1	22.9	77.4	34.0
IndoorVG	RelateAnything	closed set	44.7	52.9	24.0	29.6	—	top- <i>K</i>
	ROBIN-3B	SBERT argmax	19.4	24.2	17.5	21.5	64.3	33.1

Tab. 26 is the cross-dataset comparison summarised in § 6.3. The margin is smallest on PSG, 1.1× on micro recall, and widens to 1.9× on VG150 and 2.2× on IndoorVG; PSG is also the only one of the three on which free-text systems are routinely reported.

Tab. 27 gives the two legs omitted from Tab. 5. Of three ways of indicating the regions to a prompted model, namely pixel boxes listed in the prompt, numbered overlays, and boxes drawn on the image, every reported run uses the last. Prompting these models with the benchmark’s own 56 predicates rather than letting them answer freely changes their R@20 by between −2.3 and +1.8 points and raises the mean recall of every one of them, by +0.5 to +3.1, leaving their order on R@20 unchanged; on mean recall InternVL3.5-8B and Qwen3-VL-8B exchange places. This is the setting in which A1 places a model, but it is not the setting of a deployed open-vocabulary model, which is why the main text reports the open-vocabulary leg.

The lower block withholds the ground-truth regions. The general models fall to 0.6–1.5 R@20,

Table 27 Closed-set and localisation legs of the VLM comparison. PSG test, 2,179 images, graph-constrained, the same scorer as Tab. 5. Top: the model is given the 56 PSG predicates instead of answering in its own words, with ground-truth regions throughout. Bottom: the regions are withheld and the model must localise as well; *strict* requires the predicted object class to match, *lenient* matches on IoU alone, and mAP@50 is COCO-style detection AP over the 133 PSG object categories, defined only in this block.

Model	Setting	R@20	R@50	mR@20	mR@50	mAP@50	rel/img
<i>Closed set — the model is prompted with PSG’s 56 predicates</i>							
RelateAnything ViT-S/16 ⁺ (53M)	closed-set	34.7	40.1	28.4	30.6	—	top- <i>K</i>
Qwen3-VL-32B	closed-set	14.2	15.2	8.2	8.6	—	19.9
Qwen3-VL-8B	closed-set	12.0	12.3	5.6	5.7	—	14.8
InternVL3.5-8B	closed-set	9.1	9.1	5.0	5.0	—	10.8
GLM-4.6V-Flash	closed-set	6.6	6.8	3.3	3.4	—	13.7
<i>SGDet — the model predicts its own boxes</i>							
Qwen3-VL-8B	strict	0.6	0.6	0.3	0.3	23.5	1.1
Qwen3-VL-8B	lenient	1.0	1.0	0.5	0.5	23.5	1.7
Qwen3-VL-32B	strict	0.8	0.8	0.3	0.3	25.6	1.3
Qwen3-VL-32B	lenient	1.5	1.5	0.8	0.8	25.6	2.3

and their localisation is not the reason: 23.5 and 25.6 mAP@50 over 133 categories is genuine detection. The two facts are consistent because the pairs a model can reach scale with the square of its box recall (at IoU box recall of 0.32 and 0.38, only 10% and 14% of annotated pairs have both endpoints present), and because these models emit only 1.1 to 2.3 relations per image. This is the ceiling Fig. 4 measures for detectors, at a harsher operating point, and the reason A4 evaluates every system on a single shared detector.

F Representation probes

Table 28 Effect of relation fine-tuning on the dense backbone features, for the pre-trained backbone and for the two towers of § 4.2. *Patch similarity* (PSG val, last state, per-image mean-token subtraction) queries from subject-box patches and measures the mean cosine similarity to the relation partner, to another object of the same class, and to unrelated regions; *class* and *rel. sel.* are the corresponding selectivity margins, and $P(\text{partner} > \text{same class})$ is the per-query win rate against a chance level of 0.500. *Interaction region* (HICO-DET test, 736 hand-verb relations) queries the subject patch nearest the object within the intersection: *contact adv.* is its similarity advantage over a distant subject patch, and the 1-NN columns give cross-image retrieval accuracy for the *verb* and for the *object class* from that same feature.

backbone	patch similarity (PSG, $n = 300$)			interaction region (HICO-DET, $n = 736$)		
	class sel.	rel. sel.	partner > same-cls	contact adv.	verb 1-NN	object 1-NN
DINOv3, pre-trained	0.242	0.133	0.469	0.356	0.251	0.713
+ relation fine-tuning	0.276	0.134	0.387	0.377	0.270	0.713
+ relation fine-tuning (shipped)	0.281	0.138	0.385	0.438	0.269	0.761
chance	—	—	0.500	—	0.424	0.068

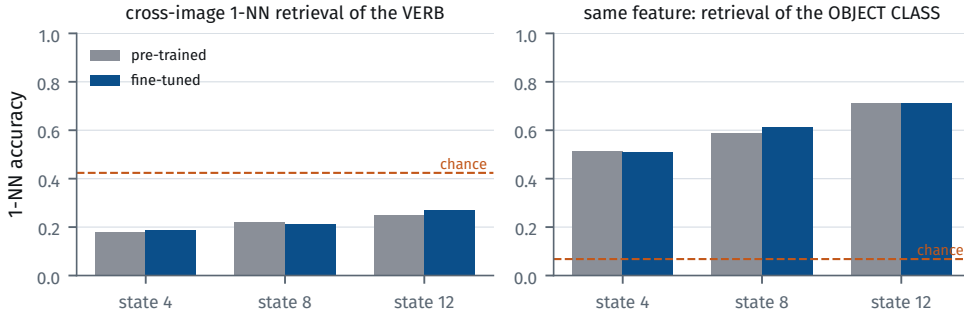


Figure 25 Verb and object retrieval from the contact-region feature. From the contact-region feature of 736 HICO-DET hand-verb relations, cross-image nearest-neighbour retrieval of the verb remains below the majority-class chance level at every depth, before and after fine-tuning, whereas retrieval of the handled object’s class from the same feature is $10\times$ chance.

Tab. 28, Fig. 25 and Fig. 26 are the probes underlying § 7.3. Beyond the findings reported there, what fine-tuning adds is small and lies elsewhere. A linear probe on frozen box-pooled tokens shows that relatedness beyond object identity is already present in the pre-trained backbone, with a within-class-pair AUC of 0.73 on PSG against 0.70 for a geometry-only baseline, and fine-tuning does not increase it (0.73 $\rightarrow$ 0.72). What increases is the linear decodability of the predicate, with macro-F1 rising from 0.445 to 0.475 on PSG and from 0.264 to 0.305 on VG150. This is the gradient of the pair head reaching the features it reads, rather than a relational representation forming within them.

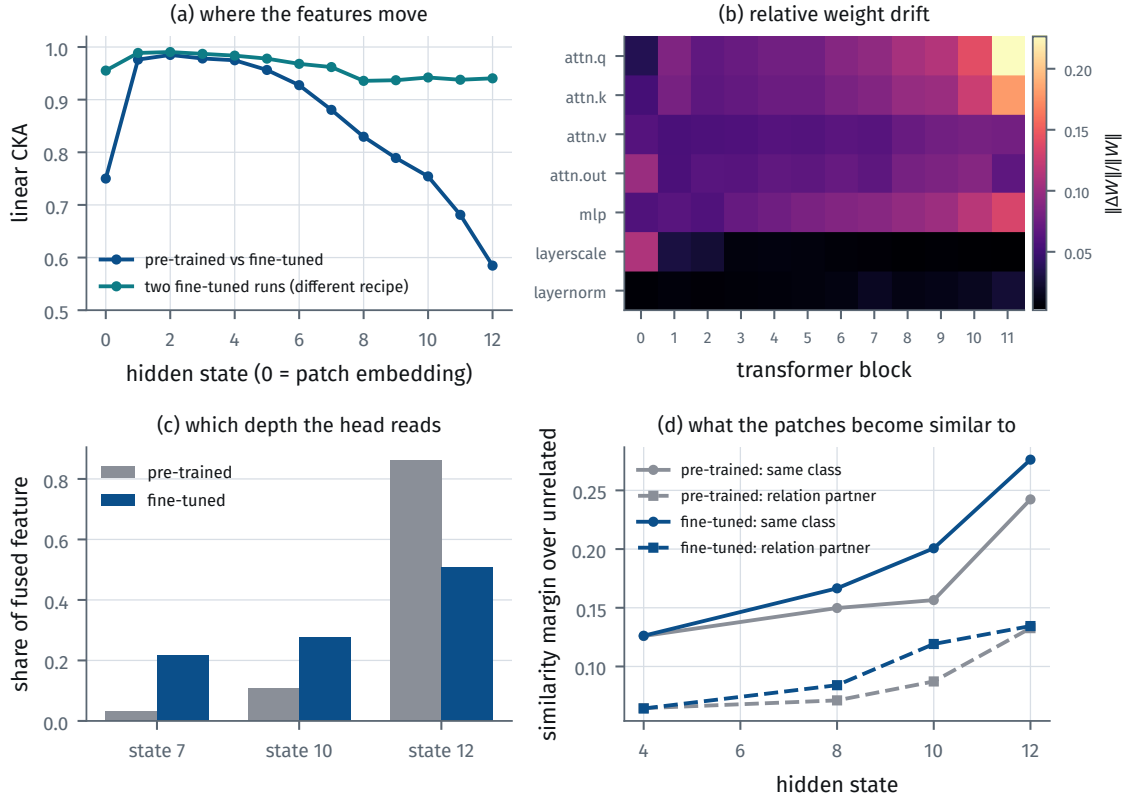


Figure 26 Effect of relation fine-tuning on the DINOv3 backbone (ViT-S/16+, 200 PSG-val images). (a) Linear CKA between the pre-trained and fine-tuned features decreases monotonically with depth, to 0.58 at the last state, whereas two fine-tuned runs under *different* recipes agree at 0.94, which indicates that the task rather than the recipe determines where the features go. (b) Relative weight drift is small and uniform (6–9% per block for attention and MLP, 1% for the normalisation parameters) except in the last block, where the query and key projections move by 23% and 18%, so the change is concentrated in last-block routing. (c) The learned fusion over the three taps shifts from 86% on the last state to 51%, recruiting the mid-depth taps. (d) Patch similarity margins over unrelated regions: the margin to *same-class* objects grows at every depth, whereas the margin to the *relation partner* is unchanged.

G Backbone scaling

Table 29 The backbone ladder, matched on corpus, schedule, batch size and every other hyper-parameter, for the released family and for the zero-shot trio trained without the HICO-DET share and without source-aware negatives. Axis scores are chance-corrected (§ 5). **The OVS column here is computed over A1, A2, A4 and A6**, not the five-axis OVS of Tab. 3: A5 costs one judge run per arm and was run for the released tower only, so the two numbers are not comparable. A3 is reported alongside the composite axes. Throughput and memory are measured on the training hardware (4×A100). The released tower is highlighted.

Backbone	OVS-F1	A1	A2	A3	A4	A6	HICO F1	params	img/s	peak GB	wall
<i>released family: RA-4M + 5% HICO-DET train, source-aware negatives</i>											
ViT-S/16	34.6	27.4	59.6	30.6	24.3	35.1	36.4	46.1M	114	25.1	4:52
ViT-S/16+	37.2	28.6	60.8	32.5	27.4	37.9	37.1	53.2M	109	30.0	5:04
ViT-B/16	37.2	28.9	64.4	33.4	26.8	37.2	37.5	113.8M	87	39.8	6:09
<i>zero-shot trio: RA-4M + raw Visual Genome, no HICO-DET share</i>											
ViT-S/16	34.7	23.2	53.0	31.1	27.0	38.7	18.5	46.1M	115	25.0	3:58
ViT-S/16+	34.2	23.4	53.1	32.5	24.7	40.8	18.7	53.2M	108	29.9	4:12
ViT-B/16	35.3	24.7	56.0	33.4	25.8	39.8	20.0	113.8M	87	39.8	5:10

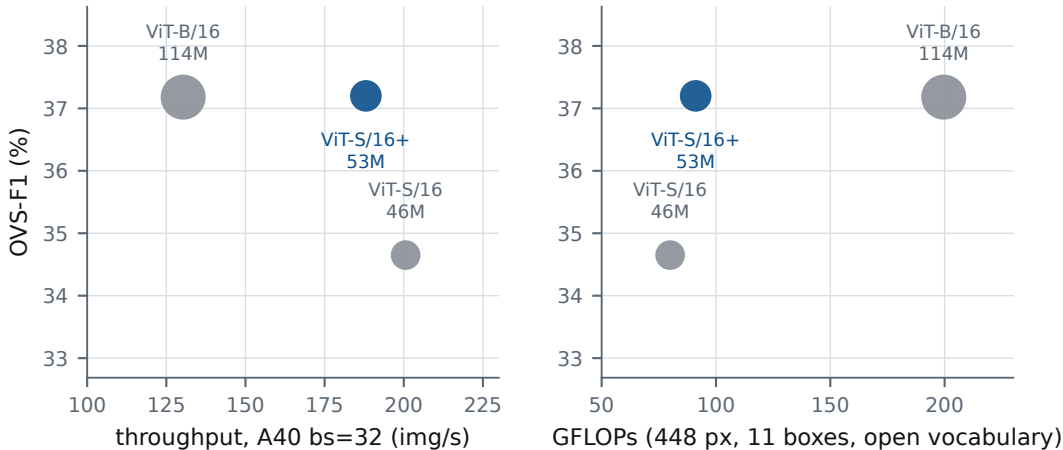


Figure 27 The released family on the cost–accuracy plane; bubble area is proportional to parameter count, and both axes come from the same benchmark as Tab. 33. Left: batched throughput (batch-1 latency is dispatch-bound and nearly identical across towers, 19.3–20.0 ms on an A40). Right: measured GFLOPs (dense operations, 448 px, 11 boxes, full vocabulary). Relative to the released tower, ViT-B costs 2.2× the FLOPs and −31% throughput for −0.02 OVS-F1, and the advantage it holds on individual axes does not survive the deployment operating point.

Tab. 29 and Fig. 27 support § 8.1. One observation belongs with them. On the zero-shot ladder the train–validation NCE gap at the final epoch is +0.083/ +0.094/ +0.159 and the best development epoch moves earlier (12 / 8 / 7), so the ladder brackets the optimum: ViT-S is still improving when training ends and ViT-B has begun to overfit, which is why “capacity-limited” applies to the S class only. Separately, a DINOv3-ConvNeXt family trained for a latency ladder places its tuned tiny variant in the ViT-S accuracy class and makes it the fastest of all at batch size one; its stage fusion is dominated by the coarse stages by an order of magnitude, and forcing the fine stages up changes nothing. It remains an option where dispatch latency dominates.

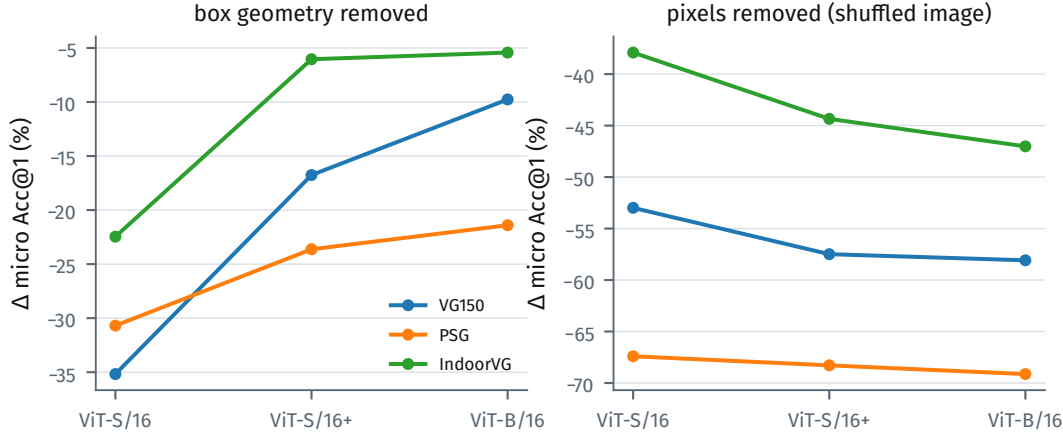


Figure 28 Lesion ladder across backbone sizes. The lesion ladder of Tab. 10 at all three sizes: the cost of removing box geometry decreases with size while the cost of removing pixels increases, in every source. Greater robustness would have moved both in the same direction.

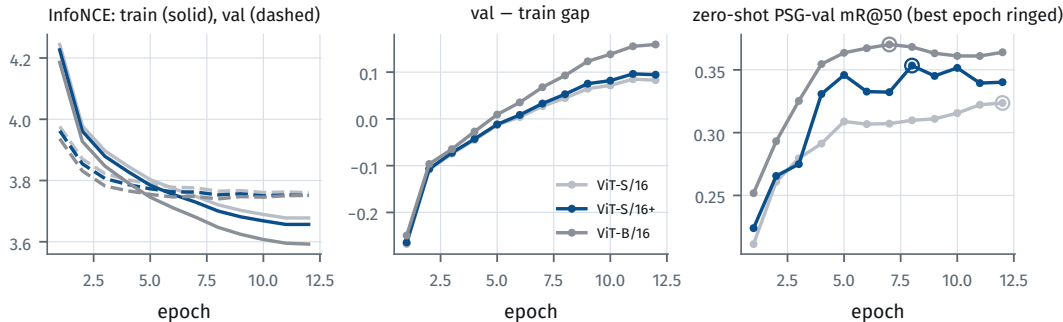


Figure 29 Training dynamics of the ladder. Train and validation InfoNCE, their gap, and the zero-shot selection metric per epoch. The gap grows with width and the best epoch arrives earlier, so the ladder brackets the optimum.

G.1 Effect of backbone capacity on transfer

The ladder of Tab. 29 was trained twice, once with the released recipe and once without the HICO-DET share, changing nothing but the tower. Capacity does not raise the composite, and on neither ladder is the composite monotone in model size. On the released ladder, $S \rightarrow S+$ costs +15% parameters and yields +7.4%, whereas $S+ \rightarrow B$ costs +114% parameters, $1.33\times$ memory and $1.22\times$ wall-clock time and yields -0.1% ; the largest tower does not outperform the released one. On the zero-shot ladder the middle rung is the weakest of the three, -1.5% below S , before B recovers +3.1%. Three observations matter more than the ordering.

Spatial axis. On both ladders ViT-S/16+ is the best of the three on A6, and ViT-B is worse than a model of less than half its size. This is the deciding evidence for the supervision-ceiling reading of § 6.6: capacity cannot substitute for supervision that is absent.

Source of the evidence. The lesion ladder of Tab. 10, run at all three sizes (Fig. 28), shows that dependence on box geometry decreases with size (from -35.2% to -9.8% on VG150) while dependence on pixels increases (from -53.0% to -58.1%), monotonically and in all three sources. The two move in opposite directions, which rules out the simpler explanation that larger models are more robust to any lesion. Larger backbones recover from pixels what smaller ones read from

explicit box coordinates: a substitution rather than an addition, and consistent with the training dynamics of [Fig. 29](#).

Deployment. Run through the deployment path, with detector boxes, its own calibration fit and its own count-matched threshold, the two larger towers tie, and S+ achieves the higher value on the only precision we can measure against adjudicated negatives (0.903 against 0.893). Benchmark recall reads 50 ranked slots per image, whereas the operating point emits 5–7 edges, about 70% of which fall on boxes that no ground-truth object matches. The capacity advantage therefore lies in a part of the ranking that deployment never emits, which is the evidence behind the choice of tower in [§ 8.1](#).

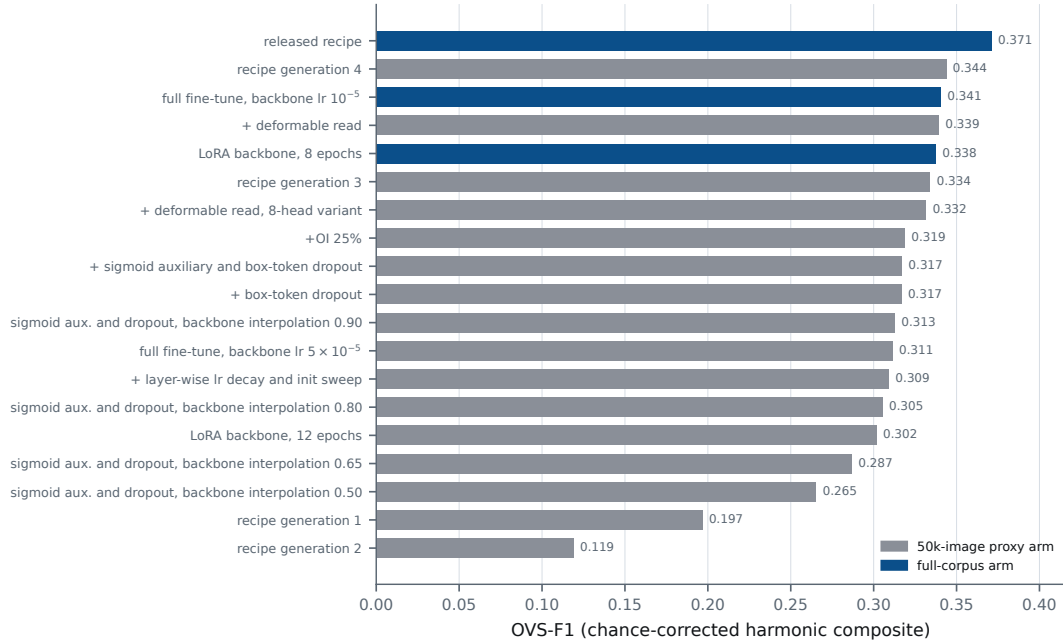


Figure 30 The recipe ladder: every arm with a complete axis set, ordered by composite. Development arms are scored on the axis set that existed while the recipe was being chosen (A1, A2, A3, A6); none of them was run in detection mode, so none has an A4 cell, and the ladder is internally comparable but not comparable with Tab. 3. Grey arms are 50k-image proxies and blue arms are full-corpus runs. The proxy winner became the full recipe.

H Ablations and negative results

H.1 Selection protocol and reproducibility

Because a full-scale arm costs several GPU-hours, the recipe was developed on a 50k-image proxy of the corpus and promoted according to criteria fixed in advance. The reliability of that proxy is itself a measured result. Its *ranking* of arms settles early: on the selection metric, the sign of the difference between two arms agrees with the twelve-epoch outcome for 82–86% of the comparisons that exceed the noise floor, at every epoch from the first. Its *levels* do not settle, the composite of the winning arm rising from 0.339 on the proxy to 0.371 at full scale. The noise floor between seed-matched reruns is 1.3%, and every difference reported in this paper should be read against it.

Two properties of the selection are worth stating, because both could have biased the outcome. The recipe was chosen on a composite over A1, A2, A3 and A6, before A4 and A5 were defined. Recomputing every selection decision on the A1/A2/A4/A6 composite changes no outcome: the same three towers pass, the same one is released, and the backbone-learning-rate comparison still resolves to the default arm, by -0.012 OVS-F1 against a $+0.003$ threshold where the earlier composite gave -0.006 . The selection is therefore not an artefact of which axes are summed. Selection also reads the final epoch rather than the best one, which predicts out-of-distribution behaviour better (§ 8.1) and inverts the ordering of some arms, including the comparison between backbone adaptation schemes.

Every number in the tables and figures of this paper is generated by script from evaluation artefacts; the only hand-entered values are literature rows, each of which carries its source. All recall is graph-constrained, evaluation is deterministic, and claims that rest on a single seed are

Table 30 The recipe ladder in tabular form: every arm with a complete axis set, the four development axes chance-corrected, and their harmonic mean OVS-dev. This is the number on which the recipe was selected, and it is not comparable with the five-axis OVS of Tab. 3, since no development arm has an A4 or A5 cell. The Open Images arm should be read against the deformable-read arm, the matched arm without that share.

arm	scale	A1	A2	A3	A6	OVS-dev
released recipe	full corpus	26.4	50.0	33.0	51.1	37.1
recipe generation 4	full corpus	23.4	49.4	31.5	46.2	34.4
full fine-tune, backbone lr 10^{-5}	full corpus	27.1	46.7	35.1	32.6	34.1
+ deformable read	50k proxy	23.7	48.4	29.5	47.4	33.9
LoRA backbone, 8 epochs	full corpus	26.8	46.2	34.2	33.0	33.8
recipe generation 3	full corpus	25.8	46.6	33.8	33.4	33.4
+ deformable read, 8-head variant	50k proxy	23.1	48.5	29.4	44.2	33.2
+ Open Images share, 25%	50k proxy	21.7	43.2	28.9	46.0	31.9
+ sigmoid auxiliary and box-token dropout	50k proxy	21.7	47.3	26.9	46.1	31.7
+ box-token dropout	50k proxy	24.3	44.7	29.5	34.5	31.7
sigmoid aux. and dropout, backbone interpolation 0.90	50k proxy	21.4	47.1	26.1	46.3	31.3
full fine-tune, backbone lr 5×10^{-5}	50k proxy	24.7	43.4	30.4	31.2	31.1
+ layer-wise lr decay and init sweep	50k proxy	21.1	46.6	25.3	47.9	30.9
sigmoid aux. and dropout, backbone interpolation 0.80	50k proxy	20.8	47.0	25.1	45.8	30.5
LoRA backbone, 12 epochs	50k proxy	23.2	43.9	27.9	32.4	30.2
sigmoid aux. and dropout, backbone interpolation 0.65	50k proxy	19.2	46.4	23.0	45.0	28.7
sigmoid aux. and dropout, backbone interpolation 0.50	50k proxy	17.2	45.3	20.8	44.3	26.5
recipe generation 1	full corpus	15.9	35.8	13.7	25.1	19.7
recipe generation 2	full corpus	13.4	33.7	5.3	22.2	11.9

identified as such. The corpus, the model weights, the benchmark tooling, and the generation and evaluation code are released.

H.2 Attributions

The following attributions come from the ladder of Fig. 30 and Tab. 30, each from a single-variable arm against a matched control, and should be read against the 1.3% noise floor.

- **Sigmoid auxiliary:** moves the projective spatial band at a small transfer cost, and makes the scale of the output head trainable.
- **Deformable scene read:** improves every axis and recovers the transfer cost of the sigmoid auxiliary; the two together constitute the full-scale recipe.
- **Layer-normalised feature taps:** free, and equivalent to a multi-level read costing $3\times$ as much, which was therefore not adopted.
- **Multi-scale sampling:** +3.7% on the development composite with 9 of 11 cells improving, and a regulariser rather than a resolution effect: it reduces the proxy’s train-validation gap by 60%, so its gain at full scale is correspondingly smaller, as that mechanism predicts.
- **Same-predicate feature mixing:** +10.1% zero-shot tail mean recall, reproduced with two seeds, whereas the identical mechanism with *random* partners yields -2.8% . The class-conditional matching rather than the mixing is the mechanism; both mixing arms have a lower training loss than the control, which rules out an augmentation effect and points to prototype smoothing.

- **Source-aware negative masking:** +12% on the spatial axis when corpora of different annotation scope are mixed (§ B.1), and the best composite of the family.

H.3 A prior violation traced to one loss term

One diagnostic is reported here because no axis of § 5 can detect it. The corpus teaches that a person wears a bounded set of garments attached to their body. Under a gated variant of the background-suppression term, the trained model violated that prior $22\times$ more often than its training data does, asserting **wearing** between disjoint boxes at a rate of 11.04%.

Three explanations were tested and rejected. A vocabulary explanation fails, since 85% of the type errors involve body parts absent from the source ontology, so the model is not choosing incorrectly among available categories. A graph-reasoning explanation fails, since the fan-in is 97% co-reference, that is, the same garment attributed to several boxes of one person, which requires no relational inference. A spatial-shortcut explanation fails in the opposite direction, since probes show that the entity features already identify body parts correctly, with 66.9% assigned to the correct family. Removing the gate reduced the disjoint-box **wearing** rate to 0.00% at no cost elsewhere, and the released model does not use it. None of A1–A3 could have detected this failure, because the benchmarks’ own annotation never asserts the violated pairs in either direction. Checking a model’s output against a prior stated by its training data is an inexpensive measurement that is not currently made.

H.4 Negative results

Each direction in Tab. 32 was evaluated and not adopted on the basis of the measurement listed. None supports a claim in the main text, and every number is a single arm read against the 1.3% noise floor. Tab. 31 reports one of them, evaluation resolution, in full.

Table 31 Evaluation resolution for the released tower, trained multi-scale at $[0.5, 1.5] \times 448$, without retraining. Test splits, ground-truth boxes, graph-constrained.

eval px	VG150			PSG			IndoorVG			A6
	R@50	mR@50	rare	R@50	mR@50	rare	R@50	mR@50	rare	AUC
448	52.2	28.1	36.2	39.2	28.4	18.7	51.1	29.7	22.0	0.689
560	52.3	27.7	29.0	39.5	28.7	18.7	51.5	29.9	22.2	0.686
672	52.3	28.5	39.7	39.7	29.5	20.6	51.7	30.8	24.1	0.686

Table 32 Directions evaluated and not adopted, with the measurement that decided each.

direction	measurement	conclusion
<i>modelling and training</i>		
stochastic depth [Huang et al., 2016] WiSE-FT [Wortsman et al., 2022] toward the pre-trained backbone	train-validation gap reduced 8.8×; benchmark change approximately zero monotone loss at every interpolation weight	the generalisation gap and the transfer gap are different quantities the transfer cost is not backbone drift; a jointly trained head invalidates the premise
model soups over seeds	averaging same-recipe, different-seed checkpoints halves A1	a new seed re-randomises the head, so the arms lie in mismatched basins; shared initialisation untested
masks as training input	−18% under the released box input; +6.6% only when masks are supplied at evaluation to a box-trained model	train on boxes, accept masks at inference
multiplicative (Tucker) pair queries	used by the network (norm grows), composite-neutral over two seeds; one robust cell, zero-shot PSG macro recall +4 to +7%	not adopted
head width 768	−2.7% on the largest backbone; the predicate bank has an effective rank of 43.7	12M head parameters already suffice for a target of rank about 44
<i>text space and decoding</i>		
CSLS hubness correction [Lample et al., 2018] at the synonym matcher	negative; the cosine threshold it was calibrated against accepted 0.6% of true synonyms in the released space	every similarity threshold is calibrated to one embedding space
commit-late decoding	failed with three aggregators for one cause	no certified cross-lemma synonym resource exists to aggregate over
predicate box embeddings	entailment geometry on frozen queries trades against ranking: VG150 mR 0.37 / 0.18 / 0.13 against Spearman 0.17 / 0.55 / 0.60 as the entailment weight rises	a Pareto trade-off rather than an improvement; joint training untested
<i>evaluation and deployment</i>		
evaluation-resolution tiering	flat, and shrinks with model size (Tab. 31); 1.5× resolution yields +10% rare-bucket recall and nothing on micro recall or the spatial axis	multi-scale training already provides the robustness
letterboxing	corrects a genuine 5.2° angle error; projective spatial axis +0.1%	the model does not use precise angles
CUDA-graph capture	3.45× faster and silently incorrect; graph compilation moves evaluation metrics by 40× the noise floor	no reported number uses either
per-predicate calibration	fits in-domain and does not transfer	two global Platt parameters transfer, per-predicate ones do not

I Inference cost details

Cost was measured on three GPU classes under one protocol: batch 1 on real VG150 test images, CUDA-event timed, median of 200 frames after 20 warm-up frames. Batch 1 is the quantity relevant to deployment; batched throughput is reported alongside it because the two disagree.

Table 33 Cost of the released family, relation head only, with an open vocabulary of 19,103 strings, in bf16. GFLOPs count dense operations at 448 px with 11 boxes and are GPU-independent. The vocabulary matrix used here is a shape-matched stand-in for the distilled student. Cost depends only on the shape of that matrix, so the timings are exact, but nothing in this table is an accuracy claim; [Tab. 35](#) uses the actual bank.

Backbone	GFLOPs	batch 1, ms			batch 32, img/s		
		A40	A100	H100	A40	A100	H100
ViT-S/16	80	19.5	27.6	18.9	201	387	708
ViT-S/16+	91	20.0	28.4	19.4	188	367	678
ViT-B/16	200	19.3	27.5	18.9	130	261	504

Table 34 Cost of the released tower on three GPU classes, with the actual 243-predicate release vocabulary for the end-to-end rows and the full 19,103-string vocabulary for the relation-head rows, batch 1 unless stated.

Stage	Configuration	A40	A100	H100
relation head, batch 1 (ms)	bf16, 19,103-string vocabulary	19.3	28.1	19.3
relation head, batch 32 (img/s)	same	188	367	678
detector + head + decode, batch 1 (ms)	PyTorch eager	30.5	43.1	32.1
detector + head + decode, batch 1 (ms)	PyTorch, <code>torch.compile</code>	20.3	24.4	18.1
detector + head + decode, batch 1 (ms)	ONNX Runtime, CUDA	29.7	27.6	21.2

Dispatch-bound latency at batch 1. [Tab. 33](#) gives the cost of the released family and [Tab. 34](#) the cost of the released tower across three GPU classes. The three towers cost 19.3–20.0 ms on an A40 despite a $2.5\times$ range in parameters and FLOPs between the ends of the ladder, so at batch 1 the model is bound by kernel dispatch rather than by arithmetic. The ordering appears only at batch 32 (201 / 188 / 130 img/s), which is why the argument for the smaller tower rests on throughput rather than on batch-1 latency.

Cost of the open vocabulary. Scoring against the full 19,103-string vocabulary instead of a 50-predicate one costs 0.7–0.8 ms per frame on an A40 and, within noise, nothing on the A100 and H100. At batch 32 the same head costs 11–22% of throughput on every tower. Supplying the vocabulary as an input therefore has no cost on a live frame and a moderate cost on an offline batched workload.

Host dependence of batch-1 latency. The A100 is the slowest of the three GPUs at batch 1 (27.5–28.4 ms against 18.9–20.0 ms on the A40 and H100) while delivering $1.9\text{--}2.0\times$ the throughput of the A40 at batch 32. This is not contention, since it reproduces on an idle second A100 node (27.2 / 28.1 / 27.2 ms). The A100 is attached to a Zen3 host and the A40 and H100 to Zen4 hosts, and a dispatch-bound workload is a host-CPU workload; the same pattern appears in the host-bound graph-inference stage of the baseline. At batch 1, a latency on a named GPU therefore does not

identify an operating point unless the host CPU is also named, which is why the figures here are given per GPU.

Table 35 The deployment path of the released tower, end to end with an open-vocabulary detector, batch 1, with the actual 243-predicate release vocabulary. The compiled rows are deployment figures only: compilation changes the reduction order and moves evaluation metrics by far more than the noise floor, so none of the accuracy tables uses compiled inference.

Configuration	A40		A100		H100	
	p50 ms	FPS	p50 ms	FPS	p50 ms	FPS
eager, sequential	30.5	33	43.1	23	32.1	31
eager, concurrent backbone	27.9	36	41.9	24	59.3	17
torch.compile, sequential	20.3	49	24.4	41	18.1	55
torch.compile, concurrent	22.3	45	31.0	32	55.1	18
CUDA graphs, concurrent	21.5	47	29.9	33	50.8	20
ONNX Runtime, CUDA EP	29.7	34	27.6	36	21.2	47

Compilation and concurrency. Tab. 35 times the same model end to end. Compilation contributes more than any other change evaluated, at 20.3 ms on an A40 and 18.1 ms on an H100 against 30.5 and 32.1 ms for eager execution, and it is the fastest configuration on all three GPUs. Since the relation backbone reads only pixels, it could in principle run concurrently with the detector. Measured, that overlap recovers 2.6 ms on an A40 under eager execution and costs 27 ms on an H100, and costs time on both once the model is compiled; the pattern held across five configurations, two thread settings and three detectors, so the sequential path is used. Tab. 36 reports the same pipeline on each deployment device.

Table 36 Deployment targets for the released ViT-S/16+ tower, detector + relation head + decoding, batch 1; *agreement* is the top-1 / top-20 Jaccard agreement of the emitted triplets with the fp32 graph. Every row names its device; the CPU rows are measured on a cluster node.

Target	Runtime	Device	ms	FPS	agreement
GPU, A40	PyTorch 2.13, torch.compile, bf16	NVIDIA A40 (host AMD EPYC 9334)	20	49	—
GPU, A40	ONNX Runtime, CUDA EP	NVIDIA A40	30	34	—
CPU	OpenVINO IR, fp16	Intel Xeon 6338, 8 threads	137	7	0.955 / 0.971
CPU	OpenVINO IR, int4 weights	Intel Xeon 6338, 8 threads	126	8	0.727 / 0.771

J Generation prompts

The three passes of § C use the prompts below verbatim. They are reproduced in full because the calibration study that selected them, coverage over verify-before-assert, can only be interpreted against the actual text.

J.1 Semantic pass

You are a precise visual scene analyst. Extract visually grounded relationships between the {n}
↪ objects marked with coloured numbered dots in the image.

Objects (number → label):
{object_list}

Follow these steps carefully:

STEP 1 -- Observe what each object IS DOING: Before reasoning about pairs, note the activity,
↪ state, or role of every object.

What is it doing? (grazing, running, sitting, flying, eating, carrying a load, casting a
↪ shadow, etc.)

What is it interacting with in the scene? (the ground, another object, a person, food, etc.)

STEP 2 -- For each pair, find the MOST INFORMATIVE relationship:

Priority order -- stop at the first that applies:

(a) Activity / locomotion -- what is one object doing relative to the other or its environment?

Open vocabulary: use any precise verb phrase (grazing on, eating from, running across, standing on, walking through, casting shadow on, drinking from, pecking at, chasing, herding, pulling, pushing, swimming in, flying over, climbing, digging in ...)

(b) Physical interaction -- direct manipulation, support, containment:

holding, wearing, riding, carrying, sitting on, resting on, leaning against, attached to, part of, mounted on, hanging from, covering, surrounding, inside

(c) Directed attention / social:

looking at, watching, talking to, pointing at, following, leading

Critical rules:

- DO NOT output any spatial or directional predicate -- spatial layout will be handled
↪ separately
- Prefer open-vocabulary action phrases: describe WHAT objects are doing, not WHERE they are
- Use DIRECT predicates -- no "is", "are", "was", "were" prefix
- Each (subject_id, object_id) pair may appear AT MOST ONCE in your output

STEP 3 -- Select relations:

- Aim for 6-10 relations total -- quality over quantity
- Every numbered object should appear in at least one relation
- If an object genuinely has no informative relation to any other marked object, skip it
- DO NOT pad with spatial or directional predicates

STRICTLY BANNED predicates -- your output is invalid if you use any of these:

has, near, next to, alongside, by, with, on, beside, and, same scene, overlapping, adjacent, associated, related, interacts, features, involves, appears, is positioned, is situated, is located, faces, supporting, displaying, above, below, behind, in front of, to the left of, to the right of, between, standing behind, standing in front of, standing beside, standing next to, sitting behind, sitting in front of, sitting beside, parked behind, parked in front of, parked beside, walking behind, walking in front of, flying above, hanging above, positioned, located, placed, arranged, oriented

STEP 4 -- Output: After your reasoning, output a single JSON block with no extra text outside it:

```
```json
{"relations": [{"subject_id": <int>, "subject_label": "<str>", "predicate": "<str>",
↪ "object_id": <int>, "object_label": "<str>"}]}}
```
```

J.2 Grow pass

Here is an image and a PARTIAL scene graph for it -- relationships already found between the
↪ numbered objects.

Objects (number → label):
{object_list}

Relationships already found:
{draft}

Look carefully at the image and ADD relationships that are genuinely present but MISSING from the
↪ list above -- find what the previous pass overlooked:

- Objects that appear in FEW or NONE of the relationships above -- what is each doing, or what is done to it?
- A genuinely DIFFERENT relationship for a pair already listed (e.g. a rider both "sitting on" AND "holding" a horse) -- but NEVER a synonym or rephrasing of one already
↪ there.
- Any action, physical contact, support, use, or attention you can see that is not yet listed.

Rules:

- Output ONLY NEW relationships. Do NOT repeat any relationship already in the list, and do NOT output the inverse of one already there.
- Use the MOST PRECISE open-vocabulary predicate; direct form (no "is/are"); never use vague words (near, next to, with, on alone, has, positioned, located, situated).
- Report only what is genuinely visible. If nothing is missing, output an empty list.
- The dots, numbers and outlines are markers added to the photo -- never describe them.

Output a single JSON block with ONLY the new relationships, nothing else:

```
```json
{"relations": [{"subject_id": <int>, "subject_label": "<str>", "predicate": "<str>",
↪ "object_id": <int>, "object_label": "<str>"}]}}
```
```

J.3 Spatial pass

Here is an image with numbered objects and their SEMANTIC relationships already found.

Objects (number → label):
{object_list}

Semantic relationships already found:
{draft}

Your task: describe the SPATIAL LAYOUT of the scene -- where the objects are relative to each
↪ other. Choose the pairs whose arrangement best DEFINES the scene's layout, most informative
↪ first.

Spatial vocabulary (pick the most precise fit -- close natural synonyms are also fine):

depth in front of / behind → who is closer to the camera, who
↪ occludes whom
contact on / resting on / leaning against / hanging from / attached to

| | | |
|-------------|-------------------------------------|---|
| proximity | next to / beside / near | → physical support or contact you can SEE |
| containment | inside | → clearly close together in the scene |
| | ↪ (food inside a bowl) | → one object genuinely within another |
| vertical | above / below / beneath / on top of | |
| horizontal | to the left of / to the right of | |

Rules:

1. PREFER depth, contact and proximity -- they say the most about a scene. Use "to the left of" / "to the right of" only when the horizontal offset is the single most informative thing about that pair.
2. COVER the scene: every major object should appear in at least one layout relation. Skip only pairs whose arrangement is genuinely ambiguous or adds nothing.
3. One relation per pair, judged from the VIEWER's perspective. Never also output the inverse (if A is in front of B, do not add B behind A).
4. Do NOT use "between" (it needs two reference objects). Do NOT use action verbs -- actions were covered above; this pass is layout only.
5. Do NOT place body-part objects (human face, hand, arm, leg ...) -- their position is implied by the person they belong to.
6. Do NOT repeat a pair already covered above unless the layout adds genuinely new information (e.g. "man riding horse" already implies he is on it).
7. Target 6-12 relations; fewer only for genuinely sparse scenes (2-3 objects).
8. The dots, numbers and outlines are markers added to the photo -- never describe them.

Output a single JSON block with ONLY the new spatial relationships, nothing else:

```
```json
{"relations": [{"subject_id": <int>, "subject_label": "<str>", "predicate": "<str>",
↪ "object_id": <int>, "object_label": "<str>"}]}
```